Collaboration-Based Learning Environments Using Augmented Reality

by

Sebastian Gil Parga

Submitted to the Department of Creative Technologies
in partial fulfillment of the requirements for the degree of

DOCTOR OF PHILOSOPHY IN CREATIVE TECHNOLOGIES

at the

AUCKLAND UNIVERSITY OF TECHNOLOGY

November 2025

Authored by: Sebastian Gil Parga
School of Future Environments, Department of Creative Technologies
November 30, 2025

Certified by: Stefan Marks
Associate Professor, Thesis Supervisor

Certified by: Jairo Gutierrez
Professor, Thesis Supervisor

Collaboration-Based Learning Environments Using Augmented Reality

by

Sebastian Gil Parga

Submitted to the Department of Creative Technologies
on November 30, 2025 in partial fulfillment of the requirements for the degree of

DOCTOR OF PHILOSOPHY IN CREATIVE TECHNOLOGIES

ABSTRACT

The developments of the "kinetic theory" of gases made within the last ten years have enabled it to account satisfactorily for many of the laws of gases. The mathematical deductions of Clausius, Maxwell and others, based upon the hypothesis of a gas composed of molecules acting upon each other at impact like perfectly elastic spheres, have furnished expressions for the laws of its elasticity, viscosity, conductivity for heat, diffusive power and other properties. For some of these laws we have experimental data of value in testing the validity of these deductions and assumptions. Next to the elasticity, perhaps the phenomena of the viscosity of gases are best adapted to investigation.[1]

Thesis supervisor: Stefan Marks
Title: Associate Professor

Thesis supervisor: Jairo Gutierrez
Title: Professor

[1]Text from Holman (1876): doi:10.2307/25138434.

Acknowledgments

Write your acknowledgments here.

Contents

List of Figures

List of Tables

List of Acronyms

VR Virtual Reality

AR Augmented Reality

TEL Technology Enhanced Learning

MR Mixed Reality

CSCW Computer Software for Cooperative Work

CL Collaborative Learning

CoP Community of Practice

VLE Virtual Learning Environment

LMS Learning Management System

BRT Bloom's Revised Taxonomy

SLR Systematic Literature Review

Chapter 1

Introduction

In the Netflix animated series Cyberpunk: Edgerunners (Imaishi et al., 2022), the main character attends courses at a prestigious academy that uses advanced technology to boost education and performance. The students gather in a room with Virtual Reality (VR) headsets and connect to the lesson, which is taught by a holographic (and probably powered by artificial intelligence) tutor. The episode does not show the lesson itself (the entire system fails catastrophically when attacked by malware), but the brief scene highlights the vision many people have of a futuristic classroom, one guided by advanced technology and automatization.

Forsler et al. (2024) propose the term “post-digital classroom” to identify the characteristics and trends in education present in a world beyond the adoption of digital technologies. The purpose of this classroom is not anymore to introduce new technologies to students, but to implement them as an essential factor of the learning process. The post-digital classroom is interconnected, social and global. In this scenario, learning goes beyond the physical classroom because it is based on creating relationships between concepts and developing valuable skills rather than acquiring knowledge. It is also a classroom where information is detached from the physical learning institute, and access to facts and sources is, ideally, immediate, ubiquitous and democratic.

The classroom has also become hybrid. There is little distinction between network-based lessons and the face-to-face spaces associated with schools and universities (Goodyear et al., 2004). Hybrid and blended approaches blur the distinctions between learning in an online setting and learning in a classroom. In both instances, information is at hand in the network, students can easily connect to peers and mentors, distance and time constrains are less meaningful and knowledge is presented in a multimedia format that needs a technological backbone to be created and shared (Rivoltella, 2008; Shank, 2005).

This thesis explores the use of technology in such post-digital classrooms, specifically, AR within the context of mobile computing. Moreover, this research will analyse the creation of effective learning spaces, those places where students can build, personalise and control all the activities they engage with (Gourlay & Oliver, 2016). Learning spaces are interesting tools at the disposal of students because these are spaces that can exist outside the norm of a typical classroom, they are more fluid, less structured and can easily connect to other similar spaces (Nørgård & Hilli, 2022). The proposal of this research is that AR can be used to boost the way students build and interact with their own learning spaces and can also

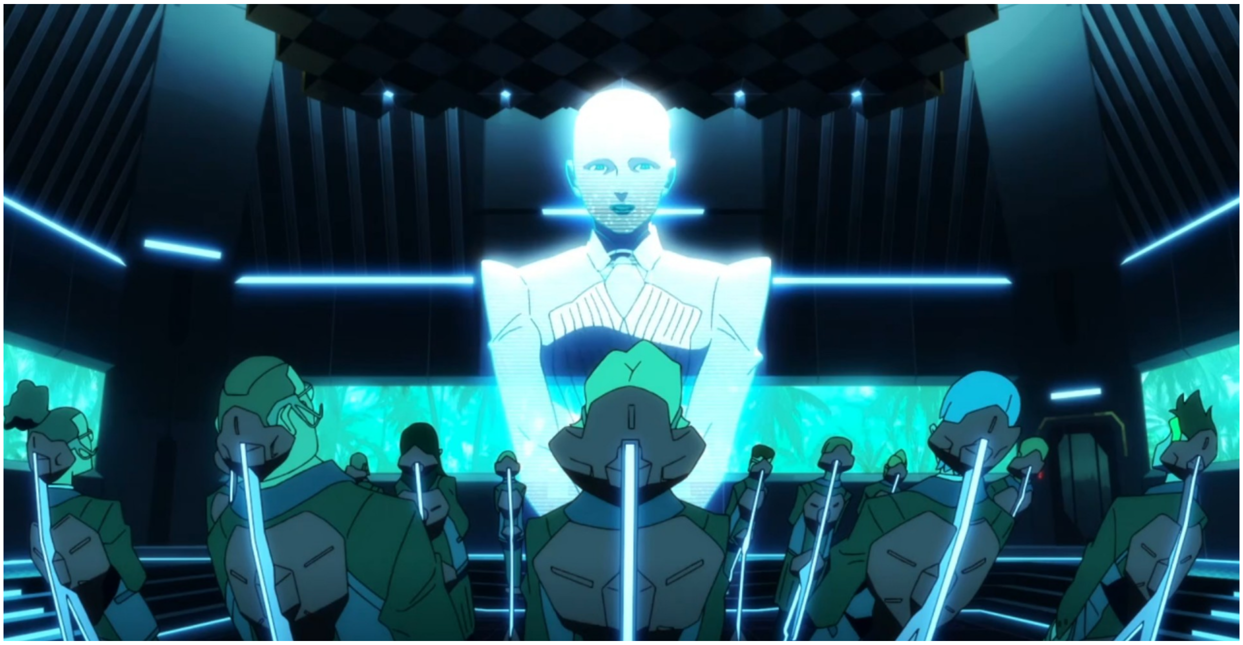

Figure 1.1: Virtual Classroom in Cyberpunk: Edgerunners

offer educators another option to be integrated in the construction of a lesson that capitalises the characteristics of the aforementioned post-digital classrooms.

Within the field of Technology Enhanced Learning (TEL), AR and related technologies were chosen as a focus of interest because of their intrinsic relationship with space. The initial example scene of Cyberpunk: Edgerunners serves to illustrate the collective idea of what could be achieved in the future with technologies like VR and AR: the possibility of creating an entire new space, suited for the needs of the classroom or the students. It is a space that is shared, in which information and interactions flow between users and that can be controlled at will by students and teachers. In reality, the technology is not quite there yet, especially in terms of fidelity and immersion (Checa & Bustillo, 2020; Hamad & Jia, 2022). What can be explored right now is the creation of interesting synthetic environments that facilitate or allow experiences that are not possible in the common classroom. This process is limited more by the creativity and expertise of the developer rather than by the technology, and therefore can be explored easier and in more detail.

AR, compared to VR, has a more direct relationship with the *real*, physical space and with the context of the user. Instead of an isolated immersion, AR creates a connection with the environment based on image detection, data and visualizations. This natural expression of the technology is where the value of AR can be explored in relation to the creation of innovative learning spaces. The technology not only has interesting capabilities for visualization and interaction, it also allows an easier implementation of shared, social components that can be used to build activities based on cooperative simulations, sharing knowledge and the creation of communities, all crucial elements for the construction of effective learning spaces (Bligh & Crook, 2017).

The research is positioned then in the conjunction of these three fields: Learning spaces, TEL and AR. The guiding goal is to understand how students are using technology to learn, and how AR can be used to boost that process. AR shines at creating immersive experiences that can be shared with others and that consider the physical context of the user. I want to use this collaborative space as an educational tool that takes advantage of the inherent social aspects of learning, building and sharing knowledge.

No modern classroom is free of technology, the simple act of taking notes on a paper already implies a set of technological developments needed to put knowledge into written form. Networking and information technologies are just another set of tools at the disposition of any student. A personal learning space can incorporate different software tools, many of which are now ubiquitous and almost indispensable in any learning activity: text processors, multimedia editing tools, web browsers and document readers, plus any specialised tool that the learning subject requires. But that learning space is also built around more elements, some of them equally indispensable for some students. People need a space to work, access to different amenities, access to textbooks, information and teachers. Some may need a quite space to concentrate, or, on the contrary, may need access to music, to snacks or to a separate place to periodically take a break. The construction of a learning space involves many different elements, some more technological than others, and its effectiveness is based more on the integration of those elements than anything else (Lu & Churchill, 2013; Reinius et al., 2021). This research aims at identifying how to properly integrate AR in the construction of a learning space, which is already composed by a complex web of many other objects, tools and circumstances.

As a researcher. I have explored in the past the way different technologies can be used as part of those learning spaces, either with tools for online learning scenarios or using VR to create a digital space to explore multimedia content (Gil Parga, 2015). This current project can be seen as a way to expand on the work done before, identifying ways to adapt a different technology, to apply lessons learned and to explore new avenues of research. In particular, it is an opportunity to expand from personalized learning environments into social learning spaces, and explore the role of the community in teaching and learning.

It was clear in the work done previously and through some notable literature like Richards (2023) and Long and Ehrmann (2005) that the construction of learning spaces is always mediated by the community around us, even when creating a space as personal as possible. Even nowadays that we live in a post-pandemic world where we seek remote, personalised and secure ways of learning, it has become clear that the social aspect of learning is a crucial element in the equation (Baber, 2021; De Felice et al., 2023). The aim of this research is to delve deeper into the social elements that influence the construction of a learning space, how the students negotiate the process of learning with others and the benefits of collaboration, as well as the challenges it conveys.

This is also a technological enquiry. Having explored VR in the past, it was clear that this family of technologies offer interesting advantages by using complex visuals, new forms of presenting information and more natural interactions suited for the exploration and experimentation with 3D spaces. These technologies also present curious challenges, especially in the usability front. For instance, the public has not yet developed a strong relation or consistent literacy with the technology (Özgen et al., 2021), which creates accessibility problems. And those technologies that leverage common and accessible platforms, like mobile devices, risk carrying with them problems associated with the excessive and pervasive presence of such technologies (Forsler & Guyard, 2025; Petrucco, 2021).

The following introductory chapter will expand this brief exploration of the context and motivation for the research proposed, outlining the main research goal and the reasoning behind the formulation of the selected research questions. There will be a brief exposition of the methodology followed to build possible answers, the significance of this research and why the answers to the questions proposed can provide important insights about the way we use technology to learn and how we design technological tools for the classroom. It will be followed by a summary of the results obtained and the contributions achieved, as well as the scope and limitations of the work done. The chapter will conclude with a description of the structure of this thesis document.

1.1 Context

AR technologies are positioned somewhere in the middle of the reality-simulation spectrum proposed by Milgram et al. (1995), a model for display technologies used to show digital or physical data in virtual or synthetic realities. The main characteristic of AR in this spectrum is the incorporation of the real world as part of the experience, making use of different image recognition and composition techniques to identify features in the physical space and overlay them with digital information.

Schmalstieg (2016) adds another layer of analysis to the description of AR-like technologies by using the ubiquitous computing spectrum proposed by Weiser (1991). The model positions different implementations of AR depending on the use of more static or more mobile deployment platforms. In one end of the spectrum are applications that can only be used in singular static terminals, while the other side contemplate platform-independent mobile technologies and ubiquitous systems. Figure 1.2 shows examples of this distribution as proposed by the authors.

In this bi-directional spectrum we can find mobile AR as a family of technologies that offer an in-between mixture of virtual and physical elements in which location, place and context are additional inputs used for interaction with the synthetic reality. Any technology in the spectrum will offer a variety of characteristics that align with one or more descriptions at a time. Mixed Reality (MR), for example, are seen as an extension of both VR and AR, offering seamless interaction and connectivity between the real and the simulated world. It is a blurry line that often must be analysed in a case-by-case approach. No matter what label is used to describe a technology, some agreed upon characteristics must be present for it to be categorized as AR, as stated by Azuma (1997):

- A combination of real space and digital information.
- Interaction with the simulation in real time.
- The simulation is done in a 3D space.

With these descriptors it is possible to position the used of the technology in different fields, independent of the labels used to identify it. Based on the themes introduced in the opening statement of the chapter, the goal is to analyse AR as CSCW. For this task, a useful analytical tool is the classification of CSCW technologies proposed by Rodden (1991)

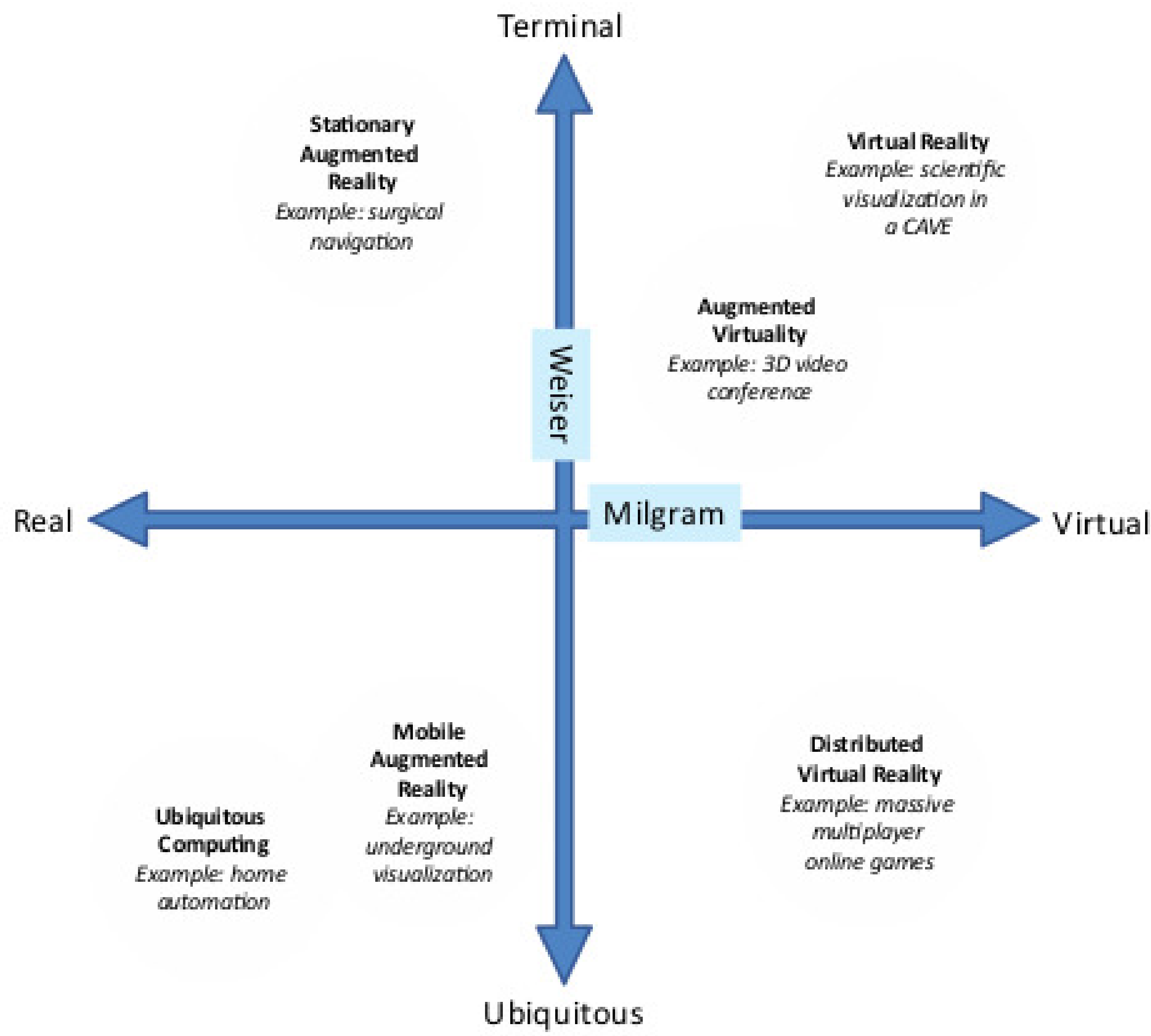

Figure 1.2: AR in the Milgram-Weiser Spectrum (Schmalstieg, 2016).

in terms of the point in time and space where the collaboration is taking place. Figure 1.3 shows examples of different systems that can be found in each quadrant of the classification, which indicate whether the activity is synchronous or asynchronous, remote or co-located.

Complementarily, Kiyokawa et al. (2002) offer two important elements present in a collaborative activity: communication and tasks spaces. In the communication space, people exchange information and can sustain formal and informal conversations. In the task space, the actual cooperative work is done. Those spaces can be unified or separated, and they can be physical, virtual or in a mixed state. Independent of the nature of each individual space, both need to be present for collaboration to succeed.

Based on these classification models, it is possible to define the main field of interest of this research as the use of AR in a mixed task space for face-to-face collaborative learning, which implies synchronous activities in a co-located space. In this setting it should be possible to use the capabilities of AR for visualization and interaction with the physical space to

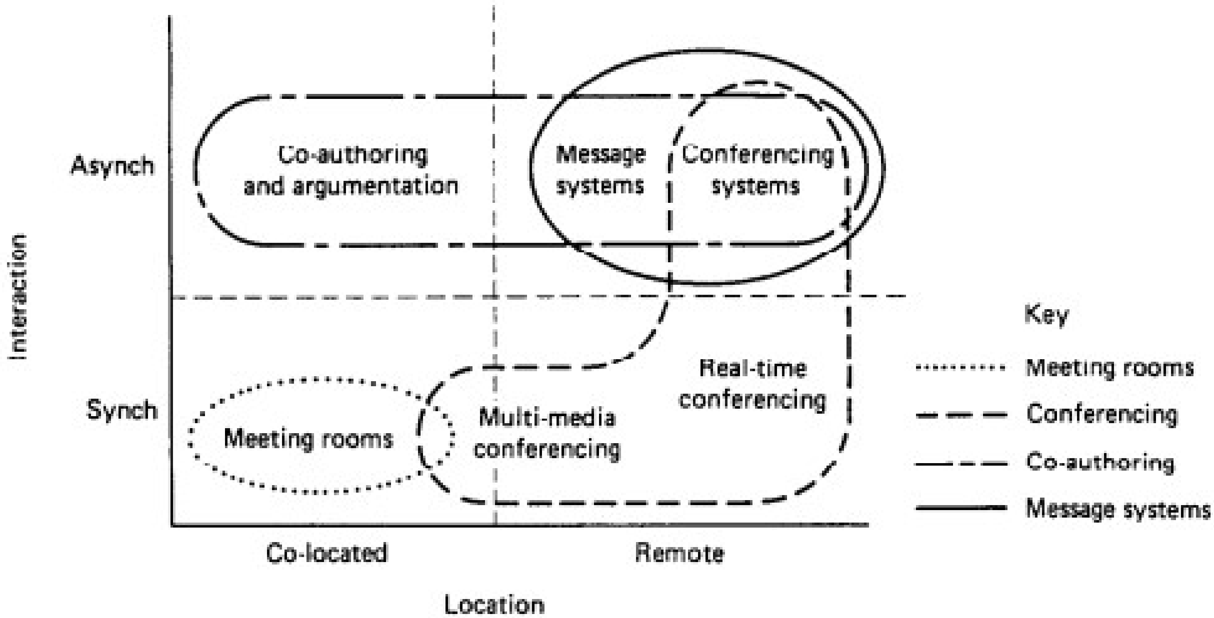

Figure 1.3: Classification of CSCW Systems by Rodden (1991).

augment the collaboration process in terms of:

- Activities related to the communication space, the flow of information and the creation of knowledge in the group.
- The use of the physical space as an asset in the collaborative process.
- Management of the collaborative process itself and the structures that can be created to facilitate effective teamwork and cooperation.
- Guidelines to acquire and refine skills related to teamwork and collaboration.

The main proposal is that the unique characteristics that AR provides in the field of MR are well suited to experiences in co-located collaboration, and that a learning design that is using a collaborative approach can benefit from an AR tool mediating in the communication and tasks spaces of the activity.

The interest in collaborative learning derives from the analysis of the use of AR in education and the tendencies identified in the available literature. It was possible to note that AR is often used as a tool to explore concepts and practices in hands-on approaches that apply previous knowledge in different situations and contexts. Unfortunately, the technology is seldom used in cooperative scenarios, despite the tendency to use mobile platforms for deployment (Gil Parga et al., 2023).

Collaborative learning is a subject rich in discussion. The conversation can focus on the benefits of groupwork in the classroom like promoting different points of view, facilitating complex understanding of a topic and creating the space for an integral development of the student (Smith et al., 1992; Van Wyk & Haffejee, 2017). On the other hand, the challenges associated with collaborative work are also amply explored (Baloche & Brody, 2017; Buchs et al., 2017; Popov et al., 2012), and there is plenty discussion around issues related to classroom management, inclusivity, effective assessment and good practices.

Many studies have identified and compiled the advantages of using AR and MR as learning tools. For instance, Penn and Ramnarain (2023) offer an analysis of the educational foundations used to design learning interventions. They conclude that most projects converged into constructivist foundations and embodiment theory, opting to create spaces for students to experience and explore the learning subject, with an emphasis on immersion, practice and hands-on interactions. Despite these commonalities, the authors also remarked that most reports do not mention explicitly or do not rationalize the theoretical background of the intervention. They also note that a lot of researchers still hold the idea that introducing a new technology in the classroom is enough to reap its benefits, without caring to build or follow a proper integration plan.

This issue acquires even more relevance now that the mediation of different technologies in the collaboration process has grown due to the Covid-19 pandemic, which forced a quick and often poorly planned implementation of remote learning (Drueke et al., 2021; Nuere et al., 2021). Over time, this prompted the re-design and re-thinking of how to use technology to support the sudden disruption in learning methodologies and approaches (Adi Badiozaman et al., 2020; Fischer et al., 2020). The forced isolation also highlighted the importance of social spaces in the constructions of learning environments, and the need to provide those social interactions even in online spaces (McInnerney & Roberts, 2004; Walas-Trębacz et al., 2025).

The post-pandemic classroom is dealing now with the need to balance the advantages of online spaces with the problems associated with socialization solely mediated by technology, which tends to be asynchronous and impersonal. There is value in the physical space for the learning process (Paechter & Maier, 2010; Petchamé et al., 2021; Thai et al., 2020), and a technology that is able to connect the physical and the digital is worthy of a deeper exploration to identify proper approaches and successful use cases.

The idea is not to oversimplify the presence of digital technologies in the classroom, nor follow a misguided tendency of selling all the benefits of a technology without acknowledging its problems. If the proposal is to identify the effects of a collaborative learning design mediated by AR, it is important to also identify all the problems that derive from this approach, either because of the issues that naturally come from teamworking scenarios in the classroom or the negative behaviours that the excessive use of digital technologies promotes, especially in the specific environment of mobile platforms.

With that context just detailed, it is possible to summarize the goal of this research as: The construction of an AR tool that serves as aid in a collaborative learning setting by helping students to communicate and develop teamworking skills. This can be achieved by means of interactivity, visualization, immersion and augmentation of the physical space.

1.2 Research Structure

The goal of the research can be justified by an abundance of literature showing the elements of AR used in the past to offer distinctive value for learning, and the open opportunities to explore and understand better the development of multi-user AR experiences for collaboration.

The nature of the proposal implies that the research has two main fronts:

- In the *technological* front, the goal is to build a software solution that considers the development challenges of an AR tool in a multi-user and multi-platform environment, and the specific requirements derived from the learning goals proposed in the educational front.
- In the *educational* front, the goal is to identify, describe and develop a case study based on collaborative learning, and to analyse the behaviours that students develop when learning in a collaborative environment mediated by technology, comparing the use of AR with other tools or approaches.

1.2.1 Research Questions

The guiding research questions of this project are aimed at obtaining information of what constitutes an effective collaborative learning design using AR. The answers can be stated in terms of learning goals, interaction design and integration plans with the classroom. The questions also seek information about how AR is creating an effect in the behaviour of the students while using the tool for learning. These behaviours can be related to communication, socialization, planning and engagement with the learning material. The main research question will be stated as follows:

> MQ: How does the behaviour of a group of students changes when using an AR tool designed to aid in the achievement of a learning goal framed in a collaborative learning scenario?

To guide the implementation of the tool through a strong design, a lot of information can be extracted from current literature and from previous projects using similar approaches. Information gathered from real and incremental usage off the system also offers valuable insights in terms of usability, viability and end-user perception. With this in mind, an iterative development process is proposed to facilitate the exploration of ideas regarding interaction design and a proper fit to the learning scenario where the tool is going to be used.

The implementation of the tool alone is insufficient to answer the main question. For this purpose, a case study research methodology is proposed with the objective of identifying and describing a strong context in which collaboration is used for learning and in which the implementation of the AR tool can be contrasted and compared to other approaches. In this context it is also easier to observe more natural behaviours from the students and the observations will show results closer to how the tool could be used in a real deployment.

It is very important to carefully select a case study that not only offers unique circumstances valuable enough to be analysed in detail but also has the proper learning context to explore the themes of interest of this research project. Although the specific learning field of a prospective case study is not that important, it needs certain characteristics, independently if it is developed through a single activity, a section of a course or through the whole course. The case study must incorporate all the following elements:

- A clear learning goal, either as a whole or as a step to achieve a more complex goal. Assessment activities alone are not enough because the research objectives require the observation of collaborative learning, not the development of a single teamwork activity.

- A collaborative approach to the learning activity. Students must be expected to work in groups to achieve the task, or at least, to have access to a space for collaboration in the development of the activity, even if it is individual in nature.

Ideally, the case study also contemplates the process of learning to collaborate, either as its own learning goal or willing to implement such an objective as a secondary goal. This is not essential, but the current proposal borrows a lot of elements from learning designs about developing collaborative skills. If the selected case study already seeks learning objectives in this venue, then the proposed intervention could offer more value.

The selected case study offers an excellent opportunity, not only to experiment with the proposed thematic of collaboration and technology mediated learning, but because it is also a unique and interesting context in terms of learning design and the opportunities for coordinated teamwork, project development and innovative use of technology.

The Industry Project course of the Built Environments program is a complex scenario that provides students with opportunities to develop industry-level skills through a project-based approach. It presents the opportunity to intervene in the whole course and to use the software solution proposed across different stages of development of the course and different learning objectives, an ideal scenario for varied observations and a rich space for experimentation and discussion.

A set of secondary question can be described that will help to focus the data gathering stages of the research, especially those related to the observations and activities in the context of the case study:

> SQ1: What differences in behaviour can be observed between students collaborating using the AR tool and those that are not?
>
> SQ2: What differences the students report in their perception of the activities done in the course between those using the AR tool and those who are not?
>
> SQ3: How does the collaboration and communication process changes between students using the AR tool and those that are not?
>
> SQ4: What differences can be observed in the end results of the course between all the different collaborative approaches observed, especially of those that incorporate the AR tool?

The focus of the observation process should be in identifying behaviours and processes rather than only differences in the end results, that is, the final assessment of the course project. Relying solely on possible improvements to grades or assessments can be a valid but flawed tool to understand the effects of a technology in the classroom (Dalziel, 1998; Katz et al., 2017). Additionally, the selected case study proposes a complex course to analyse, where several and diverse factors influence the work of the students. Taking this into consideration is why a positive or negative result does not tell the full story and why is more important to observe the whole learning process, from planning to development, documenting the behaviours developed to fulfil tasks and reach goals. The objective is to identify if the AR tool proposed is an influence, either positive or negative, and if it provides a more meaningful impact to the learning process as a whole and not only to the end assessments.

Assuming there are actual substantial differences between the students using the AR tool and those that are not, then it is also important to ask:

> SQ5: What elements of a technological intervention using AR create the most impact in changing the behaviours, perceptions or results of the students?
>
> SQ6: What elements of a multi-user AR tool creates the most impact in changing the behaviours, perceptions or results of students collaborating and communicating in a learning scenario?

These last questions are meant to provide an evaluation to the decisions taken in the development of the AR tool, especially the multi-user interactions, and how the tool was integrated to the learning design of the course. It also offers the possibility to highlight elements of the tool that could be polished or reframed in future work.

1.2.2 Scope and Limitations

Using a case study methodology, it is possible to approach the research questions through a qualitative lens, which is better suited for understanding the information gathered about behaviours, relationships developed between students and their subjective perceptions of the course, the activities developed, and the results obtained. All those elements would be hard or contra productive to try and quantify. In this light, it is important to acknowledge the limitations in scope and reach derived by the selected methodologies.

In relation to the software development aspect, it is important to establish the scope of the product as a prototype that explores the ideas of collaboration and multi-user AR. A thorough analysis of the design and architecture of the software was conducted to describe different possible approaches and solutions suited to the context and the needs of the tool, keeping an analytical mindset for all the ramifications and consequences derived from the implementation of a particular idea or design.

In terms of the concrete implementation of the software, the work was undertaken by a sole developer and limited in time and resources related to access to the case study. Most of the design and architectural proposals were tested via mock-ups or wireframe prototypes (Coleman & Goodwin, 2017; Sutipitakwong & Jamsri, 2020) in pseudo-structured test scenarios, aiming primarily for quick feedback and impressions rather than structured data. This allowed constant iteration and exploration of different ideas. The final implementation of the software prioritized showing the most critical use cases and the most relevant interactions with the tool, most of them tailored for specific observation sessions.

This approach had an important influence in the structure of the observation process and the nature of the data gathered, fomenting a more guided use of the tool and a more structured response from the participants rather than the free observation initially intended.

This was an acceptable concession made to facilitate the development process and to give the participants early and consistent access to the tool, albeit in a more incomplete state. This also minimized the disruption to academic work while maximizing the opportunities for meaningful observation in the available time.

There is also an inherent problem in the development of a case study in relation to the applicability of the results, generalization and subjectivity. The Industry Project course was chosen because it fulfilled all the characteristics needed for the research, but convenience aside, the course also provides a unique opportunity to analyse a collaborative learning design in action, with scenarios that could be extrapolated to other similar contexts, but also a unique take on the design of project-based learning and assessment, as well as a complex context for implementation and challenges that had to be tackled, all elements worth of a detailed analysis.

Nonetheless, it is important to identify the limitations of using one sole case study, which can be summarized as issues with the transferability and applicability of the data gathered. Other factors that impact heavily the nature of the data are the overall seniority of the students, which were all close to finish their studies and the lack of a concrete study plan of the course, which focused more on the development of the project as its main teaching instrument, offering just a relatively vague and general set of learning goals.

Chapter 3 will describe in more detail the scope and limitations of the case study, as well as the measures taken to minimize or acknowledge the flaws and restrictions of the chosen framework. For now, these issues can be summarized the main limitations related to focusing the software development into a gradual an iterative progression of a prototype and the validity of the case study in terms of the characteristics needed for the research and the unique value it offers for analysis and discussion.

1.3 Significance and Contributions

The technological environment in which we found ourselves right now evolves rapidly and at large steps. Despite critics of stagnation, the classroom evolves and adapts in response and constantly tries to understand how the available technologies can improve the way we learn (Watters, 2021). It is not always a positive relationship, it is often marred by overselling products and forcing a technological implementation without a plan and a proper consideration of the context and needs of the classroom (Cochrane, 2012; Odunaike et al., 2013). Currently, it is also a field that experiences a lot of social scrutiny as we react to the idea that we have become somewhat addicted to digital content and to the devices used to consume it (Gui & Büchi, 2021; Liu et al., 2022).In a context in which we are banning the use of smartphones in the classrooms and obtaining somewhat positive results because of it (Rahali et al., 2024), why, then, rely on a proposal that fundamentally uses that technology?

As Murray and Perez (2014) stablish, exposure to technology does not equal comprehension. In an age in which almost all aspects of life are mediate by digital activities, it is important to create a positive, critical and informed relationship with the technology that we have to use in our daily lives, and that relationship does not develops naturally (Margaryan et al., 2011; Selwyn, 2009). This is a task that can and should be executed in the classroom at different levels of schooling, in different contexts and at appropriate age and field depths.

The classroom is the ideal scenario to create and nourish this relationship in a structured and positive way, yet this is a factor that is seen as lacking in education institutes, especially in middle and high school (Burton et al., 2015; Jeffrey et al., 2011).

This project is an exploration of that philosophy, and an experimentation with a technological design that uses mobile devices in a healthy approach that promotes the development of skills needed today. Although the capacities of AR as a learning tool are evident, this endeavour needs to follow a due process. It is important to properly understand how to use the technology and what elements of the design and implementation process need to be highlighted and worked on to create that intended positive impact. The technology does offer a big opportunity to enhance and reframe the way we communicate and work together, but it also has proven to impose certain challenges that have to be analysed, understood and dealt with.

This document will describe the proposal, design, implementation, observation and analysis activities done in the development of an AR tool for collaborative learning and the test deployment conducted focused on developing skills related to teamwork and effective communication. The development of this research was carried in two interlocked steps. In one hand is the design and implementation of the software WorkshopAR, an AR tool used as a companion in the face-to-face work that was part of the activities proposed for the Industry Project course. Specifically, the software was used during the weekly group sessions of work that the students held for the development of three different assessment milestones in the course-long project.

The development of WorkshopAR consisted in solving two main issues, both related to the technological front of the research. First, there was a design challenge that required identifying the most common interactions performed in an AR environment for cooperative work. Different use cases were identified that were unique for this context and that posed an interesting design challenge, like sharing information, sharing and coordinating resources, managing the status of the team and coordinating the development of a task. A user-centred interaction design process was used to outlined solutions for the identified use cases based on important restrictions and particularities of the problem.

This stage was accomplished through a detailed exercise of user-centred design, a process meant to identify ideas and decisions born from the direct interactions of the users with prototypes and mock-ups, a structured process for gathering feedback, integrating the user in the design loop and refining ideas in an iterative process. The results obtained highlighted interesting issues and ideas related to multi-platform interactions, the expectation of users when presented with familiar and unfamiliar controls, complications related to interactions in a 3D space and the integration of the tool in the overall workflow of the students.

The second focus in the development of WorkshopAR was the technical challenge of creating a multi-user, shared AR space. Several commercially available frameworks and APIs were explored, such as the AR Foundation suit provided by the Unity game engine and the Lightship API provided by Niantic. It was possible to extract valuable insights about the current state of the development of AR tools, especially in the field of education, and it was possible to describe the lessons learned while integrating the development process with the educational goals of the course. An effort was made to relate the results obtained to those reported in similar studies by identifying common and divergent results and issues, especially those related to finding the technology distracting and the technical development of the tool too detached from the role of the educator.

Parallel to the construction of WorkshopAR was the development of the case study and the design of the intervention of the Industry Project course. In the design phase, the course was analysed based on the written material available and on interviews with the teaching staff. This information was used to guide the development of WorkshopAR and to identify the instances in the progression of the course in which the intervention could create more impact without disrupting too much the academic activities of the students. It was also an opportunity to detail and analyse the learning design of the course and to understand the context in which the observations were going to take place.

During the observation phase it was possible to obtain detailed observations of the work done by the students with and without the AR tool, about the development of the project through the conception, design and implementation phases, to adapt elements of the tool to better suit the circumstances of the course and to observe the role of the teaching staff in the development of the course and as part of the usage of the tool. This helped to create a rich record to report in the case study and plenty points of analysis and discussion.

The analysis phase used the recorded information from the observation stage plus all the material that could be recollected from the execution of the course. This material included the written communication given to the students via the official channels of the course, the workshops executed during the semester as material to help in the development of the project and the final products created by the students, such as presentations, reports and designs. The observations were also complemented by a set of semi-structured interviews to the students and the teaching staff. The interviews focused on gathering better records of issues related to the personal perceptions about the use of technology and the collaborative activities of the course.

From the observations obtained and the analysis of the results it was possible to formulate some possible and satisfactory answers to the research questions. The data points to interesting behaviours promoted by the intervention, such as students creating a more structured work process when the tool prompted or encourage them to do it. It was also observed that this enforced structure helped the students with the organization of activities, prompting good practices for communication and the organization of ideas and plans.

But the structure also prompted some unintended behaviours, like a tendency to create a hierarchical organization of the group around the student managing the tool, which was not in tune with the objectives of the course. The app also forced the students to be aware of their phones more than they were planning or willing to do, highlighting a current trend among students of regulating their phone consumption. It was also noticed that the tool was most useful for teams struggling with their organization and offered little value for those already with a clear work plan.

Data and observations also provided a lot of insights related to the design of collaborative interactions in virtual 3D environments, to the relationships of students with their phones and how students use different resources, properly or not, to sort out the difficulties of a learning project. It was evident that the social aspect of the collaborative work resonated the most with the students and with the general satisfaction they reported with the activity, giving more importance to the tools created for them to communicate, express ideas and personalize their experience. The case study also opened an unexpected opportunity to analyse the role of the teaching team in the use of the tool, offering a clear path for future work.

1.4 Document Structure

The purpose of this document is to provide an organized recount of the development of this research work, the theoretical background that supported the proposals given and the methodology followed to gather data, analyse it and build a set of answers for the research questions.

This introduction chapter was meant as a summary of all these elements, highlighting the personal motivations behind the chosen research field and the general reasoning that accompanied the design process, the hypothesized and desired results, as well as the methodological pathway taken. Chapter 2 of this document will describe in detail the literature explored that helped in shaping the research questions and that provided the theoretical basis for the educational methods used as anchors for the software implementation.

Chapter 3 details the methodological framework of the research. It briefly explores the research paradigms picked to guide this project and explains the reasoning for choosing them and their relevance to the learning and technical fields this project is based on. The chapter will also offer a complete explanation of the educational front, detailing all the instruments used in the development of the case study. In the technical front, it details the software development methodology followed, especially the reasoning and procedures used for the interaction design process and the methods used to ensure usability, flexibility, expandability and a general user-centred approach.

The description and analysis of the results have been divided in three chapters. The technological front has been described in Chapter 4, where a detailed explanation of the requirement analysis can be found, including use case scenarios and quality objectives. The chapter also details the most relevant elements of the architectural solutions proposed for the tool. Chapter 5 continues with the description of WorkshopAR, now focused on the implementation process followed. The chapter describes the design phase to the development cycles followed to build the software, the testing strategies used to ensure validity and the feedback received by the users during testing and the observations in the case study.

The case study itself is described in Chapter 6, explaining the context of the Industry Project course, the teaching objectives guiding its design and the points of interest for the intervention using WorkshopAR. The chapter will also provide a narrative description of the observations taken over 20 weeks of development of the course, and a preliminary analysis of the data based on the secondary material extracted, like the written documentation, the work produced by the students and the information gathered from the interviews.

The final analysis is presented in Chapter 7 along with the necessary conclusions gathered from the data and the process described in the previous chapters. The chapter will detail what elements of the development of WorkshopAR and the case study support the proposal of AR as a tool to promote positive learning behaviours and effective collaboration, what important technological hurdles were identified and what surprises were found during the development of the project. The chapter will also offer and outline for future work to address the most outstanding issues derived from the research limitations and the problems encountered during the development of the project.

Chapter 2

Literature Review

This project is centred around the concept of Collaborative Learning (CL): the process of acquiring or creating knowledge as part of a group, and the dynamics that emerge when a team needs to manage participation, the overall success of an activity and the learning objectives that need to be fulfilled (Laal & Ghodsi, 2012). It is also possible to analyse the concept as learning to collaborate, to acquire the skills needed to be an effective team member.

The purpose of this literature review is to find the benefits of adopting a CL approach in a learning design and the challenges that any educator needs to consider when implementing a strong focus on collaboration. It is also important to understand why collaboration and teamwork are such important skills to develop, why is it so valued in most professional environments and why is there a general feeling that educative institutions are failing in teaching those skills.

This research will approach the field of CL using the lens of TEL: the use of technologies to support processes or management of teaching and learning (Passey, 2019). Understandably, this is still too wide of a subject, so, a particular focus is set on AR technologies. The objective will be to understand how AR can be used in the classroom, and the particularities that it offers in terms of benefits, challenges, common implementations and relevant success stories.

AR can be a valuable collaborative learning tool based on the unique abilities it possesses. Specifically, the technology excels at incorporating strong visuals into the surroundings of the user(s), which provides an opportunity to design enticing experiences for both learning and collaboration. For this purpose, it is important to identify what collaborative experiences have been created in the past with AR, what learning goals align with the technology and what challenges in design and development can be expected.

This chapter explores these questions and themes across three sections. The first section tackles the topic of collaboration, how it is defined and measured, different implementations in the classroom, perceived benefits, common challenges and lessons learned from previous projects. The second section presents a systematic literature review aimed at identifying the use of AR as a tool for education, how the technology has evolved, which elements of the technology have derived in success stories, and which have become challenges. The final section will combine both subjects and explore the literature for previous experiences using collaborative AR, both in education and in other fields.

2.1 Collaborative Learning

2.1.1 Collaboration as a Transversal Skill

The subject of CL is wide and can be tackled through different angles. For instance, we can talk about collaboration as an isolated term, and relate it to similar concepts like teamwork, cooperation and intrapersonal skills. In need of a definition, the approach taken by Andriessen and Baker (2020) is very complete. The authors state that "(...) collaboration means developing, in an equal and mutually respectful way, a shared view of a situation". Is a simple definition that implies two important things. First, collaborators find themselves in equal terms, there is a sense of equality and mutual correspondence. Authoritative or hierarchical relations hardly produce collaborative settings. Second, collaboration goes beyond cooperation, beyond working together to fulfil a goal. Collaboration implies building, through time, a shared view. The group has a shared set of goals, purpose and values. The classical definition of collaboration proposed by Roschelle and Teasley (1995) also complements this idea by stating that cooperation involves the division of labour among participants, of individual responsibility, while collaboration is a coordinated effort to solve the problem. Echoing this distinction between collaboration and similar ideas like cooperation or teamwork, Salmons (2019) states that collaboration is a process that needs to constantly be managed, while teamwork is a more *natural* or *improvised* way of working together when needed.

Andriessen and Baker (2020) later expand all these ideas by stating how collaboration is built through time in three different dimensions. In a personal sense, we collaborate because we gain something from it, or in the worst-case scenario, because we need to do it. Our goals align with those of others, creating a second inter-personal dimension, in which we also care about the relationship we create with others and the social advantages we obtain from the collaboration. Finally, with strong inter-personal interactions, we create a group dynamic, in which decisions are taken beyond the individual level and align with the necessities of the group. This evolution can be fractal, which means that after a strong group has been created, it can be developed in a fourth dimension, an inter-group relationship in which several tight-knitted groups collaborate in a shared purpose within a bigger system.

Now is possible to build an initial definition to be use as conceptual backbone. Collaboration will be understood as: the process in which a group of people build a set of values, shared views and coordinated efforts to fulfil a purpose or reach a goal. This definition considers the most important aspects highlighted previously by other authors: collaboration goes beyond just coordinating tasks because is strongly linked to building relationships and communities, therefore, it is a very social skill. Collaboration is not a single action, is a process that needs time and effort to be constructed. Finally, collaboration is aimed, it needs a purpose or a goal. More important, that goal needs to be shared among the collaborators instead of being imposed. Collaboration can only be developed when all the members of the group find themselves as equals and share the perceived value or the necessity of working towards that aim.

It is possible to add more elements to the definition, creating a wider and stronger framework, if we analyse the concept of collaboration from other perspectives and in different contexts. For instance, collaboration can be viewed as a skill to be developed within the context of professional development and employability.

As stated by Crant (2000), it is common for a workspace to seek in a professional a proper balance between theory and knowledge, that is, the information gathered through practice and experience. This position can be read as employers seeking people that have built a career rather than only have a profession. For employees, it means that they must build that career identity: the right set of skills, experience and personal traits that show value beyond a profession. Additionally, employers seek candidates that can properly response to a variable environment, either due to high pressure or a fast-moving market. Employees must then demonstrate personal adaptability, a willingness to explore, and life-long learning values. Finally, employers seek for the ability to properly construct and use the surrounding social context, the employee needs to build social capital by cultivating abilities in cooperation, networking and teamwork.

This framework shows the importance on adding value to the professional profile, and highlights skills related to adaptability and collaboration as key value pointers. The framework also relates easily to other popular concepts like social intelligence, personal development and problem-solving attitudes, all part of what has been known as transversal skills. The UNESCO defines transversal skills as those that can be used in a wide variety of situations and work settings (UNESCO, 2014) and are valuable for their general applicability and transferability. These skills can be organized in five broad categories:

- Critical and innovative thinking.
- Interpersonal skills.
- Intrapersonal skills.
- Global citizenship.
- Media and information literacy.

To understand better the type of abilities that the previous categories encompass, we can extract some ideas from diverse sources in the literature, a set of studies focused in understanding the work market in different contextual variations of geography, professional field, years of experience and even nature of the job-hunting process (Abir et al., 2024; Boahin & Hofman, 2013; Garst et al., 2019; Huynh et al., 2024; Llorens et al., 2017; Nadarajah, 2021; Ng et al., 2021; Smaldone et al., 2022; Zhang et al., 2024). This sample of literature shows the type of skills most demanded by employers around the world. Although skills that are considered *hard*, or better described as field-specific are understandably diverse, there is almost a complete coordination related to *soft*, transversal skills. These skills are common to all contexts and independent of the region or work field. Figure 2.1 shows a visual distribution of the most common transversal skills sough after in potential candidates, extracted from the studies mentioned above.

Despite being a small sample, the data shows a clear emphasis on relational skills, and it seems to portray the added value of hiring and developing people with high communication abilities, teamwork, good social interactions, and flexibility (Al Asefer & Zainal Abidin, 2021).

Another important sentiment that can be extracted from all the literature mentioned is a perceived discordance between the skills and abilities focused on by high educational

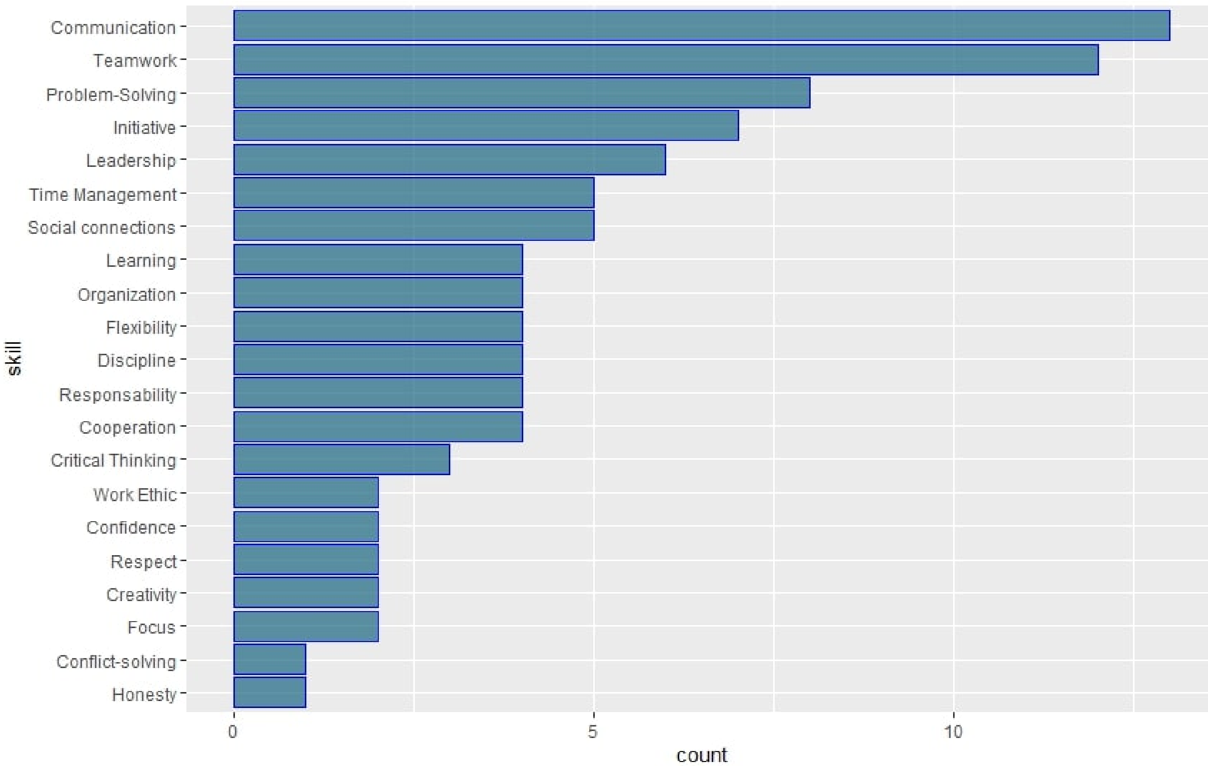

Figure 2.1: Most Common Soft Skills Identified

organisations and those sought after by companies and employees, as identified so far in this section (Damoah et al., 2021; Ng et al., 2021).

Tulgan (2023) identifies specifically three soft skills missing in young workers: professionalism, critical thinking and teamwork. Tulgan argues that this gap is closely related to a generational and technological shift where there is a lower need to develop communication skills because we have offloaded most of our communicational needs into electronical devices (mails, meetings, conversation, social interactions), forgoing the opportunity to develop such skills in person.

Tulgan keeps explaining that in recent years *good old-fashioned* teamwork became an issue for young people, who have developed a different set of values that clash with those of previous generations. Where older generations built a long career as part of a company or institution, newer generations have little to no loyalty to a single employer, and they are more likely to search for better deals. Young professionals find little value in adapting to the ways and stablished institutions of a group when they know very well, they won't be there for too long.

Secondary and tertiary education programs are often seen as the main responsible agents in the development of skills deemed crucial for the general professional life (Kechagias, 2011) and is in this light that modern curricula in almost all disciplines now have a higher emphasis on building those transferable skills (Earl et al., 2021). This is not without problems, and there is a noticeable consensus that teaching such skills is not a trivial task (McCale, 2008). Several factors influence the ability of a curriculum to provide such learning pathways. Most common problems are related to how little educators are prepared to incorporate those skills in their classrooms (Kemp & Seagraves, 1995), the costs of adapting the curriculum, training the personnel and changing old mindsets to effectively provide the space to create such learning (Chadha, 2006).

Considering these extra contextual elements, it is possible to modify the previous definition of collaboration as follows: Collaboration is the skill to work as part of a group by identifying, building and acting upon a set of values, shared views and coordinated efforts to fulfil a purpose or reach a goal. This modified definition not only states a set of characteristics of what the process of collaborating implies, it frames it as a skill to be executed by each member of a group, and not as something that happens naturally when people gather. It also acknowledges the need to learn and develop such skills.

The theoretical framework around collaborative learning can be expanded by exploring the definition of a second concept, that of Communities of Practice, with the intention of later linking this idea with the general definition of collaboration built so far and the approach proposed by this project to use it as an educational tool.

2.1.2 Communities of Practice

A term proposed by Lave and Wenger (1991), a Community of Practice (CoP) can be defined as a set of mechanisms of situated learning used by communities to transmit knowledge internally, especially to newer members. These communities are formed around a focused domain and were initially used to describe the learning process used in different trades like craftmanship and health practices. The meaning of the term has been broadening with time and can be applicable to any group that gathers towards a common goal over an extended period, that is, enough time for several generations of members to come and go.

Strongly linked to CoP is the idea of legitimate peripheral participation. This can be defined as the process in which newcomers learn mostly through observation and gradual participation with their senior peers (McDonald & Cater-Steel, 2017). CoP are used to give some structure to the process of informal learning, especially the type of learning found outside of academy, and which is more linked to professional practice, personal development and the relationships forged in the day-to-day activities with peers and a community in general.

A healthy CoP is one that is generated naturally due to the interaction of a community around a specific domain. It is different from a task force or team of people gathered around a goal or a problem in the sense that the general objective of a CoP is more undefined in terms of time, is focused more on the practice rather than the goal or product itself and is constituted by a social community rather than by just a collaborating team of people. These are the three fundamental elements that define a CoP: a domain, a community and a stablished practice, as described in Figure 2.2.

The domain is the knowledge base in which the community chooses to work because it offers value or just interest in general. New members come voluntarily and interact with elder members gaining experience in the practice, with the expectation of eventually gaining mastery.

The community is a group of people that can be driven by complex social dynamics. The community will create norms, stratifications and behaviours over time. The community is

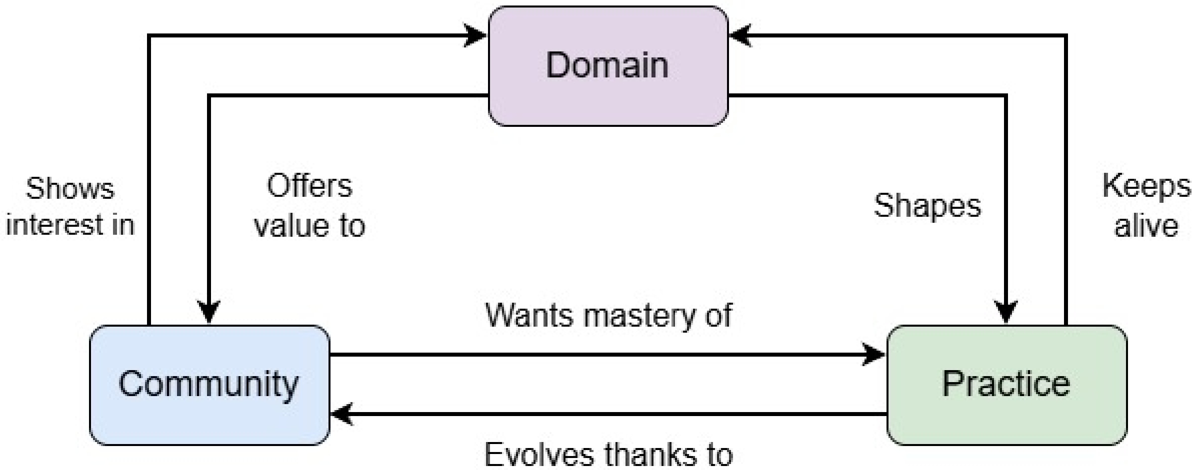

Figure 2.2: Elements of a Community of Practice

expected to be in a constant flux of members and to evolve with time, in terms of social interactions and the relationship it has with the domain.

The practice is an integrated set of activities and knowledge executed and transmitted by the community to create mastery over them. Some CoP focus more on nurturing members into the mastery of the practice, while other focus more on the transmission and preservation of the knowledge generated, as noted by Klein et al. (2005).

CoP have shown to be the cornerstones in the development of informal and social learning, leadership development and organizational change (C. Anderson & McCune, 2013). Often, high indexes of performance and employee satisfaction are present when a CoP has been identified and nurtured within the informal structure of an organization (Blackmore et al., 2010). Also, they have proven to be valuable integration, communication, and identity reinforcement tools in communities with a high degree of disparity or diversity among members (Hildreth & Kimble, 2004).

A particular emphasis on the idea of nurturing a CoP is important. A community can perish due to lack of proper nourishment or caused by artificial intervention. Previously it was noted that it is not possible to force the creation of a CoP, new members must voluntarily engage in legitimate ways with the practice. But even when a community develops and grows in members, and its value is acknowledged and sough after, forced practices that run in contradiction with the spirit of a CoP can end up killing it. Authors like Ardichvili et al. (2003) and Brown and Duguid (1991) have analysed the lifecycles of successful and failed CoP inside the organizational practice of many and varied industries, from insurance brokers to high education institutes. Of all the key factors identified in the failure or disappearance of a CoP, it is useful to highlight the following ones:

- Not allowing enough time for members of the community to form relationships and sustain them. This can be due to frequent administrative reorganization or the prevalence of temporary or part-time staff.
- A culture of tight management, where every aspect of the practice is highly standardized

and supervised, and there is no possibility for a natural development of the practice due to the relationships formed in the community.

- A culture of highly individualized work, either due to the very nature of the practice or because of other factors, like a very competitive environment. This inhibits collaboration and discourages collective engagement.
- Practices with extreme time or performance pressures, which leaves little time to develop a collective or communicative environment.
- Spatially fragmented work, so that there is no opportunity to establish a relationship, or because there is not a common, unsupervised space for the community to gather.
- Communication and activities are heavily mediated by technology or even other people. This makes interactions (arguably) less immediate and intense, therefore, lacking importance.

So far, all the elements that have been illustrated paint the concept of a CoP as a natural-occurring social structure formed around the need of preserving, communicating and nourishing knowledge within a well stablished community. It is a manifestation of informal education that thrives and offers the best results when acknowledged at an organizational level but not tampered with (Brown & Duguid, 1991), and when given enough resources to grow and evolve (Klein & Connell, 2008).

Given the importance of naturality and informality within the definition of a CoP, it is possible to see how these fundamental elements can be at odds with the expectations and common practices found in more traditional and institutionalized education, where learning is more structured, participation is hardly voluntary and collaborative processes are more task-focused rather than long-term approaches to the practice. These issues will be acknowledged in the following section.

2.1.3 Communities of Practice and Education

A common image of what could be considered *collaborative learning* in a classroom setting can be of a group of students working together to fulfil an assignment, say, a presentation on an assigned topic. Although it is possible to define this group of students as a community that came together with the goal of learning something, it fails several other criteria needed to be called a CoP.

In first instance, the example scenario breaks the idea of *natural occurrence*. The group was not formed by the voluntary interest of the students to engage with the domain, but as a necessity to complete a particular task. Also, it would not be uncommon for the composition of the group to be a forced assignment, either by chance or by the designation of the teacher, which would not constitute a community formed by the natural connection of its members.

The community itself also lacks perpetuity because its existence is linked only to the fulfilment of the task. Although the students can benefit from the work done alongside their peers, the knowledge and mastery acquired benefit only the immediate members of the group, they will not be transmitted to other members, nor is it expected for the group to grow and evolve.

Despite all these short comings, there are advantages in using the CoP approach in this scenario. For example, it can be argued that the activity does offer a genuine peripheral participation with the practice, allowing the students to engage with elements of the domain and work towards mastery. This engagement is achieved more through structured lessons rather than by informal observation of the actual practice, but it still offers the general idea of a gradual and guided approach.

To properly identify the community formed around this practice more elements need to be added to the analysis. In that way, the natural structures that are normally present in a CoP will get clearer. By including the teacher in the community, it is possible to identify expected elements like eldership and meaningful learning-focused forms of communication. By considering all the groups formed by all the students in the classroom, not only the initial single group, it is easier to see the opportunities for more meaningful relationships to be formed between peers, for more variations on the ways they approach the task or for more opportunities for that practice to evolve. The same results can be obtained by extending the timeframe in which we are analysing this activity, and considering not only the current cohort of students, but also all previous and future groups that participated and will participate in the activity. It is possible to keep adding more elements to the scenario, like all the teachers that have overseen this course and all the courses that are part of the same department, for instance. This exercise exemplifies that, although there are fundamental problems in the definition of a CoP in the scenario of institutionalized education, a lot of the core elements can be found with interesting variations.

A study group could be a better example of using a CoP as a useful framework of analysis in this context. In these circumstances, it is common for the group to form naturally, as a voluntary activity surging from the necessity of mastering a subject (for an upcoming test, i.e.) or because the students have identified that having the support of other peers helps them achieve better academical results. In any case, the core purpose of the group is learning and practice, rather than the fulfilment of tasks. The relationships between the participants extend beyond organizational roles and the focus lays on the transmission and collaborative generation of knowledge and expertise.

Cremers et al. (2008) analyse the formation of CoP in higher education in a similar way to the one proposed before. They argue that knowledge sharing and knowledge creation require a different set of skills than those needed for groupwork. A CoP becomes an ideal tool to promote those skills and to prepare students for the type of informal learning processes they will find later in life. Yet, forcing students to create a CoP or creating a lesson around the use of a CoP goes in contradiction to the spirit of the term, which is meant to be voluntary and informal. Giving continuation to the initial example, a study group work the best when the students themselves want to be part of it, and it is because they recognize that they need the help of their peers or because the collaborative activity is beneficial, not necessarily because it is part of a lesson proposed by the teacher, or even worse, because is a mandatory part of an assessment.

It is then necessary to come back to the idea that a CoP must be cultivated rather than created (Wenger et al., 2002). When designing a lesson, a course or a curriculum, several steps can be taken to promote the natural formation of CoP. Klein and Connell (2008) propose a series of principles used to nourish CoP that can be followed to create proper spaces for them to form naturally. For example, students should be allowed to propose and develop their own activities, with possibilities to grown on their own and to have active channels of communications with external sources, like teaching staff . It is also important for students to have access to physical spaces to develop such activities, encouraging both public and private work. The teaching staff should also be aware that they are stakeholders in this scenario, being an integral part of the *value* that the students create by engaging with the CoP and the natural *rhythm* the creation of this CoP follow related to the activities proposed by teachers, like due dates and exam weeks.

Cremers et al. (2008) go a step beyond and propose that students should actively learn to be part of a CoP, arguing that it is no different to learning the proper skills to be a good team member. Speaking in terms of learning outcomes, they stablish that a student should be able to:

- Reflect on the state of the practice and the necessities for future development and evolution.
- Be able to communicate and make explicit the results of their engagement with the practice.
- Be able to use the resources of the community in their engagement of the practice.
- Construct, organize and create new knowledge in collaboration with the community.

Rather than asking students to create a real CoP or forcing their participation in one, there is more value in giving the students the tools to understand the dynamics found in a CoP and the proper way of participating and creating value in one. Keay and O'Mahony (2016) echoes this idea by arguing that the significance of a CoP lies in the process rather than in the product. The process of interacting and collaborating with the community has an inherent value, even if a CoP is not properly constructed, and that is what a student needs to learn and be exposed to.

It has been noted by several authors that the dynamics found in a CoP can also emerge and be fostered in online environments, such as Riverin and Stacey (2008) in the perspective of the students and Kirschner and Lai (2007) in the perspective of the educators. These online CoP can emerge and be effective provided that similar circumstances allow the natural growth of the community, like an open participation and clear opportunities for mentorship. CoP for online environments have been found to boost participation and engagement in students, especially young ones (Thomas, 2005), which reinforced the idea that the creation of a community is crucial for a successful remote learning experience (Archambault et al., 2022; Swan, 2002), which circles back to the importance and usefulness of collaboration as a learning tool.

In summary, a Community of Practice is an analytical framework better used to understand processes of informal learning strongly linked to the social dynamics of its members (Wenger, 1998). These characteristics often find themselves at odds with the more formal and structured forms of learning in educational institutions but still offer valuable insights and working tools when analysing the activities done by students outside the structure of a classroom, and more related to their practice of being students.

As discussed in Section 2.1.1, collaboration is as skill that needs to be developed, not only because it is a well sough-after *soft skill*, but also because it is an integral component of professional development, community integration and life-long learning skills. As students, we become part of a particular Community of Practice that works around the development of useful skills for learning and academical success. The development of these skills is, at risk of proposing a circular definition, a collaborative process, one that is hard and unproductive to create in a forced or artificial manner, but than can be promoted and grown given the proper resources and the correct nourishment.

2.1.4 Learning Teamwork and Collaboration

A lot of literature can be found that try to create a model of how individuals come together as a group to achieve a goal, identifying all the factors (internal, external, personal, emotional or circumstantial) that contribute to the success and failure of a group. For example, Bion (2003) offers a good explanation of the Tavistock approach, a systematic view of the psychological dynamics present in the conformation and life cycle of a group. This model identifies important structures that appear inside a group like authority figures, responsibilities, boundaries (both internal and external), crisis responses and organizational structures. Participants form and engage with these elements even if they do not properly identify and name them.

A group can also be analysed in terms of the behaviours adopted by its members. Bales (1970) categorizes the possible interactions performed by team members into task-oriented and emotional. The latter category can be charged with a positive, neutral or negative tone. These behaviours can be expressed through every form of communication that the group uses, including non-verbal cues like body posture or voice tone.

Following the behavioural line of thought, Knowles (1982) identifies some factors that strongly influence the unique identity that a group develops through time. Factors like time constraints, physical environments, group size, composition, overall goals, patterns of participation, communication tools and the set of norms, procedures and structures that emerged within the group.

All these elements create a valuable toolbox of concepts that are useful to systematically describe the work, behaviours, relationships and experiences of a group. They create an essential common language that describes problems within a group, propose solutions, set goals or creates awareness of the elements that are expected to emerge or the issues that must be addressed when working with others. These elements shouldn't be the focus in the process of teaching collaboration but do offer a solid structure in the design of a solution.

A better approach to the learning process is to understand why we collaborate, not only in terms of the employability necessity that has been exposed previously, but also in terms of finding value in teamworking, even if the approach was not voluntary or when collaboration becomes a necessity rather than a goal. When framing our group relationships through the idea of creating value for the group and extracting value for ourselves, we can focus on identifying all the aspects that develop an effective collaboration, those that create more value (DAWSON, 2017; Wagner et al., 2010). When identified, these factors can be replicated and taught in controlled and varied scenarios so that students become aware of their value, build their own experiences of success and failure in a safe space and gain proficiency in identifying and replicating those same elements by themselves.

Hibbert and Huxham (2005) use the terms collaborative advantage and collaborative inertia. We navigate towards collaboration because we find a tangible advantage in the process (for example, the task cannot be accomplished by an individual). But even with a concrete advantage, no interaction can be initiated if the initial collaborative inertia is not surmounted. This inertia is defined as all the barriers that impede collaboration, things like:

- A personal unwillingness to seek input or to learn from others.
- Hoarding expertise, defined as an unwillingness to help others, share knowledge or skills.
- Inability, due to lack of skills or resources, to transfer knowledge.
- An impulse to compete.

The role of competition, especially in academic settings, has an interesting antagonistic nature to collaboration. A dose of competitive elements in an academical setting creates interesting results in terms of providing motivation and building a sense of belonging (Bergin, 2016). Yet, it has also been analysed that focusing too much on the competitive aspects of schooling can derive in undermining the intrinsic value of learning (Tauer & Harackiewicz, 2004) or creating an environment in which the competition is more important than the learning outcomes (Kowalski & Christensen, 2019). This pervasive sense of competition in schools is one of the main issues behind the generalize reticence of students to participate in collaborative activities, which can also derive in lack of motivation, loss of focus and general disinterest in the goal of the collaborative activity itself (Driscoll, 2025; Kloepper, 2023).

One of the first steps to implement a framework for learning how to collaborate is to identify which of these barriers are at play and provide tools to deal with them and not just hope for people to continuously stumble into these problems and finding solutions by chance or by trial and error. Salmons (2019) provides an extensive analysis of what collaboration is and how to create an environment that teaches and promotes it. One first effort is to acknowledge and identify different types of collaboration, defined by the overall achievements gained and the value obtained by participating in the collaboration. Salmons identifies these four degrees of collaboration:

- *Co-creation:* Collaborative generation of new knowledge, processes or products.
- *Acquisition:* A team effort to acquire or develop new skills or knowledge that none of the participants had prior to the collaboration.
- *Transfer:* Sharing expertise or knowledge between the members of a group to create a wider and more complex pool of knowledge or skills.
- *Exchange:* Sharing physical and human resources to fulfil a goal that could not be achieved individually.

Salmons also offers a complementary approach to the idea of value generation explored in other literature. She stablishes that before any form of valuable collaboration can be generated, it is important to reinforce two cohesive elements of any group trying to work together: trust and communication. These building blocks are the main forces used to create meaningful connections inside the group, which in the future leads to the creation and perception of value.

Trust refers to feeling comfortable while working towards and sharing goals with other people. *Trust* can also be seen as a form of social capital (Addison & Teixeira, 2020) that determines how willing is someone to cooperate with others and accept relationships and situations of dependence (Zand, 1972), risk, uncertainty (Frederiksen, 2014) and hope for mutual benefit (Kramer & Tyler, 1996). On the other hand, a lack of trust can be evidenced trough feelings of dread and exhaustion within the group, forcing collaboration only when strictly necessary. Vangen and Huxham (2012) also identify the ability to create a safe environment for all the members of the group as an exercise of trust, where the social risks associated with dealing with other people are negated or mitigated, so it is possible to concentrate in creating results consistently. A group that can create trust within is more resilient to external issues, to errors and to conflicts. It also shows a better capability for self-reflection and faster course correction (Shinya et al., 2016). A high level of trust is also correlated to safe environments for innovation and risk-taking (Bichard, 2005).

The second building block proposed by Salmons is to develop proper channels, skills and habits of communication. The author describes two types of communications in a collaborative setting:

- *Dialogue:* Defined as formal and structural communication best used for planning and coordination. It is also the best form for stablishing rules and initiating collaboration itself in new groups.
- *Conversation:* Defined as informal and loose communication used to create relationships. It is also the best tool to build and convey trust in the group.

Proper and structured dialogue is also the main venue for students to learn and practice collaboration. Essential aspects of teamwork can be evidenced and assessed through activities guided by dialogue that students engage with. The structuredness of dialogue also helps in creating activities and resources that show collaborative skills that students can develop and provide evidence of, such as:

- Proper use of interpersonal skills.
- Understanding of the task or project.
- Work design.
- Task allocation.
- Quality assurance.
- Coordination.
- Communication of progress.
- Accountability.

- Conflict resolution.
- Decision making.

With these elements is possible to create a roadmap that guides a lesson, a project or an assessment in which collaboration is an important learning goal.

Through dialogue it is also possible to create a third communication skill fundamental for a successful collaboration, *review*. This is defined as a process of self-reflection focused on the work or progress done by the group. A review process can also be structured, facilitating teaching and assessment activities, and can be mapped, as shown previously with the main indicators of dialogue, to key aspects like:

- Ask and give support.
- Analyse and challenge other's contributions.
- Analysis and incorporate feedback.
- Encourage peers.
- Monitoring peers.

Other authors like Keay (2006) proposes similar criteria for proper collaboration in terms of trust, respect and effective communication. Additionally. example implementations like the one showed by Lingard and Barkataki (2012) highlight some other success factors like developing a sense of self-organization, inclusivity and responsibility. Marasi (2019) proposed a series of activities that highlight important attitudes and approaches for healthy teamwork like listening before responding, being affirmative of other's ideas and being mindful of nonverbal communication. Davis and Ulseth (2013) and Asrori and Tjalla (2020) iterated over these concepts and proposed different frameworks, activities and observations that identified other important ideas like accountability, feedback, ownership, empathy and long-life learning.

Table 2.1 serves as a general summary of the concepts and ideas explored throughout this section and shows a map that links fundamental criteria of effective collaboration with activities and behaviours that acts as helpful evidence of those fundamental elements in the work and interactions of a group of students. This map provides a structured tool that will be used as an initial standpoint for all the educational and technological developments of this research project that have collaboration as the main learning goal. They are also a useful tool to frame the analysis of the work done by student in the development of the case study and to describe the learning design proposed by the Industry Project course.

2.1.5 Conclusion: Approach to Collaborative Learning

At this point this literature review has explored some of the tools needed to properly characterize the benefits and challenges around collaborative learning. From the definition built in section 2.1.1, collaboration is a skill developed by students to achieve a common goal through a shared set of values, views and efforts. The common goal here is learning.

Table 2.1: Evidence Factors and Assessment of Collaborative Work

Criteria	Evidence	Examples in Literature
A shared common goal	Work Design	Fisher (2014) and Kernaghan & Cooke (1990)
	Goal and task ownership	Kropp & Dodd (2024) and Sohmen (2015)
	Co-creation and acquisition of knowledge	Drejer & Jørgensen (2005) and Salmons (2019)
Willingness to collaborate	Inclusion of all members	Keay (2006) and Zubiri-Esnaola et al. (2020)
	Affirmation of all ideas	Marasi (2019)
	Group bonding and empathy	Blanco et al. (2017) and Widayati et al. (2022)
	Share knowledge and resources	Nissen et al. (2014)
Proper management	Task allocation	Hirshfield & Chachra (2015)
	Self-Organization of the group	Keay (2006)
	Communication patterns	Amponsah (2003) and Jahng et al. (2010)
	Flexible structure	Gross (1997) and Jacobs (2015)

Section 2.1.2 introduces the concept of a Community of Practice and all the positive implications of using such framework in terms of understanding the motivations that drive the people partaking in collaborative work. In a CoP there is a feedback loop between participants and the constantly evolving field of work in which they are immersed. A proper and healthy CoP has an antagonistic view of the type of work normally found in a classroom, with little agency from the students and a practice limited in time and scope to the fulfilment of a task.

Nonetheless, exposing students to the general dynamics of a CoP has value, especially considering both the ubiquity and the transparency of such dynamics in the workspace. It is also valuable to refocus the approaches used for a CoP in the classroom not as a fancy synonym to teamwork or classroom group work, but as a tool to analyse how students in general approach their learning tasks. This can be achieved by identifying a whole classroom (or cohort, or course, or department, etc.) as the community. The practice is not a task or a project, but the process of learning in the specific domain. In the proper environment, students develop tools to manage their learning and the relationships that naturally grow between peers and teachers.

Finally, in section 2.1.4, the information found on current literature was used to establish a set of criteria and evidence-base assessments that can guide a pedagogical design based on collaboration as a learning tool or that focuses on collaboration as the learning goal.

Collaborative learning can become a useful tool for the classroom based on the approach taken to encourage and manage teamwork. While trying to collaborate in the fulfilment of a task, students learn how to summarize and share information, co-construct ideas through dialog and debate, and improve retention and comprehension through the simple action of raising questions and answering them (Andrews & Rapp, 2015). Activities requiring teamwork also help in the development of critical thinking abilities (Zandvakili et al., 2019) and prepare students for a realistic work environment (Chowdhury et al., 2002).

Collaborative activities help students create a better understanding of the task at hand. Salmons (2019) uses the concept of *workflow*, where students must discuss, design and implement a concrete plan that help them achieve their objectives. Even if this plan involves a simple coordination of consecutive or parallel tasks, or a more complex synergetic form of work, the process of coming with ideas and solutions to tackle the needs of the task creates a stronger involvement with the team and the project. Similarly, coordinating work efforts develops proficiency in communication, especially in developing literacy (and digital literacy) around tools for remote communication, file sharing, document cocreation and asynchronous work coordination (Lo, 2009; Murphy & Cifuentes, 2001).

Collaborative activities are also often associated with creativity and transformative learning by providing the proper conditions to bring ideas together, compare solutions and expose different viewpoints (Peppler & Solomou, 2011; Turnbull et al., 2010). Rubbing ideas together offers extra value to the generation of knowledge by also providing the circumstances to make connections around that knowledge and challenging preconceptions or stereotypes (César & Santos, 2006; Eden & Ayeni, 2024).

But working in a collaborative setting is not without its challenges and hurdles. Just the act of being in a group is not enough to boost the educative process of the students. If the proper tools are not provided and major barriers are not acknowledged, the situation can flip around, and teamwork can become more of a liaison than a boost.

One of the most common issues is power imbalance, a natural occurrence whenever there is a group of people gathered towards a common goal (Essabbar, 2015). Elements related to gender, social status or economic issues, among other things, are always factors to consider when using teamwork as the main driver of a lesson. When a strong imbalance is present in a group, it can be evidenced is several behaviours and outcomes that could hinder the learning process of the students. Different authors have identified some of these patterns in different contexts, like inconsistent exposition to the learning material, coarse relationships forming between students, lack of agency in work distribution (Griffin et al., 2015), stress, downgrading performance and lack of response to feedback (Van Der Vegt et al., 2010).

Educators need to develop the necessary foresight to acknowledge these difficulties and provide the students the tools to solve them. In general, it has been shown that self-reflection, an effective mediation and clear communication methods work the best when dealing with issues of imbalance. A closely related analysis in search of solutions for this situation is related to academic imbalance, especially when it affects the educator and different expectations in terms of performance and achievements formed for each group member. A lot of studies have worked on strategies to identify these discrepancies and how to work around them, proposing tools like peer-assessment (Rezaei, 2018) participative rotation, close time management (Davies, 2009) and smart grouping (Akhrif et al., 2019; Q. Wang, 2009). The important note is to be aware that issues with academic imbalance are common, and that there are several tools to manage them and even take advantage of them.

Management chaos is also always associated with groupwork, not only from the point of view of the students but also from the educators. It is expected that students will have problems with managing elements associated with groupwork, like distribution of work, problem solving and self-reflection. It is also expected that the students will find the educational material needed to learn how to cope with those management issues in the educators or the course itself. Nonetheless, it is surprising how often it is expected that students collaborate without giving them the tools to learn how to do it. Bruffee (1973) provides an interesting record of his experiences implementing a collaborative approach with his students, and a lot of information can be extracted from this case study in terms of the work needed to set-up and manage the tools, environment and mindset that allows a proper collaboration. This extra management time is noted to be present both for the students and for the educator.

Jaques and Salmon (2007) also identify other behaviours that arise when students don't feel comfortable working in groups. Poor participation and general distress can be associated with cultural differences and expectations around how to behave in a group, or, as Biggs et al. (2022) also noted, around personalities that inhibit themselves easily when in a *formal* or *high stakes* form of participation. Other issues reported by students themselves are related to frustration around low performance groups, previous bad experiences or having to deal with personalities that are too outgoing, and which completely dominate the whole group in detriment of other students.

The myriads of advantages present in building a learning experience around collaboration can be attained when the proper tools and the proper environment is provided to the students. It is also important to acknowledge and provide pathways to deal with inevitable issues around conflicts and overhead management time. With all this material is possible to propose a guiding conceptual framework for the educational and technological design that is going to be undertaken in this research. This guideline consists of four principles:

1. Collaboration is a skill that students need to learn and develop. This learning offers value to the employability profile of the student, to the development of lifelong transversal skills and as a useful tool to facilitate more learning.

2. Students develop a form of CoP when learning together, with emerging relationships with their peers, their environment and their learning goals. This communities can be nurtured, which in reciprocity generates a rich, heavily knitted and unique development, both as learners and professionals.

3. It is possible to aim for the ideal characteristics that help the development of collaborative attitudes and use them to design and develop learning tools that take advantage of a collaborative learning approach.

4. The advantages inherent to collaborative learning can be minimized or negated if proper attention is not given to the equally inherent problems that immediately arise. Any learning tool needs to be designed around these issues and provide resources for both the students and the educators to propose solutions fitted to their context and particularities.

This set of guidelines is meant to inform the design and development of learning objects, tools or experiences. Therefore, it is now necessary to analyse all the ideas so far exposed about collaborative learning in the context of technology-enhanced learning, and particularly around augmented reality as a tool for education. With this approach, we will have all the theoretical foundations that inform this thesis, and this is the objective of the next section.

2.2 Augmented Reality in Technology Enhanced Learning

Throughout chapter 1, it was briefly explained how the interest in using AR as a tool for education is born from an inherent interest in the technology and a perceived value that the characteristics of AR could offer unique solutions to educational problems related to motivation, engagement and communication. It is now necessary to explore the current use of the technology in the education sector to identify advantages, disadvantages, common issues and, in general, the current state of the technology as related to the field of education.

2.2.1 Context

The idea of technology-mediated education has often been linked with that of collaborative learning. Lehtinen et al. (1999) highlight key ideas about the communication process developed in learning scenarios and how we use technology to facilitate and manage such activities. Many of the main concepts explored by the authors echo those already presented in the previous section . Ideas like the importance of nurturing a community, selecting proper tools for communication, structured process of review and mediation are explained in similar detail as before, as well as other common issues identified, like excessive competitiveness and factors for social friction.

Dillenbourg and Fischer (2007) propose some recommendations to incentivise effective collaboration in students, like physical spaces that facilitate interaction instead of individualisation, a common but flexible structure for collaboration that is transmitted to the students and techniques that the educator can use to orchestrate the technology used in classroom. It is key to acknowledge that the process needs constant moderation and motivation to create consistent results. These examples reinforce the idea that collaboration needs an extra effort to transform the work performed by teams into knowledge and learning. Coordination between the technology mediating the interactions and the expertise of the educator building the experience is crucial to achieve that goal.

Complementary, Kirkwood and Price (2014) offer a systematic review which aim is understanding what is being enhanced when talking about TEL. When categorising the data, the authors found that interventions often sought changes in the current state of an educational material, and that the chosen technology could offer positive quantitative or qualitative changes, either in terms of markings, assessments results, motivation or deeper understanding of the teaching material.

The literature in the realm of TEL is rich and it is possible to identify some form of evolution in the topics analysed when following the jump from teaching machines (Watters, 2021) to computer-aided learning, to online and mobile learning. Among all the information that can be found it is easy to get lost or confused in all the definitions and terminologies proposed. A good example is the term Virtual Learning Environment (VLE), which will often be used to describe the approach of this research. It tends to be used interchangeably with another concept, Learning Management System (LMS), which refer to the tools used in the administration of educational content and other administrative functions common of educative institutes, which is not the aim of this research. The term VLE is often abused a little to try and give more value to a system or set of tools more akin to an LMS.

To stablish more precise use of the term VLE, we can use the characteristics given by Warburton (2009) like immersion, a sense of presence, visualisation, contextualisation of content and simulation of impossible or dangerous experiences. It is interesting that the author also emphasises these elements:

- Tools for affective, empathic and motivational interactions.
- Intelligent interactions between individuals, communities and objects.
- Opportunities for individual and collective identity and role-play.

The emphasis on the social aspects of the interaction with any form of educative technology is crucial when analysing the enhancement provided by the tool in question. That emphasis can be summarized in three important elements: Visualisation of content, sense of presence and immersion, and a sense of collaboration or participation with other people.

With a a more precise definition of a VLE at hand and a set of tools to draw comparisons between technologies, in terms of visualisation, immersion, interaction and collaboration. The next question would be, why focus on AR? What makes AR an interesting technology for education? What makes it special or different enough from other technologies in the same spectrum, like VR?

AR saw a surge of popularity between 2011 to 2013 (Akçayır & Akçayır, 2017; Fidan & Tuncel, 2019) due to many factors like a wider adoption of suitable technologies for deployment. Successful implementation of applications in diverse scenarios has also been documented and reported (Javornik, 2016), and access to developer-focused tools like ARCore and ARToolkit prompted and facilitated development efforts. In general, and very similar to the case of VR some years prior, successful applications and explorative research was boosted thanks to new toolkits and technologies that facilitated learning and development.

In contrast to VR, the resources needed to develop and deploy an AR solution are less complex, and generally cheaper (Ivanova, 2018). This relative ease of development makes AR a popular technology to try out without the risk of committing too many resources into one single solution.

In terms of comparing the usage of AR with other similar technologies, different studies have shown numerous perceived advantages in the use of AR at different levels of education, and they are mostly consistent between studies: enhanced engagement in the learning activity, high levels of immersion and cost efficiency in terms of deployment and usage (Challenor & Ma, 2019; Chen et al., 2017; Saidin et al., 2015).

Common problems found are also consistent throughout literature and they are mostly related to usability issues: technology instability, steep learning curves and time-consuming usage. The term *cognitive overload* is also commonly found through the literature, normally associated to the unnecessary stress derived from the complications of using the technology in addition to the demands already present in the learning activity.

It is noticeable that most reported results are focused on validating the engagement of a particular development rather than its educational role in the learning activity and the educational objectives associated with the project in the first place. To broad the analysis of why AR is a valuable technology for education, a complementary question can be asked: What educational goals are being fulfilled with the use of AR technologies?

The idea behind this question is to identify actual pros and cons of AR technologies through the lens of educational achievements rather than usability or feasibility, as is common in the current literature. To facilitate an answer, four supporting questions can be proposed:

- Are there other advantages or positive outcomes beyond a spike in the student's engagement?
- Are there circumstances in which AR technologies work better than other technologies?
- What are common practices and mistakes emerging from the reviewed literature?
- How do advantages, mistakes and general results change when the field of study or the educational level changes?

Given that AR is the main focus of this research, it was important to propose a deeper review of current literature in order to give substantial answers to the questions just proposed, and to position the technological development of this research in a proper educational basis that both uses well established educational practices and explores avenues, solutions and ideas missing or seldom touched by current literature. Sections 2.2.2 through 2.2.4 will then discuss in detail the systematic literature review performed for this purpose.

2.2.2 Systematic Literature Review Methodology

One common way to analyse learning objectives in any given scenario is by using Bloom's Taxonomy (Bloom et al., 1964). This classification proposes 6 categories in which a learning activity can be positioned depending on the cognitive complexity it requires from students. From easiest to more demanding, these categories are: Remember, Understand, Apply, Analyse, Evaluate and Create. Each category is meant to be seen as actions the students must take to achieve any educational goal.

The taxonomy is a flexible tool in education, useful for both design and evaluation processes. It can be used to understand how a course or lesson is trying to achieve a goal, and if it is doing it with the correct approach (L. W. Anderson & Krathwohl, 2001). It can be seen as the initial blueprint for the design process of any educational material (Healy et al., 2011) and it has also been used as the basis for establishing standards and policies (Guskey, 2001).

Nonetheless, it can be argued that Bloom's taxonomy offers only a one-dimensional approach to education, relegating knowledge of any subject as the simplest aspect of cognition. Krathwohl (2002) proposes a revised taxonomy in which knowledge is a second dimension that gives context to all the other cognitive categories, considering all the ways in which different domains at different levels of education approach learning. This knowledge dimension proposes four categories: Factual, Conceptual, Procedural, and Metacognitive. These categories are

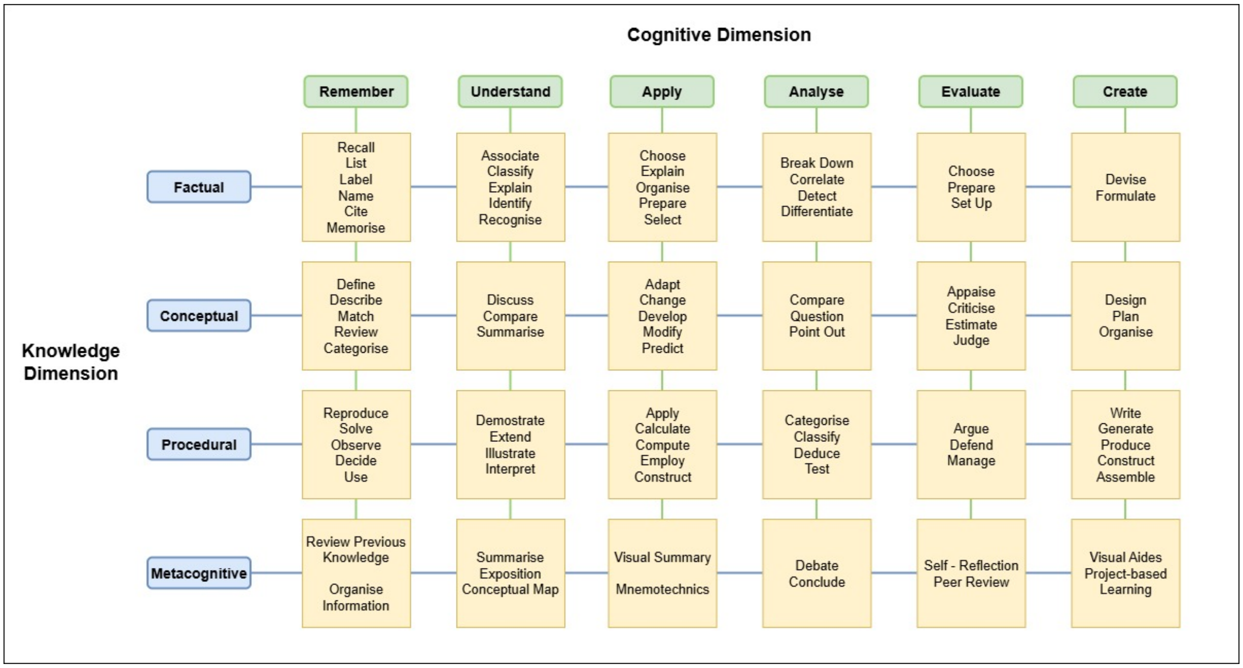

Figure 2.3: Bloom's Revised Taxonomy with Example Verbs

not meant to be hierarchical like in the cognitive dimension, just a representation of the type of knowledge that is acquired. In conjunction, the knowledge and cognitive dimensions can give an educational goal a more pinpoint description of what it is trying to achieve, and a more substantial tool for comparisons.

Figure 2.3 shows a visual representation of the taxonomy as the intersection of the complexity of a task with the type of knowledge that is asking the students to acquire. The figure shows some examples of action verbs that can be used to describe each intersection. These examples were constructed using as reference the works of Newton et al. (2020), Radmehr and Drake (2017), and Stanny (2016), which perform an amazing job at compiling, synthetising and organising reported lists in the literature of common action verbs used at each level.

This visualisation will be used as a reference to analyse the academical material found in the web to determine the pedagogical goals of a particular project. The Bloom's Revised Taxonomy (BRT) framework is valuable because it can offer insights about the complexity of a goal, the nature of its educational setting, and a concise and easy description of the goal itself to use during the analysis of the information gathered.

A SLR methodology is used due to its main premise of being replicable (Siddaway et al., 2019), to serve as a tool to encourage discussion around the type, quality and quantity of the sources used for the review, and to enable the comparison of results given different search, filtering, or codification parameters. This SLR process consisted of four distinct phases: selection, filtering, codification, and a final analysis.

Selection Phase

This phase focused on finding relevant work in AR tools for education at different learning levels and fields as possible. The main selection criteria in this phase were the year of publication and the technology used. The timeframe selected was from 2017 to 2021 due to the amount of work already done in the 2011 to 2016 period (Altinpulluk, 2019; Arifin et al., 2018; Chen et al., 2017). This timeframe also helped to focus the analysis on recent projects with a more contemporary understanding of AR technologies and access to modern development tools and products.

The technology had to be reported as AR in the title, abstract or key words list. The term had not to be exclusive or opposed to VR or MR, this with the purpose of not excluding works that could add valuable information for the analysis prior to a more detailed reading. It is also very common to use the terms AR, VR, and MR interchangeably and are common umbrellas for more specific technologies that could be used to define projects of interest for the SLR.

Google Scholar was the selected search engine using the following queries:

- (augmented reality OR mixed reality OR AR OR MR) AND (education OR training OR learning) AND (technology enhanced OR technology assisted)
- (augmented reality OR mixed reality) AND (education OR training OR application) AND (review OR evaluation OR case study)
- (augmented reality OR mixed reality) AND (industry OR job) AND (training OR instruction OR practice)

Due to the nature of the Google Scholar engine, more than 10000 results were found and could not be realistically managed in a reasonable timeframe. Nonetheless, Google Scholar was maintained as a search engine due to the diversity provided by the tool in comparison to other engines and aggregators. Only the first 10 pages of indexation were used to limit the initial selection of papers. At this point, sources were excluded if any of the following elements were identified in the abstract or summary:

- A merely technical approach to MR technologies with no identifiable educational component.
- A systematic review or a collection of case studies that would make it difficult or impossible to access information about each individual project.
- A duplicate of previously surveyed articles.

Filtering Phase

The amount of literature to be reviewed was reduced to truly relevant material with enough reported information for any analysis to be possible. A first set of criteria was used to exclude projects that:

- Use VR rather than AR or a form of MR that did not fit well into the AR definition, as proposed by Milgram et al. (1995).

- The AR implementation assisted or facilitated a task rather than to educate or train, i.e., surgery planning instead of anatomy education or procedure education.

- Show a general discussion of AR as a tool in education or other context, and not a particular application or case study that could be coded and analysed.

It was possible to identify these elements by quickly surveying sections like the abstract, the introduction or the methodological description. A second set of criteria was used to determine if the project in general provided enough information to process it in the coding phase:

- The article had a clear enough explanation of the tool and its pedagogical purpose.

- The article had a description or discussion of the outcomes of the project or experiment.

For these elements, it was necessary to make a deeper reading of the results, discussion and conclusions sections, and because of that, it was done in parallel with the codification phase.

Codification phase

In this phase, two researchers extracted the relevant information from the selected literature and classified it into two categories:

- *Analytical information:* Year of publication, target demographics, target field, particular AR technology used, success of the project (either reported or inferred) and specific experiences during the development or deployment of the tool. Special attention was given to identifying, if possible, reported positive and negatives outcomes with the tool or its usage.

- *Pedagogical information:* The concrete pedagogical goals of the project as enunciated by the text itself or inferred by the researchers. Educational goals were coded in the format *action verb* + *statement* as specified by Fuchs and Deno (1982).

Determining whether a project was successful or not was very subjective and there are no general criteria to give a pragmatic evaluation of this type. It is also debateable that publishing a report of an unsuccessful project is not a common practice either (Cook & Therrien, 2017; Dawson & Dawson, 2018). A couple of criteria were used to determine if a project could not be considered a success:

- A considerable part of the objectives or hypothesis for the project were not met or proven wrong.

- The discussion, results, conclusions, or any similar section heavily focused on problems and negative outcomes without mentioning significant positive or successful ones.

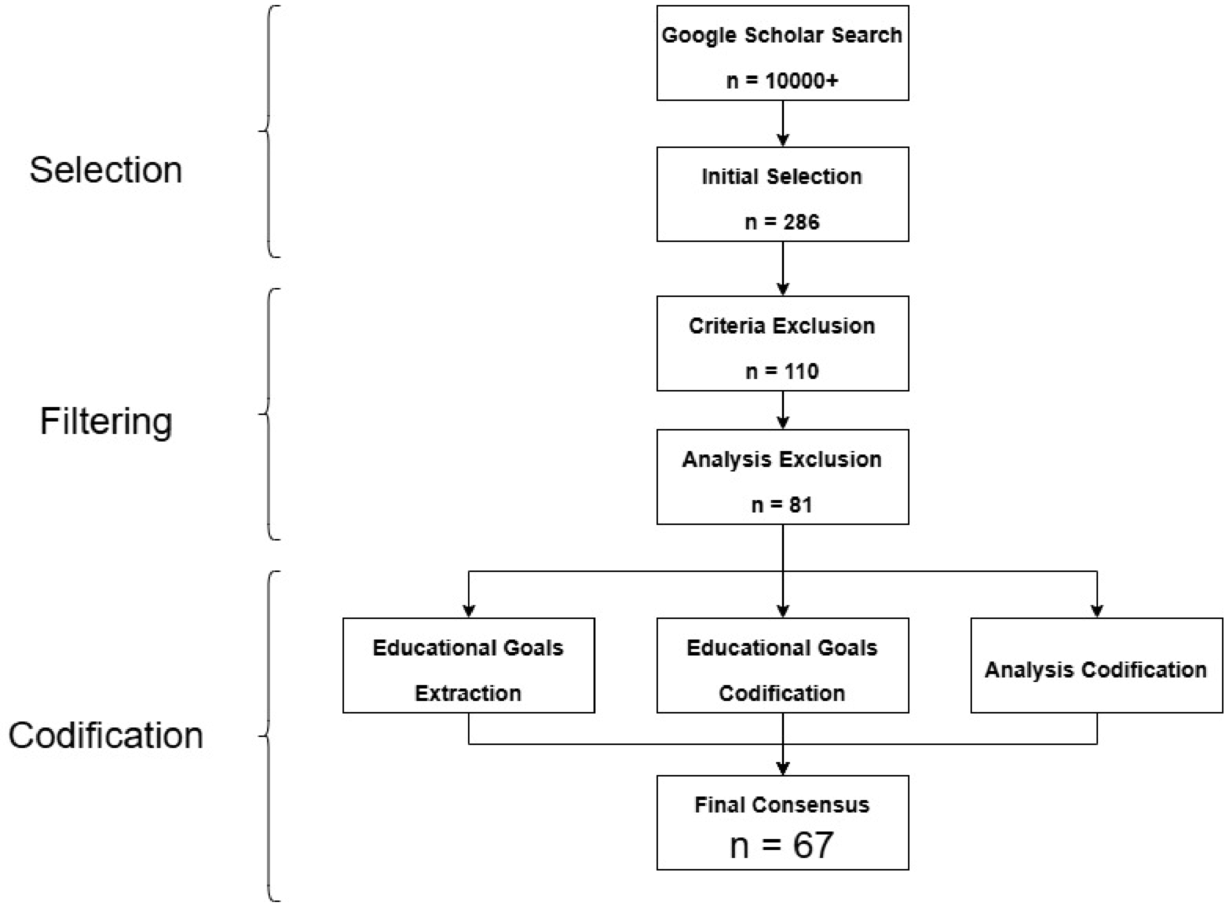

Figure 2.4: SLR Process

- It was not possible to build or test the intended project, or the design had to be significantly altered, especially for technical reasons.

The last stage of codification consisted in positioning each pedagogical goal into the BRT given its perceived cognitive and knowledge complexity. In all stages of the pedagogical codification, each researcher gave its own analysis and classification for every text individually, then reached the final codification by consensus. The final codification for every text surveyed can be consulted in Appendix A.

Figure 2.4 shows the general flow of the process, and the number of documents filtered in each phase until the final count used for the analysis. The process follows the standard recommendations given by the PRISMA statement (Page et al., 2021).

2.2.3 Analysis Phase

In the final analysis, content and thematical analysis was used to extract useful insights of the behaviour and tendencies of the surveyed data, as well as some statistical descriptions used to find categorical distributions, modes and tendencies. Data was organised to identify the general distribution of the surveyed projects among different demographics, educational levels and educational fields. More qualitative data was also organized and categorized such

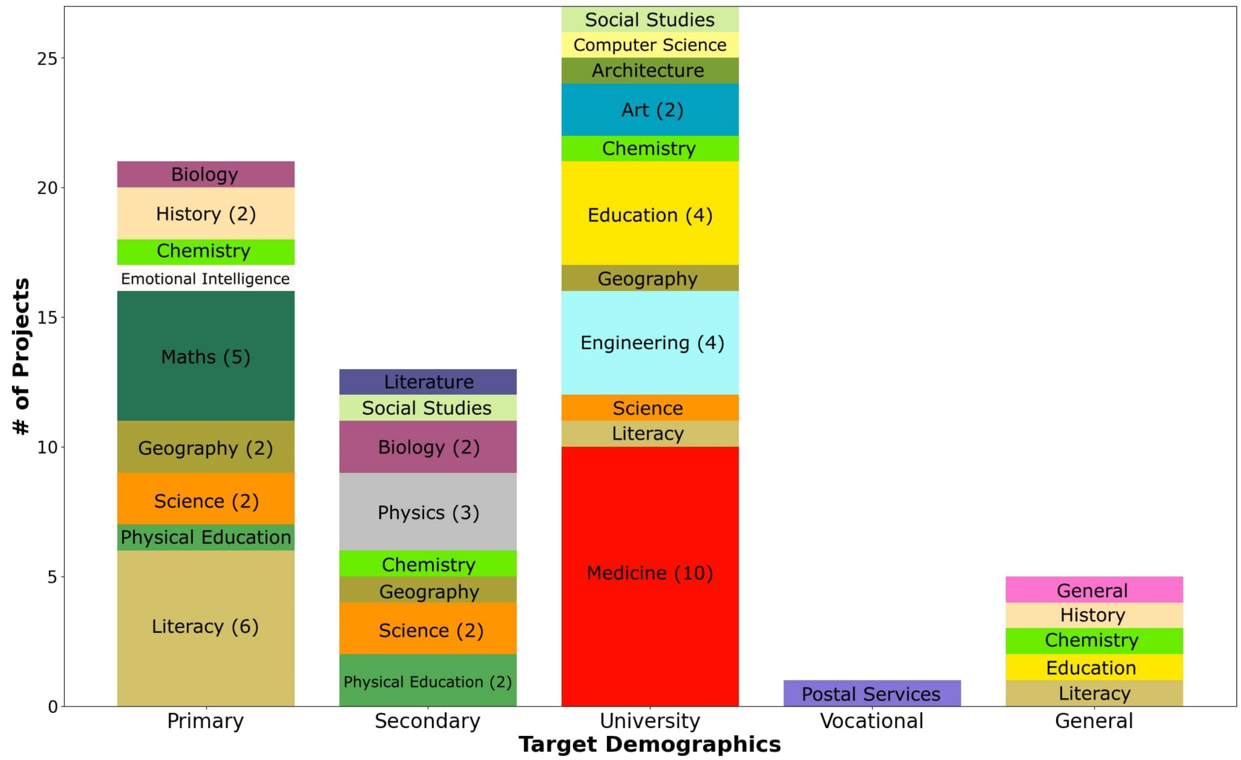

Figure 2.5: Target Demographics vs Target Field

as the experiences reported by the researchers and the users, common trends and positive and negative outcomes. Finally, there was an important focus on identify the pedagogical goals targeted by each project, their distribution and significant trends.

Demographics, educational levels and target fields

Figure 2.5 shows the distribution of projects between the target educational level of the students and the field in which the project is providing the educational material. Demographics were divided into *primary school*, *secondary school*, *university*, and *vocational training*. Target fields where coded as general as possible to avoid too much granularity in the data.

University-level education comprises most of the surveyed population and has the most variety of target subjects. Yet, K12 education in general surpasses higher education if the division between *primary* and *secondary* school is ignored. The division was kept because, as the figure shows, there is little overlap of learning fields between target demographics. This suggests that the focus fields for each educational level are well identified and most worked on. Math, science, and language are the predominant focus for K12 projects, with more complex applications of mathematics, like physics, or new subjects like chemistry being more prevalent in secondary education. University projects either focus on field specialisations (medicine in general, surgery training specifically, for example) or in common learning abilities such as critical thinking or problem solving. Little was found in vocational training using AR.

A fifth category emerged from the data and was labelled as *General*. It refers to projects aimed at creating a tool or experience focused on a particular field but not for a particular demographic, instead meant to be accessible by anyone or to present a broad approach to the field with a loose or no focused learning goal at all. One project was classified with a *General* field and *General* demographics, a metacognitive tool meant to help in the process of studying itself with issues like concentration and motivation.

An effort to create tools for learning languages is notable, used mostly as a tool for heavily ideographic languages, like Mandarin, or as an aiding tool to boost retention or even creative skills. Other projects were related to areas that could not be classified by the BRT, like physical education or emotional intelligence. Those projects were kept in the survey because it was possible to code all other data not related to the classification in the taxonomy and could provide some interesting insights.

Fields inside the STEM spectrum are a major focus for any educational research or development, often seen as a priority for a rounded education, career success or community development (Madden et al., 2020). Fields outside STEM, on the other hand, are often seen as underrepresented in research and development endeavours and not given the same importance or focus as STEM fields. In Figure 2.6, fields in red are part of the STEM classification, 52% of the identified fields. This shows almost a half and half distribution between STEM and non-STEM subjects. In terms of volume of projects per field, STEM fields have a bigger volume, taking almost 60% of all the fields identified. Non-STEM fields are present, but in a lesser quantity. Education, or learning to educate (in any field, including STEM) is the biggest representation of non-STEM fields, followed by tools focused on history, general literacy, and alphabetization.

Positive and Negative outcomes

A list of positive and negative outcomes was coded for each surveyed project based on what was reported in the findings, results, or discussion sections. Outcomes were then coded using a theme codebook to identify recurrent ideas and problems. Figure 2.7 shows the recurrences of all the coded themes for positive outcomes. Figure 2.8 shows the recurrence for negative ones. A summarized label was given for each theme, and it was assigned to a broader category to help readability. A full description of each category can be consulted in Appendix A.

Tendencies identified in other studies can also be seen here for both positive and negative outcomes: Enhanced performance and motivation are the most common results, while technology issues in general are the most common reported problems. Aside of these expected results, more surprising findings were also identified. Providing an autonomous educational experience was a positive outcome for different projects. Control over the pace of the learning process, access to a tailored experience or providing alternative tools for different types of learning were common positive feedback. In contrast, positive results associated with socialisation or collaboration were fewer, showing a high tendency to create individual experiences.

Regarding negative outcomes, the most recurring theme is *None Reported*. There is a general tendency to not publish or report any kind of negative results, either to not undermine the general validity of the research or because it is not considered important information to be reported. Is more common for projects derived from engineering or computer science to be more thorough in describing or commenting on the design and implementation process of the tool, as well as reporting problems or negative results, while projects that are closer

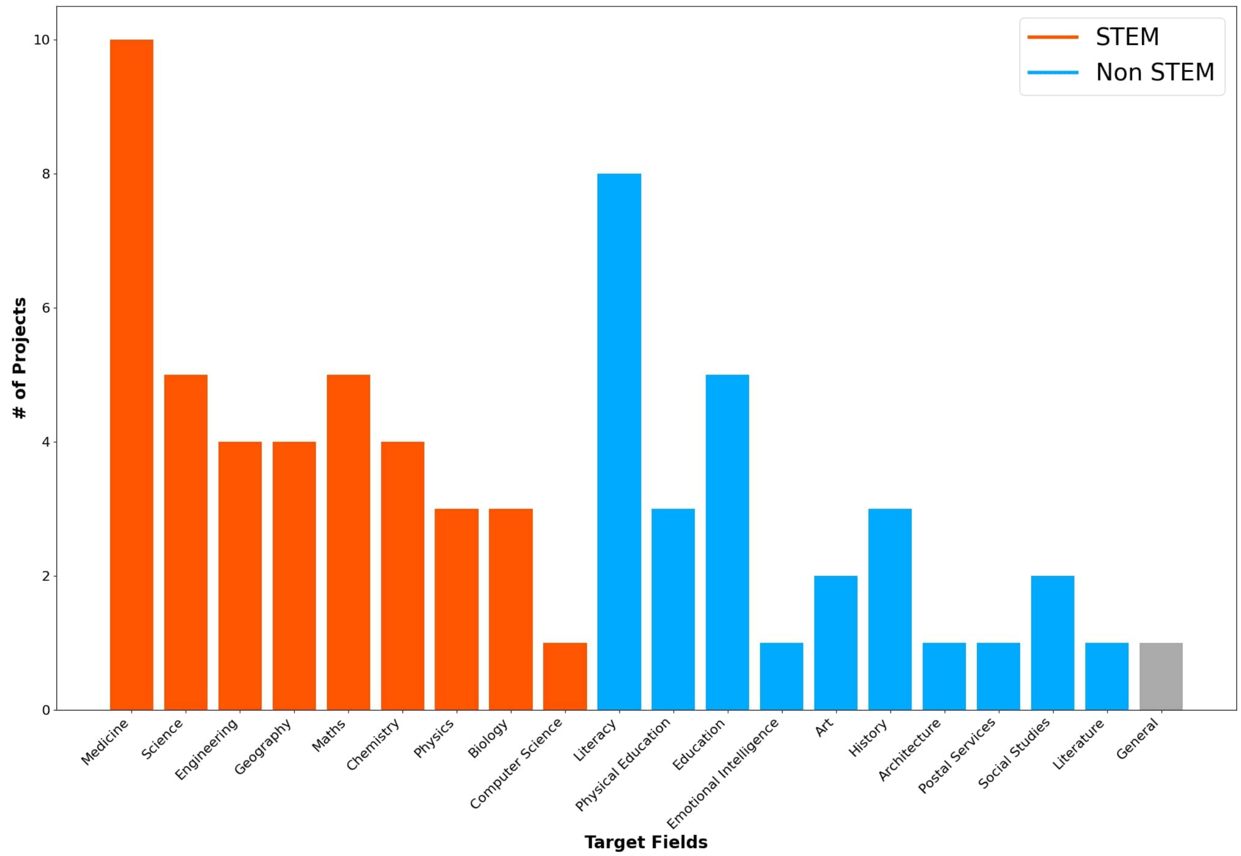

Figure 2.6: STEM vs non-STEM Fields Distribution

aligned to education research or humanities tend to focus more on the results and discussion derived from the experimental design.

Other negative outcomes are related to unintended behaviours caused by using the AR tool. Most popular undesired behaviours reported are constant distractions, hinderance of other skills or excessive time consumption. Another reported outcome is the users perceived lack of quality or quantity of content, or researchers having problems to create content for the tool.

An interesting fact that can be inferred from Figure 2.7 is that most projects guided the validation of their proposed solution by comparing performance gain through the usage of AR. This conclusion is derived from all the positive outcomes related to increased performance, and their absence in Figure 2.8. User validation is another common comparison tool present in discussions and conclusions, commonly in the form of user's perception and acceptance of the tool.

The reported negative outcomes coded were paired with the success classification given to each project by the researchers. Figure 2.9 shows this cross-reference for projects coded as successful, while Figure 2.10 does the same for projects that were coded as unsuccessful.

It can be noted that the density of negatives experiences reported by projects categorised as successful is lower than for projects that were not classified as such. Also, projects categorised as successful are more likely to report no negative outcome at all (the *None*

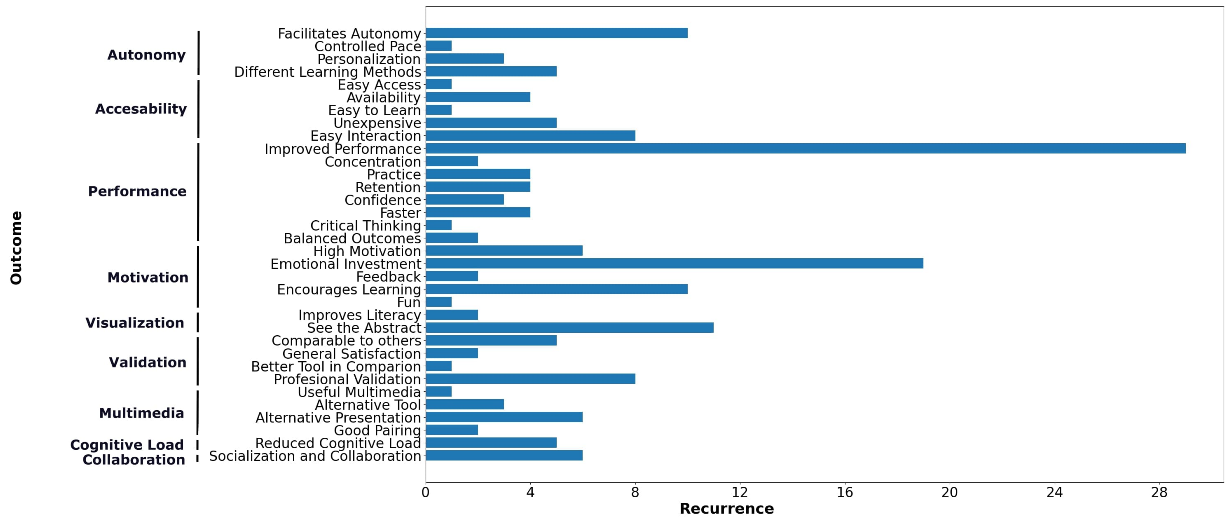

Figure 2.7: Categorisation of Positive Outcomes

Reported category), while projects not categorized as successful where more prone to report technical issues. It could be argued that when technical issues where present, it was harder for users to engage with the tools, resulting in worse perceptions or worse performance results. In contrast, when an immersive, fun, or simply good experience was provided by the tool, technical issues could be glossed over with more ease by users and researchers alike.

Distribution of Learning Goals

Information about the pedagogical goals of each project was extracted and classified using the BRT framework. The cognitive axis is hierarchical by design, and each project was classified with only the highest category analysed for it. For the knowledge axis, several of the possible categories could apply to the same goal and thus were coded like that. Figure 2.11 shows the distribution of points in the BRT matrix for each project. If the project belonged in more than one knowledge category, each pair formed with the corresponding value in the cognitive axis was plotted separately.

The biggest cluster of projects can be seen in the *Understand* and *Apply* cognitive dimensions. *Factual* and *Procedural* knowledge are the most paired with (37% of the total pairings). Most goals tackled by AR educational projects can be found then in the mid to low cognitive complexity, with a preference in applying *Procedural* knowledge. Examples of this can be training in a practice-oriented process, like a surgical procedure, or learning to perform a task, like reading or arithmetical methods. Understanding factual knowledge is the second most common cluster, in which the visualization advantages of AR technologies can help to comprehend concepts difficult to visualise or to get an example of in real life. Concepts in physics or biological processes are common examples.

Several projects were classified in higher cognitive complexity levels like *Analyse* and *Evaluate* throughout all knowledge dimensions, and some efforts were noted to create apps focused on creativity tasks. Another interesting cluster is the *Metacognitive* knowledge dimension, with several projects at all cognitive levels trying to use AR more as a complementary

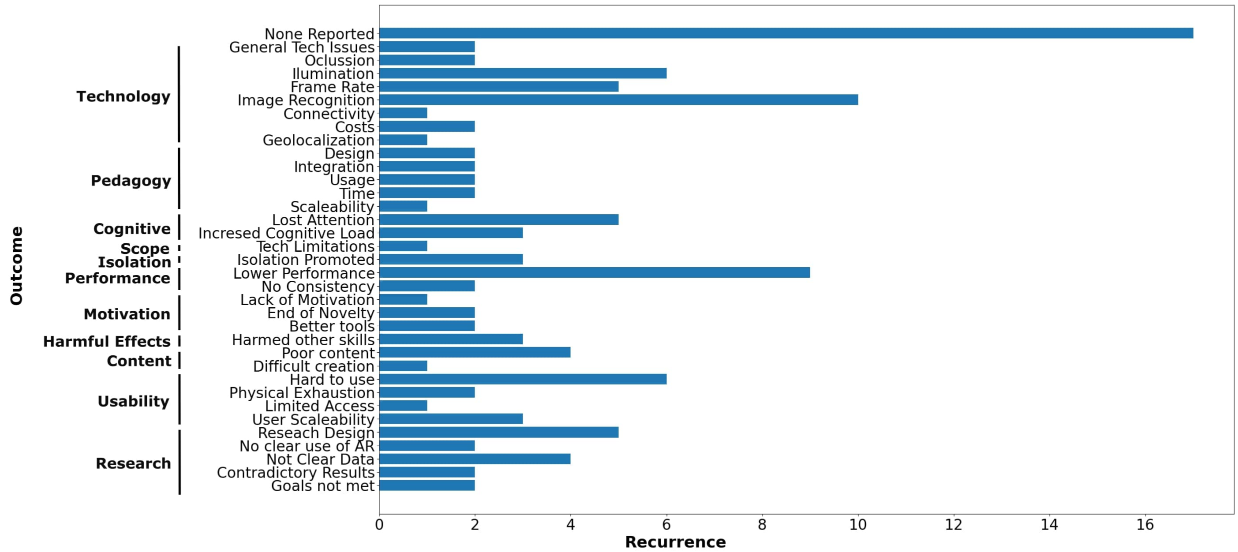

Figure 2.8: Categorisation of Negative Outcomes

tool rather than the main transmission tool of the pedagogical design. This tendency is of notable interest due to metacognitive tools being in line to modern approaches to education like distance learning, long-term learning, learning how to learn and other similar topics. The only gap in the distribution is in the *Metacognitive-Remember* zone, which could be interpreted as mnemonic tools for boosting memory or retention.

Figure 2.11 also cross-references the cognitive-knowledge data with the target demographics of each project. Primary and secondary-level projects coincide with lower cognitive complexity, while university-level projects tend to be in higher complexity levels, though all demographics levels can be found at all cognitive levels in general. University-level projects are more clustered in the *Apply* dimension while K12 levels are more clustered in the *Understand* dimension, which correspond to the common approach each level of education has. The same tendency holds when analysing the knowledge axis: K12 levels are more prone to *Factual*-type knowledge while university level projects are more common in *Procedural* knowledge. The previously identified tendency of projects being in the STEM spectrum can also be seen as a reason for a bigger cluster in the *Apply-Procedural* quadrant in comparison to the lower presence of projects in the *Analysis* or *Evaluation* areas. Figure 12 shows the same distribution of Figure 11 but cross-referenced with a more generalized categorization of the target fields.

STEM related projects are noticeable clustered at the *Apply* section of the distribution. Non-STEM projects are more noticeable clustered in the *Understand-Factual* quadrant and the *Analyse*, *Evaluate* or *Create* regions. In general, the *Remember* region could be considered of little interest or impact for most subjects, hence the lack of points, while higher level cognitive activities get less projects due to the complexity of creating pedagogical designs or tools for those goals. It could be argued that, for those same reasons, no project in the STEM spectrum was found along the *Create* dimension.

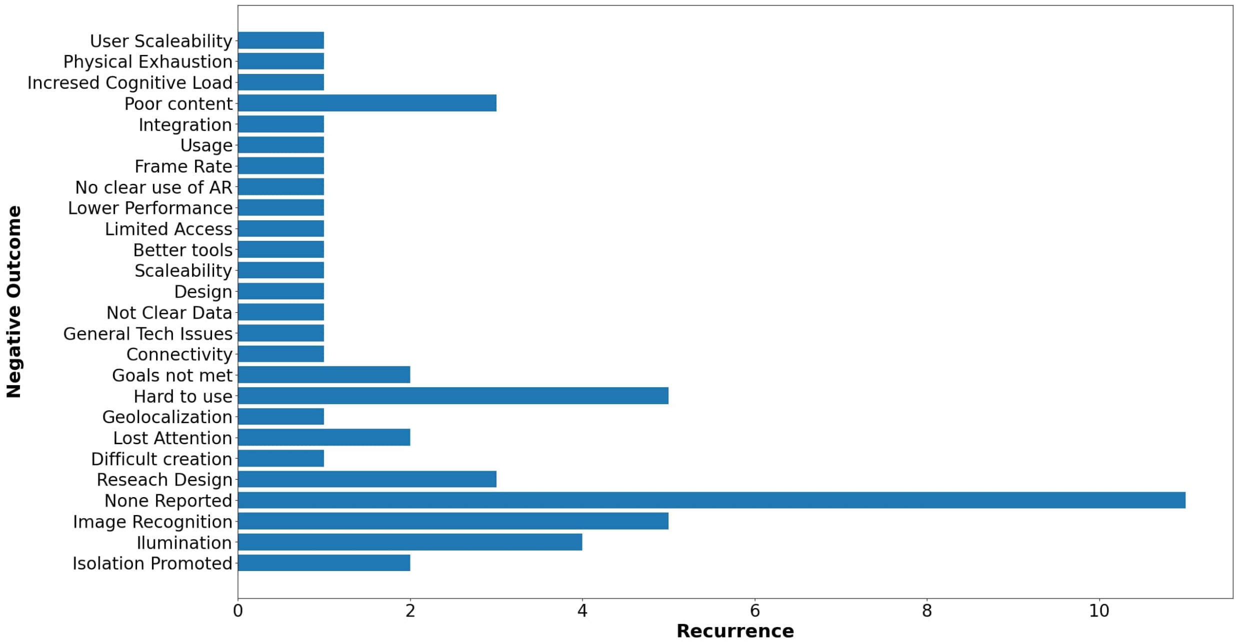

Figure 2.9: Negative Outcomes of Successful Projects

2.2.4 Discussion

The data shows that most projects either focus on *Understand* goals, that is, goals aimed at learning a concept and how to use it, or *Apply* goals, those focused on using obtained knowledge in concrete situations. Primary and secondary level education leaned more towards *Understand* goals while universities and vocational training preferred *Apply*-style ones. When used in a professional context, the prevalent use of AR seems to be as an aiding tool rather than as a learning tool. VR technologies are the main option when training is needed in the workspace, which was outside the scope of this review.

Metacognitive tools were present at all levels, showing an increasing interest in learning-how-to-learn approaches. More complex cognitive goals like *Analyse*, *Evaluate* and *Create* were present in smaller numbers, showing an interest in tackling this kind of endeavour but being deterred by both design and technological difficulties. A small effort to create experiences for the public and with a more product-focus in mind is also noted in a couple of fields.

The visualisation advantages provided by AR technologies are commonly considered as one of its biggest strengths, especially suited for complex concepts, abstract information or elements difficult or impossible to come by. The augmentation element of AR technologies was used often to complement in situ experiences, like museums or laboratories, but it was not as widely used, opting more for a simpler *scan and show* approach. Geolocalisation-based augmentation was used in a couple of cases with a variable degree of success and with several issues associated with the technology. Image recognition approaches are significantly less used and were normally associated with custom-build solutions in need of more development or research.

The primary goal for most projects was to achieve a better academical performance by

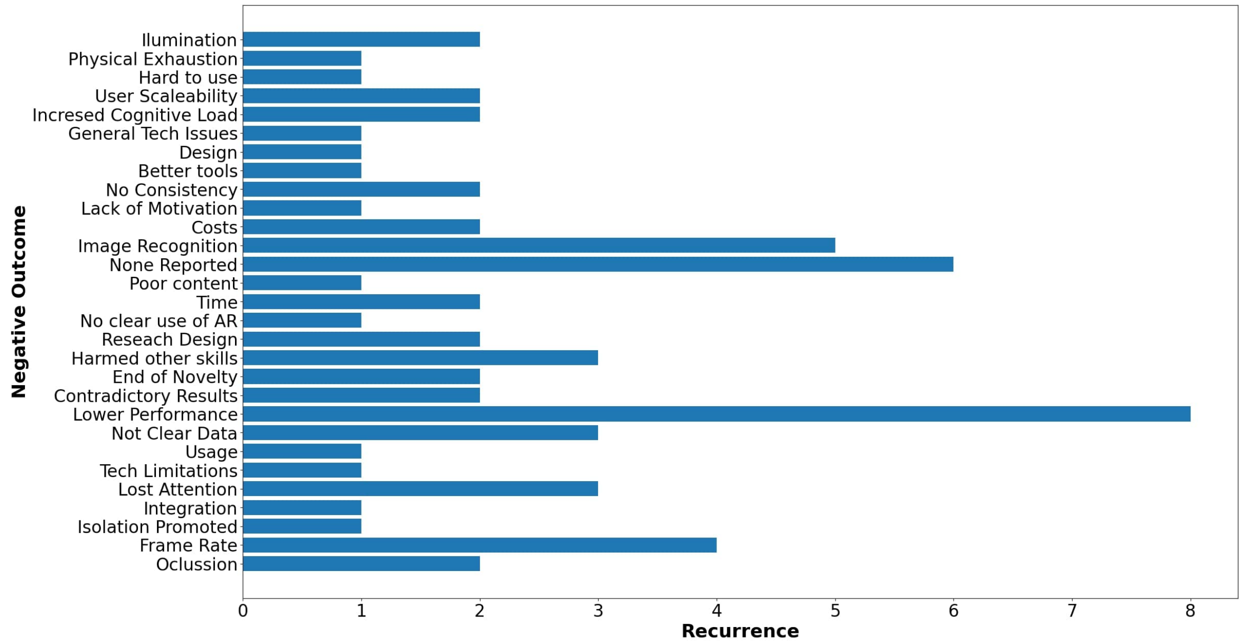

Figure 2.10: Negative Outcomes of Unsuccessful Projects

using AR technologies over other tools or approaches. AR was used to enhance the general motivation and disposition of the students towards learning and to aid the comprehension process through multimedia visualisation, interactivity and assisted practice. Improvements in academical performance were often small but present. AR technologies were commonly seen as more accessible and usable due to the prevalent use of mobile devices as the base platform for deployment and the plug-and-play nature of it.

Although collaboration was a factor in some projects, the general tendency was to create individual tools that helped students to follow an autonomous process. Opportunities for self-paced learning and the ability to have a unique or tailored experience were commonly praised outcomes. This aligns to similar results found in other studies like in Dey et al. (2018) in which the tendency for AR experiences, not only in education, is for individual interactions, with a marginal presence of a collaborative design. It is also clear that AR has been used to explore a diverse number of fields in science, technology and humanities, but is better suited for goals in need of a novel or interactive visualisation or a real-time display of information. Gamification designs that require quick access to information, movement or the visualisation of fictional or no-easy-to-access material are also a good fit for AR technologies. In these scenarios, collaboration is more common but mediated more by the activity rather than by the AR tool.

As for negative outcomes, the most common unintended result is to become a distraction from the educational goals or to impose an extra cognitive load for students, signalling a poor integration between the technological and the pedagogical design. Lack of improvement on academical performance, lack of interest, distractions and lack of content are common reported issues that can be traced to not specifying a clear goal or a clear path to achieve the learning objectives with the use of AR.

Fewer non-STEM subjects were found in the surveyed data, but when present, were often

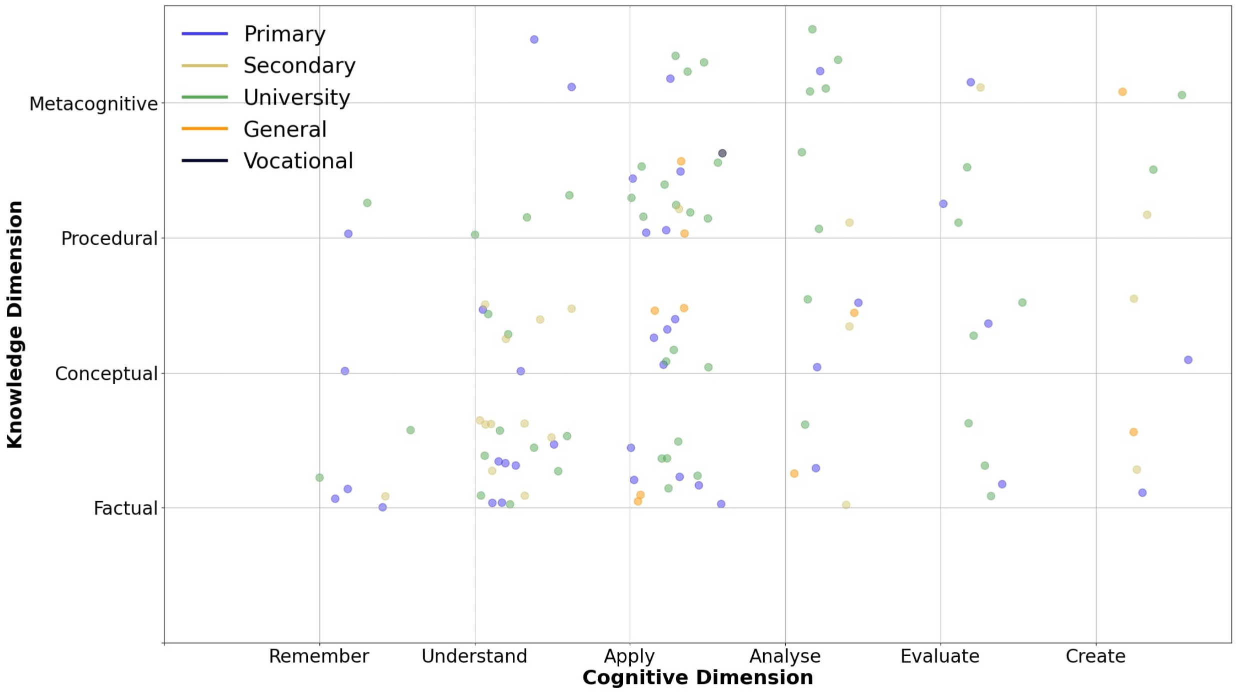

Figure 2.11: Distribution of Educative Goals per Target Demographics

associated with high level cognitive goals. A higher number of projects in subjects like arts or humanities is needed to better understand the advantages and correct uses of AR technologies in these subjects. STEM related subjects, on the other hand, could experiment with more demanding and varied goals, especially in the *Create* dimension.

Validation is mostly done through performance assessment and comparison tests. While offering a good analysis based on data, it also limits the type of discussion to have around the tool, the type of data that can be analysed, and the types of results expected and that can be considered valuable or a success. Understanding not only the statistical gain of the students, but also their experiences, the behaviours observed, positives and negatives, and how the tool relates to the general pedagogical design are also valuable sources of data to consider when analysing the impact and possible benefits of a tool.

Three main ideas can be stated as final conclusions to guide the future development of this research project. First, the identified advantages of AR technologies fit better with *Understanding* or *Applying* abilities, but many examples like Hsu et al. (2019) and Lee (2021) show the value of proposing more complex goals or testing the technologies in unconventional ways. There is an opportunity to understand how the technology behaves outside of its most trending usage.

Second, AR technologies are being used mostly as a visualisation tool, with motivation and accessibility as the main driver for the pedagogical design. Improvements in academical performance are achieved using a multimedia approach and appealing to as many different learning styles as possible. The augmentation element associated to AR has not been used as extensively in comparison. This can be explained due to the technical challenges associated with the image recognition process, and how issues in this area affect the overall stability of

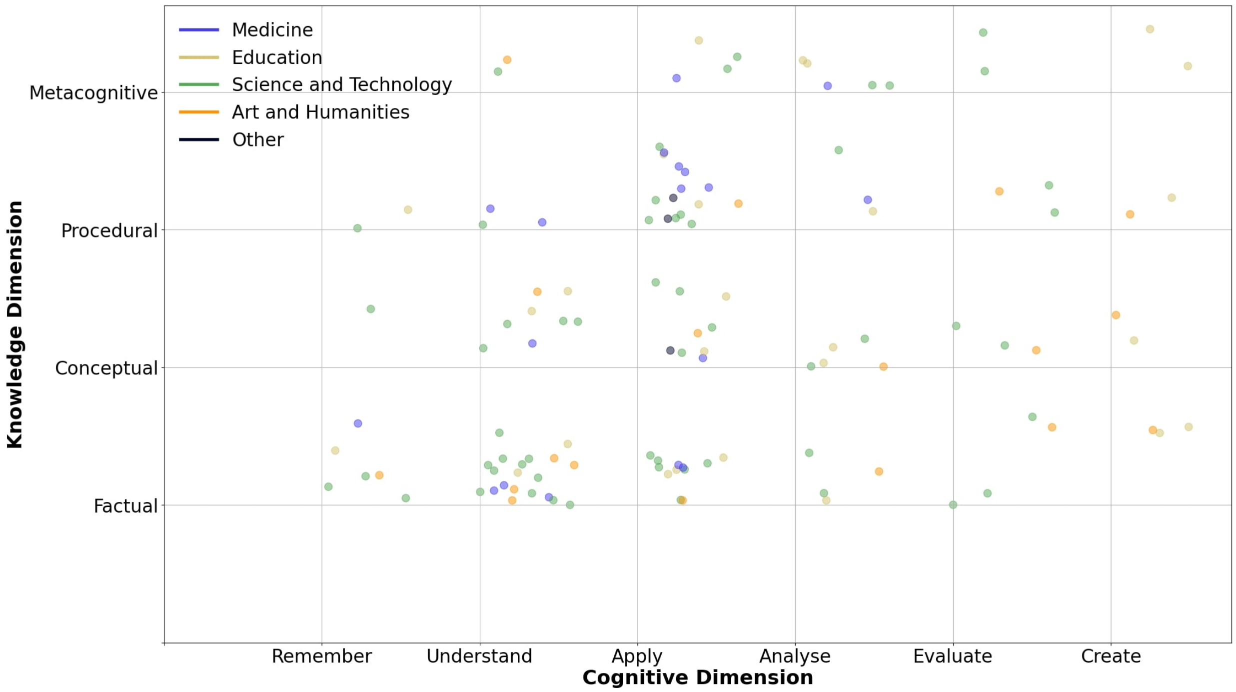

Figure 2.12: Distribution of Educative Goals per Target Subject

the tool and its perceived quality by the users. It is probable that the newest generation of consumer-ready tools will provide an easier way for designers to propose experiences more focused on augmentation rather than only visualisation. Some very interesting results can be achieved when augmentation is the focus of the experience, like in Lim and Lim (2020) or Schaper et al. (2018).

Finally, mobile devices are the most used platforms for development. This facilitates accessibility to the technology and 1-to-1 device usage in any classroom. Yet, most projects found are designed as individual experiences, prioritising student autonomy and promoting a self-pace approach, which under-uses the connectivity advantages provided by mobile platforms. Few examples like Cheng et al. (2019) and Kim et al. (2017) are a multi-user or collaborative design.

2.3 Collaborative AR

From the many insights obtained from the SLR, it is possible to highlight a potential hole in the literature related to collaborative AR experiences in education. A significant majority of the projects surveyed prioritized individual interactions, and those few using multi-user approaches perfectly highlighted the interesting possibilities that the technologies offer around collaboration, teamwork and project-based approaches to learning.

We can expand the information obtained through the SLR and focus some more on the concept of collaborative AR in general, the research done around the topic and the most notable developments found in the literature. If we analyse the topic outside the field of education and consider how AR is being used as an aide in different tasks, we can find multiple

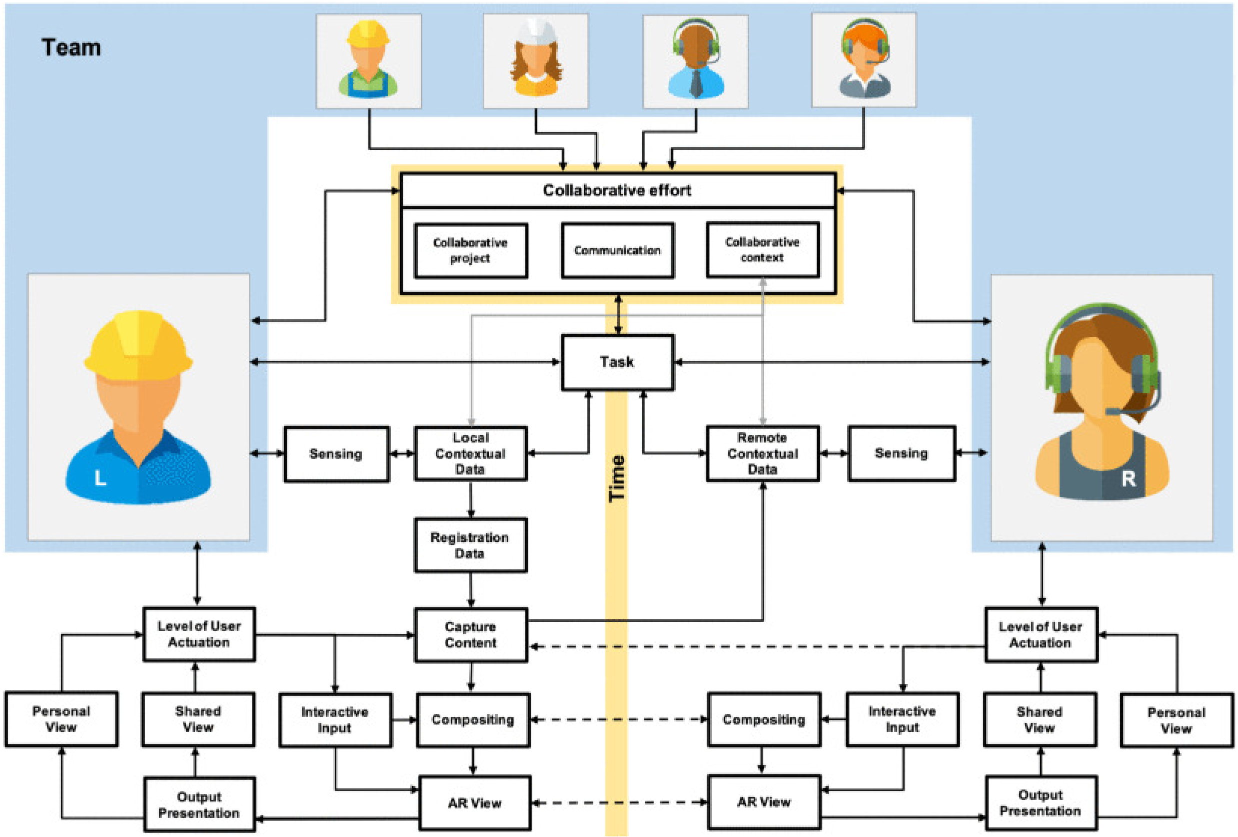

Figure 2.13: Conceptual Model for Collaborative AR as proposed by Marques et al. (2022)

tools that have been tested in diverse scenarios and have been proven to offer benefits and advantages in terms of performance, communication and workload management.

A first approach, before getting into the realm of analysing specific tools, is to understand the domain itself, identify the most important concepts in the core of a collaborative tool and their relationship with AR. Marques et al. (2022) approached the domain of collaborative AR with that same perspective and offered a detailed mapping that can help in categorising and describing any project that could fit in this field.

Figure 2.13 shows the most important components in a collaborative AR tool that interacts with the user and within each other to provide a multi-user environment. Of the key components identified, three main categories can be identified:

- *Context awareness:* Gathering data from the context, either the real space or the actions of other participants.
- *User interaction:* Sensing the actions of each user, each point of view and each flow of information.
- *Composition:* Using data from the previous categories to create the AR view of each user.

These components do not differ much from a generic conceptual model of any AR or MR interaction. The main distinction lies in coordinating all of them through time and different users. The biggest addition is the category *Collaborative Effort*, composed by the components project, shared context, communication and coordinated effort. These components resonate with the fundamental characteristics off an effective collaboration exposed in section 2.1.4. It also serves as the connective gear between users and holds the bulk of interactivity with the app.

It can also be appreciated that the model does not forces a particular architectural design, save for the general communication pathways proposed between components. Another noticeable decision is to coordinate all the users through the *Task* component, giving the collaborative effort a project-oriented nature and a more hands-on approach to the type of applications that could be designed using this model, which can be seen as a first stepstone for a more detailed proposal in the future. It is also useful as a shared language to understand, communicate and analyse the decisions and particularities of a specific implementation, such as this research project.

In terms of extracting interesting implementation cases from specific applied fields, *Collaborative Design* is a domain in which AR has been integrated into work processes a lot. Different projects experiment with tasks requiring communication of ideas, co-creation of products and balancing the workload of the team members. X. Wang and Dunston (2011) for example, proposed a comparative study that showed performance improvements when using a design pipeline that integrated a MR component. They focused on offering both a face-to-face collaboration process as well as a tangible interaction design using image markers.

Fata Morgana (Klinker et al., 2002) is a similar proposal in terms of implementation, but focuses more on the presentation of info rather than in the interactions with models. The collaboration aspect lies more outside of the tool; its main purpose is to show information while being aware of the context and giving the user freedom of mobility. The Magic Meeting (Regenbrecht et al., 2002) offers a similar concept by providing a physical setup to interact with the markers and the digital models. Again, the emphasis is more on the communication process, where the tool offers the means of interaction with the digital construct, but the main objective is what is happening between the users.

Another prominent field for the use of AR is medicine. The technology has been tested several times in diverse areas like procedure aide, data visualisation and training. When exploring collaborative instances of AR tools, we can find some modern projects focused on helping in decision-making process based on data and consensus, such as the examples shown in Friedl-Knirsch et al. (2024) and Yasojima et al. (2012). But the most prominent application is the use of the technology in procedures using telepresence, tele-assistance or tele-mentoring. AR has been used as an enhanced remote collaboration method, allowing for communication enriched with visual data and real-time interactions. Wisotzky et al. (2019) is one of the most radical examples in exploring the boundaries of the technology. The authors propose a framework for full digital reconstruction of a surgical procedure to enable remote collaboration through AR, highlighting the challenges related to image recognition and network stability. Other projects aim at helping visualise and guide complex procedures and research goals that relate more to feasibility and safety (Alismail et al., 2019; Jun et al., 2023). Other common examples, like Augello et al. (2023), Pooryousef et al. (2024), and Yu et al. (2023) focus more on the display of information, remote or collocated, using a more common marker or feature recognition approach, and providing aid in brief processes, evaluation or mentoring.

Mentoring or aiding students, specially aiding in practice scenarios for surgical procedures, is a major research venue in medicine, evidenced by most of the literature found supporting these types of projects. Just to name a few outstanding examples: Gasques et al. (2021), C. Y. Huang et al. (2018), Moro et al. (2021), and Schott et al. (2024) all show interest in providing students with tools to understand and practice learning material through visualisation and interaction, while also using a collaborative learning approach.

Other fields also focus more on using AR for communicating ideas rather than direct interaction with the immediate context. The augmentation is focused on giving form to abstract concepts or superposing non-existent elements into existent ones. The collaboration aspects focus on sharing the visualisation or interactions between users, or giving presence to remote users in the collaborative task. Some examples of note can be the project proposed by Garbett et al. (2021) in the field of construction, Shen et al. (2010) in collaborative product design, Ahn et al. (2019) in architectural design and Sanabria and Arámburo-Lizárraga (2017) in arts and creative processes.

Telepresence and remote collaboration appear to be a common goal, but it is also possible to find examples that prioritise face-to-face collaboration in diverse activities. The Collaborative Web (Billinghurst & Kato, 1999) explores the concept of collaborative browsing, sharing information and points of view. The work of Billinghurst and associates in general provides a fundamental view of the challenges and possibilities related to collaborative work in AR and focuses on identifying key design issues around interaction and the technological framework that can be used to propose a solution. Another fundamental example is The Studierstube (Schmalstieg et al., 2002), which shows the necessity of an interaction design process of tools like this. Operations in a 3D space that are shared with other people are too distant from more common desktop or mobile interactions, thus requiring extra work to properly build an interaction proposal.

Modern and early incursions with the technology offer a valuable list of considerations for common scenarios like shared interactions, communication between users and co-located activities. From the reviewed material it can be stablished that several issues need to be addressed in the design stages of any AR solution of this nature:

- A library of interactions with the physical and digital environment that is intuitive, usable, easy to learn and that accommodates the hardware being used. These interactions must consider the 3D nature of the technology.

- A protocol for communication and interaction between users. Co-located experiences should capitalise on the ability of the users to be close together, but solutions must consider issues related to coordination, change in points of view, distribution of tasks and general connectivity issues.

- A clear integration with the overall purpose of the tool. This is important when the tool is meant to be a complement to bigger processes, like and assembly line, maintenance or training. The tool should considerer the time and effort required for set up, tracking, interaction, changes of focus and priorities, establishing communications, among others issues. Good examples of projects dealing with these types of issues can be found in W. Huang et al. (2013) and Tang et al. (2003).

This small sample of projects show an interest in providing the users tools to discuss and communicate ideas, using the capabilities of AR to give a visual anchor to the concepts being exposed and to interact with them. An emphasis on offering tangible interfaces shows the importance of providing a sense of *physicality* to the digital information used in the task. It also provides an interesting analysis of different approaches to interaction design. AR solutions often need to deal with issues related to interactivity and interface problems that are unique to projects delving in 3D spaces or that want to use the user's natural body movements as a mean for interaction.

One final step in the analysis of the available literature is to situate collaborative AR in the field of education. The SLR in section 2.2 analysed analysed AR in general as a tool for education and found an important focus on single user experiences. Those projects providing collaborative designs were few and showed us a research field that offered interesting possibilities and was currently underexplored. Upadhyay et al. (2024) is a more recent SLR that not only explores AR technologies in education but also focuses on experiences using collaborative design and collaborative learning.

The review identified the importance of mobile technology for AR development and collaboration practices, being the most common deployment environment and the most common technique used for implementing collaboration. Even more notable in comparison to the results found in the SLR of this research is the educational approach used by several of the reports reviewed. Projects like Radu and Schneider (2023), Schiffeler et al. (2019), Vassigh et al. (2020), and Xefteris et al. (2019) not only explore diverse learning objectives, but they also explore AR technologies beyond engagement or performance gains, evaluating aspects like communication behaviours, social interactions, induced behaviours (positive and negative), attitudes towards the learning materials and changes in learning processes.

Overall, reported benefits align to concepts already discussed in previous sections and fortified the position of this research to focus on visualisation, interactivity and mobile communications. When analysing challenges identified in the reviewed material, different authors highlight a couple of interesting ideas. Primarily, there seems to be strong trade-off between offering a recognisable environment through mobile technologies that users can easily interact with and the difficulties the students and teachers found in managing new systems and new process. It is important to consider that any technological intervention in the classroom implies a high degree of effort in integrating it to the normal flow of the learning experience. It requires effort for the technology to become and enabling factor and not an inhibitor or a distraction.

There is also an interesting side effect when assessing the impact of the technology in learning outcomes. Section 2.1.4 briefly mentioned that collaboration is difficult to assess. This difficulty can be exacerbated when paired to other, sometimes contradictory learning outcomes, and when the assessment does not consider the raw effects of collaboration in the results, positive and negative. The same can be said about the noise in the students' performance created by the addition of an intervening or mediating technology. In conjunction, a collaborative approach that uses a mediating technology (not only AR) poses a significant challenge for evaluation, performance assessment and other similar tasks. This can be evidenced by the diverse and sometimes contradicting reports found in the review by Upadhyay et al. (2024) and the one performed for this project in relation to what works and what not.

2.4 Final Discussion

This chapter explored the theoretical framework that supports this research project and examined a varied selection of relevant literature related to the most important issues and research trends in the field of collaborative learning, TEL and AR.

This analysis helped in defining and understanding collaboration as a skill that must be built over time with proper nurture and resources. Collaborative approaches to learning can both aim at developing the necessary skills for effective collaboration and at using those skills, such as teamwork, communication and resource management, to boost individual performance, create opportunities for emergent learning and offer educators tools for better classroom management.

It is especially relevant in the exploration of different TEL scenarios to understand the diversity of configurations and tools that can be used and the short-, medium- and long-term consequences for learning, skill development and teaching. Any technology offers benefits and detriments to all these dimensions, and as researchers and educators, it is part of our responsibilities to understand them and manage them.

AR, as any other technology, has proven to offer unique advantages for education, capable of creating strong visuals, immersion and interactivity. It excels at integrating with the user's context, at providing means for exploration, at data-driven scenarios and, central for this proposal, at facilitating shared experiences. On the core of this research project is the idea that, in conjunction with mobile platforms, AR provides important advantages to the design of collaborative learning experiences by using the physical context of face-to-face collaboration and by innovating and recontextualizing the design of tools for communication.

But it is also necessary to highlight the challenges and issues posed by the chosen technology. It is especially interesting to explore solutions for problems related to connectivity, the diverse ecosystem of AR devices, the negative effects in the development of social skills and the challenges students face when adapting complex process and requirements into their learning projects. It is clear that the technology can be both an enabler and an inhibitor. The goal is to identify the proper approaches to boost the advantages while minimising or mitigating the problems.

In the next chapter, the objective will be to use the information gathered in this chapter to properly structure the research and development goals that will guide this project. The current state of the art helped at defining a useful conceptual model about collaboration, the development of collaborative skills, collaborative learning and collaborative AR development. With this information at hand, now it is possible to explain the formal methodological approach for this research and define the proper instruments that will help in the constructions of and answer to the questions formulated by the motivational background of the research and the gaps in knowledge found in the literature.

Appendix A

Pedagogical Goals for Educational Technologies using Augmented Reality

A SLR was performed as part of the literature review process guiding this research project, and the results of the codification phase of the review can be found in the following repository:

https://github.com/speregil/ar-survey-annex.git

This SLR codified a total of 67 different articles related to the use of AR in educational settings. The analysis phase of the review focused on mapping what educational goals were more prevalent in the design proposed in each surveyed project. This information was codified using the BRT framework to understand the cognitive complexity of the goals identified, while the goals themselves were codified using the verb + statement approach proposed by Fuchs and Deno (1982).

The bibliographical information of all the surveyed articles was codified using the following structure shown in Table A.1.

Table A.1: Structure of the Coded Data

Field	**Description**
ID	Unique ID for the entry
Main Author	Last name of the main (first) author of the research
Year	Year of publication
Journal/Proceedings	Name of the journal or conference in which the article was published
Author's Discipline	Main subject affiliation of all the authors of the article as best as it could be extracted by the coders. The reported affiliation to universities and faculties and simple web searches were used to determine such subject affiliation

Continued on next page

Table A.1: Structure of the Coded Data (Continued)

Target Demographics	Intended target demography or educational level of the project or tool. Demographics were divided in primary school (6 - 12 yrs. approx.), secondary school (13 - 18 yrs. approx.), university education and vocational training
Specific Demographics	Specifics about the subjects using for the research as test or final users
Target Field	Intended field of knowledge in which the research or tool works
AR Technology	The main AR engine used to create the tool
Additional Technologies	Other notable technologies mentioned in the research
Custom Build	If the tool was built from zero, tailored for the research, or if some third-party tool was used or adapted to the needs of the research
Goal Statement	Educational goal guiding the research as stated by the article or inferred by the researchers. The goal was coded in the + format
Explicit Goal	If the goal was explicitly stated in the research or if the coders had to infer and formulate a goal based on the information provided by the research
Success	If a project could be categorized as a success or not based on the criteria stated in the systematic review, and as best judged by the coders
Comments	Notable comments of the coders about the research
Knowledge Dimension	Classification given by the coders in the knowledge dimension as stated in the systematic review
Cognitive Dimension	Classification given by the coders in the cognitive dimension as stated in the systematic review
Positive Experiences	List of all the coded positive experiences extracted from the research and using the Positive Experiences Code Book
Negative Experiences	List of all the coded negative experiences extracted from the research and using the Negative Experiences Code Book

The fields *Knowledge Dimension* and *Cognitive Dimension* use the BRT to classify the identified educational goals. Data was coded using the keys shown in Table A.2.

Finally, the *Positive* and Negative Outcomes fields codify the main themes and categories

Table A.2: Coding Keys for the *Knowledge* and emphCognitive Dimensions of the BRT

BRT Dimension	Key	Value	Description
Knowledge	A	Factual	Knowledge of simple facts without context or immediate application
	B	Conceptual	Knowledge of complex concepts, relationships between facts and the context around them
	C	Procedural	Knowledge on the components and execution of a tasks, the reasoning behind them and how they affect their context
	D	Metacognitive	Knowledge about learning, how to learn and your own learning abilities, limitations and approaches
Cognitive	1	Remember	Goals associated with correctly and consistently remembering facts and information
	2	Understand	Goals associated with recognizing concepts, relationships and compositions
	3	Apply	Goals associated with correctly using previous knowledge in a concrete context or scenario
	4	Analyse	Goals associated with inferring and extracting new knowledge from other concepts, context or the relationships between them
	5	Evaluate	Goals associated with giving a value judgement based on knowledge and analysis
	6	Create	Goals associated with creating new elements, procedures or knowledge

found on the recount of experiences extracted from the results, analysis and conclusions of the reviewed works. The categorisation process compiled and used the codebooks shown in table A.3 and table A.4.

Table A.3: Codebook for Positive Outcomes

Category	**Key**	**Value**	**Description**
Student Autonomy	1.1	Facilitates student autonomy	AR makes easy for students to work in an autonomous way.
	1.2	Offers student-controlled pace of the experience	The tool advances and can be used at different paces, controlled by the desired rhythm of the user.
	1.3	Offers a personalized experience	The tool changes content or presentation elements depending on the usage of the student
	1.4	Supports different types of learning methods	The tool presents the same content or activity in different ways to foster different types of learning, like visual, memory or other similar approaches

Continued on next page

Table A.3: Codebook for Positive Outcomes (Continued)

Accessibility	2.1	Easy access to the technology	Getting access to devices need to use the tool is easy
	2.2	Use of available technology	The tool uses available technology for the students, like their own smartphones or simple tech lab's devices
	2.3	Easy to learn	It is easy for the users to learn how to use the tool
	2.4	Unexpensive or easy implementation	The cost of building and deploying the tool is low or lower in comparison to other technologies
	2.5	Easy or intuitive interaction	The tool is easy and enjoyable to use

Continued on next page

Table A.3: Codebook for Positive Outcomes (Continued)

Performance	3.1	Improved academical performance is tested	Students gain academical performance using the tool and it was tested in the research
	3.2	Facilitates concentration/attention	Students or researchers reported improvements gain in concentration or attention
	3.3	Facilitates practice/training	The tool helped students to practice or train
	3.4	Improved long-term retention is tested	Students gain improved retention using the tool and it was tested in the research
	3.5	Boosts confidence in learning outcomes	Students report feeling more confident in what they learned and the long-term retention of it
	3.6	Faster/easier learning	Learning was achieved faster or students report doing it easier in comparison to other experiences
	3.7	Enhanced critical thinking is tested	Researchers tested gains in critical thinking
	3.8	Balanced learning outcomes for different students	Results for students at different levels were balanced or the gaps were minimized

Continued on next page

Table A.3: Codebook for Positive Outcomes (Continued)

Immersion and Motivation	4.1	Higher motivation for the learning subject/activity	Students report to be more motivated or invested in the learning subject
	4.2	Facilitates emotional investment in the subject/activity	Facilitates emotional investment in the subject/activity: Students report to be more invested with the activity, the tool or its content
	4.3	Content used encourages learning	Students feel motivated to engage with the content of the tool
	4.4	Fun or interesting tool	The tool or its content is reported to be fun for the students or presents an interesting approach to the learning material
Digital Literacy	5.1	Helps in digital literacy education	The tool can be used to educate in the use of digital devices
Visualization	6.1	Helps in the visualization of abstract/difficult concepts	The tool provides visual content to explain abstract or difficult subjects.

Continued on next page

Table A.3: Codebook for Positive Outcomes (Continued)

Tool Validation	7.1	Comparable or better than other VR/AR apps	Research shows better results in comparison to other tools or technologies
	7.2	General satisfaction of the subjects with the app	Users report high satisfaction with the tool
	7.3	Better tool to accomplish an objective in comparison to others	Better results in accomplishing a particular goal in comparison with similar tools
	7.4	Tool achieved some form of professional validation	The tool was evaluated and validated by professionals in the field
Multimedia Learning	8.1	Benefits of multimedia learning are noted	The use of multimedia content in the tool greatly contributes to its success
	8.2	Sets itself as an alternative tool for learning	Students see the tool as a good alternative to achieve their learning goals
	8.3	Is an alternative or better way to present a subject	The content used in the tool does a better job to present the learning subject
	8.4	Improvement when paired with other media or tool	The tool is better used paired with other content or tool
Cognitive Load	9.1	Reduce stress or cognitive load	Cognitive load on students was tested or reported to be reduced

Continued on next page

Table A.3: Codebook for Positive Outcomes (Continued)

Collaboration	10.1	Facilitates collaboration or socialization	The tool is used in collaborative scenarios

Table A.4: Codebook for Negative Outcomes

Category	**Key**	**Value**	**Description**
None reported	1.0	None reported	No negative outcome was reported

Continued on next page

Table A.4: Codebook for Negative Outcomes (Continued)

	2.1	General technology issues	Reported failures, problems or inconveniences related to technology usage or setup, mainly of the platform or the AR tool itself.
	2.2	Occlusion	Problems related to the AR image recognition being occluded by other objects or the users themselves
	2.3	Illumination	Problems related to the AR being affected by poor or excessive illumination
	2.4	Frame rate and computational power	Problems related to the selected platform struggling to process or run the tool, creating frame rate issues or slowing down the experience
Technology Issues	2.5	Impractical markers or image recognition problems	Problems related with the selected markers for image recognition, inconvenient usage, constants tracking lost or similar relate issues
	2.6	Connectivity problems	The tool required local or internet connection, the connection was unstable, lost or the necessity for a connection was an inconvenience for the usage of the tool
	2.7	Budget/cost issues	The cost for creating or using the tool were too high

Table A.4: Codebook for Negative Outcomes (Continued)

Pedagogical Design Issues	3.1	General pedagogical design issues	Issues related to the pedagogical design, the educational goals, the content presented or the activities created around the usage of the AR tool
	3.2	AR is not integrated well with other activities	The use of the AR tool is part of a bigger design and is considered a distraction or harmful to the other elements of the activity
	3.3	AR is not correctly used as learning tool	The use of AR in the proposed project is minimal or is not related to learning
	3.4	Pacing and time consumption	Set up or usage of the AR tool consumes too much time and harms other activities in the pedagogical design
	3.5	Poor scalability of the activity	It is hard or nonviable to use the proposed tool with more users or in a more realistic scenario

Continued on next page

Table A.4: Codebook for Negative Outcomes (Continued)

Cognitive Issues	4.1	Easy to lose attention or AR takes attention from learning	The AR elements of the tool are distracting and take the attention from other learning activities or the learning core of the tool
	4.2	Increased Cognitive Load for students	Researchers reported an increased in the cognitive load of the students, either as perceived by the students themselves or tested using any other kind of tool
Scope Issues	5.1	Scope of design too big for technology limitations	The technology used did not suited correctly the intentions or ideas of the pedagogical design, either in the proposed usage or the type and/or number of users desired
Isolation	6.1	AR promotes isolation	The use of AR prompts the students to isolate themselves and not pay attention to their peers or the context in which the pedagogical activity is happening

Continued on next page

Table A.4: Codebook for Negative Outcomes (Continued)

Performance Issues	7.1	Lower or not significantly different academical results	Tests done by the research showed no academical performance increase or it was not statistically significant
	7.2	Results are not consistent between different types of users	Results show there is a significant difference in results between different types of users, either due to gender, age, academical level or any other variable
Motivation Issues	8.1	No interest or motivation in the tool	Users reported or observed to have no interest in using the tool
	8.2	Novelty of the tool ended quickly	Users reported or observed to have an initial interest in the tool but lost it quickly, before the activity ended or in subsequent sessions
	8.3	Other tools are more reliable/interesting	Users found other tools more interesting or useful than the proposed tool
Harmful Effects	9.1	The tool harmed the development of other skills	The use of the tool had an undesired negative effect in the development or acquisition of other skills

Continued on next page

Table A.4: Codebook for Negative Outcomes (Continued)

Content Issues	10.1	Insufficient or lacklustre content	Content created for the tool was reported as low quality by students or they would wish for more or different content
	10.2	Hard to create content for the tool	Researchers reported difficulties in budget, time or resources to create or procure suitable content for the tool
Usability Issues	11.1	Hard to use or requires too much training	The tool is reported by users to be hard to learn and/or use, or using the tool requires too much preparation and proper training
	11.2	Physical exhaustion	Using the tool causes exhausting even in short periods of usage
	11.3	Limited access to technology	The platform chosen for deploying the AR tool is hard or expensive to come by
	11.4	Difficulties with multiple users or different types of users	The tool was not well suited or received by a percentage of the user population

Continued on next page

Table A.4: Codebook for Negative Outcomes (Continued)

Research Design Issues	12.1	Research design has problems	Some elements of the research designs are weird or harm the overall exposition of the tool in some way
	12.2	The AR tool used is not clear	It is not clear in what AR technology is being used, what exactly the AR tool does or how it is being used in the pedagogical design
	12.3	Data presented by the study weas difficult to interpret	The research presented data in a format hard to interpret or that obscured aspects of the research
	12.4	Data presented contradicts results or presents some kind of problem	The data presented contradicts statements given in the discussions or conclusions sections
	12.5	Specific goals of the research were not met	The main hypothesis of the research was proven false or the educational goals stated for the tool were not met

References

Abir, T., Islam, R., Ullah, A., & Rahman, S. (2024). Aligning Education with Market Demands: A Case Study of Marketing Graduates from Daffodil International University. *International Journal of Learning, Teaching and Educational Research*, *23*(8), 137–157. https://doi.org/10.26803/ijlter.23.8.8

Addison, J. T., & Teixeira, P. (2020). Trust and Workplace Performance. *British Journal of Industrial Relations*, *58*(4), 874–903. https://doi.org/10.1111/bjir.12517

Adi Badiozaman, I. F., Leong, H. J., & Wong, W. (2020). Embracing educational disruption: A case study in making the shift to a remote learning environment. *Journal of Applied Research in Higher Education*, *14*(1), 1–15. https://doi.org/10.1108/JARHE-08-2020-0256

Ahn, K., Ko, D.-S., & Gim, S.-H. (2019). A Study on the Architecture of Mixed Reality Application for Architectural Design Collaboration. In R. Lee (Ed.), *Applied Computing and Information Technology* (pp. 48–61, Vol. 788). Springer International Publishing. https://doi.org/10.1007/978-3-319-98370-7_5

Akçayır, M., & Akçayır, G. (2017). Advantages and challenges associated with augmented reality for education: A systematic review of the literature. *Educational Research Review*, *20*, 1–11. https://doi.org/10.1016/j.edurev.2016.11.002

Akhrif, O., Benfares, C., El Idrissi, Y. E. B., & Hmina, N. (2019). Smart Collaborative Learning: A Recommended Building Team Approach. *International Journal of Smart Security Technologies*, *6*(2), 52–66. https://doi.org/10.4018/IJSST.2019070103

Al Asefer, M., & Zainal Abidin, N. S. (2021). Soft Skills and Graduates' Employability in The 21st Century From Employers' Perspectives: A Review of Literature. *International Journal of Infrastructure Research and Management*, *9*(2), 44–59.

Alismail, A., Thomas, J., Daher, N. S., Cohen, A., Almutairi, W., Terry, M. H., Huang, C., & Tan, L. D. (2019). Augmented reality glasses improve adherence to evidence-based intubation practice. *Advances in Medical Education and Practice*, *10*, 279–286. https://doi.org/10.2147/AMEP.S201640

Altinpulluk, H. (2019). Determining the trends of using augmented reality in education between 2006-2016. *Education and Information Technologies*, *24*(2), 1089–1114. https://doi.org/10.1007/s10639-018-9806-3

Amponsah, K. (2003). Patterns of communication and the implications for learning among two distributed-education student teams. *Proceedings of the 21st annual international conference on Documentation*, 20–27. https://doi.org/10.1145/944868.944874

Anderson, C., & McCune, V. (2013). Fostering meaning: Fostering community. *Higher Education*, *66*(3), 283–296. https://doi.org/10.1007/s10734-012-9604-6

Anderson, L. W., & Krathwohl, D. R. (2001). *A taxonomy for learning, teaching, and assessing : A revision of Bloom's taxonomy of educational objectives* [Accepted: 2015-11-06T15:44:27Z Artwork Medium: Ressource physique Interview Medium: Ressource physique]. Longman, retrieved October 9, 2021, from https://eduq.info/xmlui/handle/11515/18345

Andrews, J. J., & Rapp, D. N. (2015). Benefits, costs, and challenges of collaboration for learning and memory. *Translational Issues in Psychological Science*, *1*(2), 182–191. https://doi.org/10.1037/tps0000025

Andriessen, J., & Baker, M. (2020). *On collaboration: Personal, educational and societal arenas*. Brill Sense.

Archambault, L., Leary, H., & Rice, K. (2022). Pillars of online pedagogy: A framework for teaching in online learning environments. *Educational Psychologist*, *57*(3), 178–191. https://doi.org/10.1080/00461520.2022.2051513

Ardichvili, A., Page, V., & Wentling, T. (2003). Motivation and barriers to participation in virtual knowledge-sharing communities of practice. *Journal of Knowledge Management*, *7*(1), 64–77. https://doi.org/10.1108/13673270310463626

Arifin, Y., Sastria, T. G., & Barlian, E. (2018). User Experience Metric for Augmented Reality Application: A Review. *Procedia Computer Science*, *135*, 648–656. https://doi.org/10.1016/j.procs.2018.08.221

Asrori, M., & Tjalla, A. (2020). Increasing Teamwork Capacity of High School Students through Collaborative Teamwork Learning. *American Journal of Educational Research*, *8*(1), 46–50. https://doi.org/10.12691/education-8-1-7

Augello, A., Caggianese, G., Casoria, L., Gallo, L., Montalti, R., Neroni, P., Rompianesi, G., & Troisi, R. (2023). Augmenting Collaborative Practices: The Hololiver Case Study. *2023 17th International Conference on Signal-Image Technology & Internet-Based Systems (SITIS)*, 234–241. https://doi.org/10.1109/SITIS61268.2023.00043

Azuma, R. T. (1997). A Survey of Augmented Reality. *Presence: Teleoperators and Virtual Environments*, *6*(4), 355–385. https://doi.org/10.1162/pres.1997.6.4.355

Baber, H. (2021). Social interaction and effectiveness of the online learning – A moderating role of maintaining social distance during the pandemic COVID-19. *Asian Education and Development Studies*, *11*(1), 159–171. https://doi.org/10.1108/AEDS-09-2020-0209

Bales, R. F. (1970). *Personality and interpersonal behavior*. Holt, Rinehart; Winston.

Baloche, L., & Brody, C. M. (2017). Cooperative learning: Exploring challenges, crafting innovations. *Journal of Education for Teaching*, *43*(3), 274–283. https://doi.org/10.1080/02607476.2017.1319513

Bergin, D. A. (2016). Social Influences on Interest. *Educational Psychologist*, *51*(1), 7–22. https://doi.org/10.1080/00461520.2015.1133306

Bichard, M. (2005). Managing for Creativity. In *Leadership and Management in the 21st Century: Business Challanges for the Future* (1st ed., pp. 299–304). Oxfor University Press.

Biggs, J. B., Tang, C. S.-k., & Kennedy, G. (2022). *Teaching for quality learning at university* (Fifth edition). Open University Press, McGraw Hill.

Billinghurst, M., & Kato, H. (1999). Collaborative mixed reality. *Proceedings of the first international symposium on mixed reality*, *1*.

Bion, W. (2003, September). *Experiences in Groups* (0th ed.). Routledge. https://doi.org/10.4324/9780203359075

Blackmore, C., et al. (2010). *Social learning systems and communities of practice*. Springer.

Blanco, T., López-Forniés, I., & Zarazaga-Soria, F. J. (2017). Deconstructing the Tower of Babel: A design method to improve empathy and teamwork competences of informatics students. *International Journal of Technology and Design Education*, *27*(2), 307–328. https://doi.org/10.1007/s10798-015-9348-6

Bligh, B., & Crook, C. (2017). Learning Spaces. In E. Duval, M. Sharples, & R. Sutherland (Eds.), *Technology Enhanced Learning: Research Themes* (pp. 69–87). Springer International Publishing. https://doi.org/10.1007/978-3-319-02600-8_7

Bloom, B. S., Engelhart, M. D., Furst, E. J., Hill, W. H., & Krathwohl, D. R. (1964). *Taxonomy of educational objectives* (Vol. 2). Longmans, Green New York.

Boahin, P., & Hofman, A. (2013). A disciplinary perspective of competency-based training on the acquisition of employability skills. *Journal of Vocational Education & Training*, *65*(3), 385–401. https://doi.org/10.1080/13636820.2013.834954

Brown, J. S., & Duguid, P. (1991). Organizational Learning and Communities-of-Practice: Toward a Unified View of Working, Learning, and Innovation. *Organization Science*, *2*(1), 40–57. https://www.jstor.org/stable/2634938

Bruffee, K. A. (1973). Collaborative Learning: Some Practical Models. *College English*, *34*(5), 634. https://doi.org/10.2307/375331

Buchs, C., Filippou, D., Pulfrey, C., & Volpé, Y. (2017). Challenges for cooperative learning implementation: Reports from elementary school teachers. *Journal of Education for Teaching*, *43*(3), 296–306. https://doi.org/10.1080/02607476.2017.1321673

Burton, L. J., Summers, J., Lawrence, J., Noble, K., & Gibbings, P. (2015). Digital Literacy in Higher Education: The Rhetoric and the Reality. In M. K. Harmes, H. Huijser, & P. A. Danaher (Eds.), *Myths in Education, Learning and Teaching: Policies, Practices and Principles* (pp. 151–172). Palgrave Macmillan UK. https://doi.org/10.1057/9781137476982_9

César, M., & Santos, N. (2006). From exclusion to inclusion: Collaborative work contributions to more inclusive learning settings. *European Journal of Psychology of Education*, *21*(3), 333. https://doi.org/10.1007/BF03173420

Chadha, D. (2006). A curriculum model for transferable skills development. *Engineering Education*, *1*(1), 19–24. https://doi.org/10.11120/ened.2006.01010019

Challenor, J., & Ma, M. (2019). A Review of Augmented Reality Applications for History Education and Heritage Visualisation [Number: 2 Publisher: Multidisciplinary Digital Publishing Institute]. *Multimodal Technologies and Interaction*, *3*(2), 39. https://doi.org/10.3390/mti3020039

Checa, D., & Bustillo, A. (2020). Advantages and limits of virtual reality in learning processes: Briviesca in the fifteenth century. *Virtual Reality*, *24*(1), 151–161. https://doi.org/10.1007/s10055-019-00389-7

Chen, P., Liu, X., Cheng, W., & Huang, R. (2017). A review of using Augmented Reality in Education from 2011 to 2016. In E. Popescu, Kinshuk, M. K. Khribi, R. Huang, M. Jemni, N.-S. Chen, & D. G. Sampson (Eds.), *Innovations in Smart Learning* (pp. 13–18). Springer. https://doi.org/10.1007/978-981-10-2419-1_2

Cheng, Y.-W., Wang, Y., Cheng, I.-L., & Chen, N.-S. (2019). An in-depth analysis of the interaction transitions in a collaborative Augmented Reality-based mathematic game. *Interactive Learning Environments*, *27*(5-6), 782–796. https://doi.org/10.1080/10494820.2019.1610448

Chowdhury, S., Endres, M., & Lanis, T. W. (2002). Preparing Students For Success In Team Work Environments: The Importance Of Building Confidence. *Journal of Managerial Issues*, *14*(3), 346–359. Retrieved February 7, 2025, from https://www.jstor.org/stable/40604395

Cochrane, T. (2012). Secrets of mlearning failures: Confronting reality. *Research in Learning Technology*, *20*. https://doi.org/10.3402/rlt.v20i0.19186

Coleman, B., & Goodwin, D. (2017). *Designing UX: Prototyping: Because Modern Design is Never Static*. SitePoint Pty Ltd.

Cook, B. G., & Therrien, W. J. (2017). Null Effects and Publication Bias in Special Education Research. *Behavioral Disorders*, *42*(4), 149–158. https://doi.org/10.1177/0198742917709473

Crant, J. M. (2000). Proactive Behavior in Organizations. *Journal of Management*, *26*(3), 435–462. https://doi.org/10.1177/014920630002600304

Cremers, P. H., Valkenburg, R. A., Kimble, C., Hildreth, P., & Bourdon, I. (2008). Teaching and learning about communities of practice in higher education.

Dalziel, J. (1998). Using Marks to Assess Student Performance, some problems and alternatives. *Assessment & Evaluation in Higher Education*, *23*(4), 351–366. https://doi.org/10.1080/0260293980230403

Damoah, O. B. O., Peprah, A. A., & Brefo, K. O. (2021). Does higher education equip graduate students with the employability skills employers require? The perceptions of employers in Ghana. *Journal of Further and Higher Education*, *45*(10), 1311–1324. https://doi.org/10.1080/0309877X.2020.1860204

Davies, W. M. (2009). Groupwork as a form of assessment: Common problems and recommended solutions. *Higher Education*, *58*(4), 563–584. https://doi.org/10.1007/s10734-009-9216-y

Davis, D., & Ulseth, R. (2013). Building Student Capacity for High Performance Teamwork. *2013 ASEE Annual Conference & Exposition Proceedings*, 23.260.1–23.260.26. https://doi.org/10.18260/1-2--19274

Dawson, P., & Dawson, S. L. (2018). Sharing successes and hiding failures: 'reporting bias' in learning and teaching research. *Studies in Higher Education*, *43*(8), 1405–1416. https://doi.org/10.1080/03075079.2016.1258052

DAWSON, R. (2017). The Value of Collaboration. In *IMPROVING COLLABORATION AND INNOVATION BETWEEN INDUSTRY AND BUSINESS SCHOOLS IN AUSTRALIA* (pp. 29–39).

De Felice, S., Hamilton, A. F. D. C., Ponari, M., & Vigliocco, G. (2023). Learning *from* others is good, *with* others is better: The role of social interaction in human acquisition of new knowledge. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *378*(1870), 20210357. https://doi.org/10.1098/rstb.2021.0357

Dey, A., Billinghurst, M., Lindeman, R. W., & Swan, J. E. (2018). A systematic review of 10 years of augmented reality usability studies: 2005 to 2014. *Frontiers in Robotics and AI*, *5*, 37.

Dillenbourg, P., & Fischer, F. (2007). Computer-supported collaborative learning: The basics. *Zeitschrift für Berufs-und Wirtschaftspädagogik*, *21*, 111–130.

Drejer, I., & Jørgensen, B. H. (2005). The dynamic creation of knowledge: Analysing public–private collaborations. *Technovation*, *25*(2), 83–94. https://doi.org/10.1016/S0166-4972(03)00075-0

Driscoll, A. (2025). *Transforming Traditional Teaching for Today's College Students: Strategies for Cultivating Relationships and Community* (1st ed). Taylor & Francis Group.

Drueke, B., Mainz, V., Lemos, M., Wirtz, M. A., & Boecker, M. (2021). An Evaluation of Forced Distance Learning and Teaching Under Pandemic Conditions Using the Technology Acceptance Model. *Frontiers in Psychology*, *12*. https://doi.org/10.3389/fpsyg.2021.701347

Earl, A., Carbee, V. A., Becerra-Murillo, K., & Evans, A. M. (2021). The Four C's, Constructivism, and Digitizing Curriculum: An Inclusive Approach to Teaching Transferable Skills. In H. K. Dhir (Ed.), *Advances in Educational Technologies and Instructional Design* (pp. 65–86). IGI Global. https://doi.org/10.4018/978-1-7998-6967-2.ch004

Eden, C. A., & Ayeni, O. O. (2024). Social research in primary education management: Understanding community needs and priorities. *World Journal of Advanced Research and Reviews*, *21*(3), 503–511.

Essabbar, D. (2015). Power Imbalance in Collaboration Relationships. *International Journal of Supply and Operations Management*. https://doi.org/10.22034/2015.4.04

Fidan, M., & Tuncel, M. (2019). Integrating augmented reality into problem based learning: The effects on learning achievement and attitude in physics education. *Computers & Education*, *142*, 103635. https://doi.org/10.1016/j.compedu.2019.103635

Fischer, G., Lundin, J., & Lindberg, J. O. (2020). Rethinking and reinventing learning, education and collaboration in the digital age—from creating technologies to transforming cultures. *The International Journal of Information and Learning Technology*, *37*(5), 241–252. https://doi.org/10.1108/IJILT-04-2020-0051

Fisher, D. M. (2014). Distinguishing between taskwork and teamwork planning in teams: Relations with coordination and interpersonal processes. *Journal of Applied Psychology*, *99*(3), 423–436. https://doi.org/10.1037/a0034625

Forsler, I., Bardone, E., & Forsman, M. (2024). The Future Postdigital Classroom. *Postdigital Science and Education*. https://doi.org/10.1007/s42438-024-00488-y

Forsler, I., & Guyard, C. (2025). Screens, teens and their brains. Discourses about digital media, learning and cognitive development in popular science neuroeducation. *Learning, Media and Technology*, *50*(2), 191–204. https://doi.org/10.1080/17439884.2023.2230893

Frederiksen, M. (2014). Trust in the face of uncertainty: A qualitative study of intersubjective trust and risk. *International Review of Sociology*, *24*(1), 130–144. https://doi.org/10.1080/03906701.2014.894335

Friedl-Knirsch, J., Stach, C., Pointecker, F., Anthes, C., & Roth, D. (2024). A Study on Collaborative Visual Data Analysis in Augmented Reality with Asymmetric Display Types. *IEEE Transactions on Visualization and Computer Graphics*, *30*(5), 2633–2643. https://doi.org/10.1109/TVCG.2024.3372103

Fuchs, L. S., & Deno, S. L. (1982, August). *Developing Goals and Objectives for Educational Programs*. Retrieved October 14, 2021, from https://eric.ed.gov/?id=ED249211

Garbett, J., Hartley, T., & Heesom, D. (2021). A multi-user collaborative BIM-AR system to support design and construction. *Automation in Construction*, *122*, 103487. https://doi.org/10.1016/j.autcon.2020.103487

Garst, B. A., Weston, K. L., Bowers, E. P., & Quinn, W. H. (2019). Fostering youth leader credibility: Professional, organizational, and community impacts associated with completion of an online master's degree in youth development leadership. *Children and Youth Services Review*, *96*, 1–9. https://doi.org/10.1016/j.childyouth.2018.11.019

Gasques, D., Johnson, J. G., Sharkey, T., Feng, Y., Wang, R., Xu, Z. R., Zavala, E., Zhang, Y., Xie, W., Zhang, X., Davis, K., Yip, M., & Weibel, N. (2021). ARTEMIS: A Collaborative Mixed-Reality System for Immersive Surgical Telementoring. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 1–14. https://doi.org/10.1145/3411764.3445576

Gil Parga, S. (2015). *Ambientes virtuales de aprendizaje aplicados* [Doctoral dissertation, Universidad de los Andes]. Retrieved July 7, 2025, from https://repositorio.uniandes.edu.co/entities/publication/c5cf341b-1dfe-43bd-ac7c-61294516c8c2

Gil Parga, S., Singh, U., Gutierrez, J., & Marks, S. (2023). Pedagogical design in education using augmented reality: A systematic review. *Interactive Learning Environments*, 1–18. https://doi.org/10.1080/10494820.2023.2195445

Goodyear, P., Banks, S., Hodgson, V., & McConnell, D. (2004). Research on networked learning: An overview. In P. Dillenbourg, M. Baker, C. Bereiter, Y. Engeström, G. Fischer, H. Ulrich Hoppe, T. Koschmann, N. Miyake, C. O'Malley, R. Pea, C. Pontecorovo, J. Roschelle, D. Suthers, P. Goodyear, S. Banks, V. Hodgson, & D. McConnell (Eds.), *Advances in Research on Networked Learning* (pp. 1–9). Springer Netherlands. https://doi.org/10.1007/1-4020-7909-5_1

Gourlay, L., & Oliver, M. (2016). Students' Physical and Digital Sites of Study: Making, Marking, and Breaking Boundaries. In *Place-Based Spaces for Networked Learning* (pp. 73–86). Routledge.

Griffin, J., Brandt, C., Bickel, E., Schnittka, C., & Schnittka, J. (2015). Imbalance of power: A case study of a middle school mixed-gender engineering team. *2015 IEEE Integrated STEM Education Conference*, 64–69. https://doi.org/10.1109/ISECon.2015.7119947

Gross, T. (1997). Towards flexible support for cooperation: Group awareness in shared workspaces. *Database and Expert Systems Applications. 8th International Conference, DEXA '97. Proceedings*, 406–411. https://doi.org/10.1109/DEXA.1997.617324

Gui, M., & Büchi, M. (2021). From Use to Overuse: Digital Inequality in the Age of Communication Abundance. *Social Science Computer Review*, *39*(1), 3–19. https://doi.org/10.1177/0894439319851163

Guskey, T. R. (2001, April). *Benjamin S. Bloom's Contributions to Curriculum, Instruction, and School Learning*. Eric. Retrieved October 9, 2021, from https://eric.ed.gov/?id=ED457185

Hamad, A., & Jia, B. (2022). How Virtual Reality Technology Has Changed Our Lives: An Overview of the Current and Potential Applications and Limitations. *International*

Journal of Environmental Research and Public Health, *19*(18). https://doi.org/10.3390/ijerph191811278
Healy, W. J., Taran, Z., & Betts, S. C. (2011). Sales Course Design Using Experiential Learning Principles and Bloom's Taxonomy [Publisher: Academic and Business Research Institute]. *Journal of Instructional Pedagogies*, *6*. Retrieved October 9, 2021, from https://eric.ed.gov/?id=EJ1097018
Hibbert, P., & Huxham, C. (2005). A little about the mystery: Process learning as collaboration evolves. *European Management Review*, *2*(1), 59–69. https://doi.org/10.1057/palgrave.emr.1500025
Hildreth, P. M., & Kimble, C. (2004). *Knowledge networks: Innovation through communities of practice*. Igi Global.
Hirshfield, L., & Chachra, D. (2015). Task choice, group dynamics and learning goals: Understanding student activities in teams. *2015 IEEE Frontiers in Education Conference (FIE)*, 1–5. https://doi.org/10.1109/FIE.2015.7344043
Hsu, H.-P., Wenting, Z., & Hughes, J. E. (2019). Developing Elementary Students' Digital Literacy Through Augmented Reality Creation: Insights From a Longitudinal Analysis of Questionnaires, Interviews, and Projects. *Journal of Educational Computing Research*, *57*(6), 1400–1435. https://doi.org/10.1177/0735633118794515
Huang, C. Y., Thomas, J. B., Alismail, A., Cohen, A., Almutairi, W., Daher, N. S., Terry, M. H., & Tan, L. D. (2018). The use of augmented reality glasses in central line simulation: "see one, simulate many, do one competently, and teach everyone". *Advances in Medical Education and Practice*, *9*, 357–363. https://doi.org/10.2147/AMEP.S160704
Huang, W., Alem, L., & Tecchia, F. (2013). HandsIn3D: Supporting Remote Guidance with Immersive Virtual Environments. In D. Hutchison, T. Kanade, J. Kittler, J. M. Kleinberg, F. Mattern, J. C. Mitchell, M. Naor, O. Nierstrasz, C. Pandu Rangan, B. Steffen, M. Sudan, D. Terzopoulos, D. Tygar, M. Y. Vardi, G. Weikum, P. Kotzé, G. Marsden, G. Lindgaard, J. Wesson, & M. Winckler (Eds.), *Human-Computer Interaction – INTERACT 2013* (pp. 70–77, Vol. 8117). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-40483-2_5
Huynh, N.-T., Nguyen, P.-N. T., Nguyen, T.-N. T., Do, C.-T., Nguyen, H.-T., & Huynh, T.-V. M. (2024). Cultivating Job-Seeking Skills in Youth: Exploring Vietnamese EFL Learners' Experiences With the English for Employability Skills Program. *Journal of Language Teaching and Research*, *15*(6), 2020–2033. https://doi.org/10.17507/jltr.1506.27
Imaishi, H., Kaneko, Y., Pondsmith, M., Sztybor, B., & Ôtsuka, M. (2022, September). Cyberpunk: Edgerunners (episode 1).
Ivanova, A. V. (2018). VR & AR TECHNOLOGIES: OPPORTUNITIES AND APPLICATION OBSTACLES. *Strategic decisions and risk management*, (3), 88–107. https://doi.org/10.17747/2078-8886-2018-3-88-107
Jacobs, G. M. (2015). *Collaborative Learning or Cooperative Learning? The Name Is Not Important; Flexibility Is* (tech. rep.) (ERIC Number: ED574149). Retrieved May 26, 2025, from https://eric.ed.gov/?id=ED574149
Jahng, N., Nielsen, W. S., & Chan, E. K. H. (2010). Collaborative Learning in an Online Course: A Comparison of Communication Patterns in Small and Large Group

Activities. *International Journal of E-Learning & Distance Education / Revue internationale du e-learning et la formation à distance*, *24*(2). Retrieved May 26, 2025, from https://www.ijede.ca/index.php/jde/article/view/647

Jaques, D., & Salmon, G. (2007, January). *Learning in Groups* (0th ed.). Routledge. https://doi.org/10.4324/9780203016459

Javornik, A. (2016). The Mainstreaming of Augmented Reality: A Brief History [Section: Technology and analytics]. *Harvard Business Review*. Retrieved September 27, 2021, from https://hbr.org/2016/10/the-mainstreaming-of-augmented-reality-a-brief-history

Jeffrey, L., Hegarty, B., Kelly, O., Penman, M., Coburn, D., & McDonald, J. (2011). Developing Digital Information Literacy in Higher Education: Obstacles and Supports. *Journal of Information Technology Education: Research*, *10*(1), 383–413. https://www.learntechlib.org/p/111528

Jun, E. K., Lim, S., Seo, J., Lee, K. H., Lee, J. H., Lee, D., & Koh, J. C. (2023). Augmented Reality-Assisted Navigation System for Transforaminal Epidural Injection. *Journal of Pain Research*, *16*, 921–931. https://doi.org/10.2147/JPR.S400955

Katz, I. R., LaMar, M. M., Spain, R., Zapata-Rivera, J. D., Baird, J.-A., & Greiff, S. (2017). Issues and concerns for technology-based performance assessments. In *Design recommendations for intelligent tutoring system* (pp. 209–224, Vol. 5).

Keay, J. (2006). Collaborative learning in physical education teachers' early-career professional development. *Physical Education and Sport Pedagogy*, *11*(3), 285–305.

Keay, J., & O'Mahony, H. M. a. J. (2016). Improving learning and teaching in transnational education: Can communities of practice help? In *Transnational and Transcultural Positionality in Globalised Higher Education*. Routledge.

Kechagias, K. (2011). Teaching and assessing soft skills.

Kemp, I. J., & Seagraves, L. (1995). Transferable skills—Can higher education deliver? *Studies in Higher Education*, *20*(3), 315–328. https://doi.org/10.1080/03075079512331381585

Kernaghan, J., & Cooke, R. (1990). Teamwork in planning innovative projects: Improving group performance by rational and interpersonal interventions in group process. *IEEE Transactions on Engineering Management*, *37*(2), 109–116. https://doi.org/10.1109/17.53713

Kim, H., Matuszka, T., Kim, J.-I., Kim, J., & Woo, W. (2017). Ontology-based mobile augmented reality in cultural heritage sites: Information modeling and user study. *Multimedia Tools and Applications*, *76*(24), 26001–26029. https://doi.org/10.1007/s11042-017-4868-6

Kirkwood, A., & Price, L. (2014). Technology-enhanced learning and teaching in higher education: What is 'enhanced'and how do we know? a critical literature review. *Learning, media and technology*, *39*(1), 6–36.

Kirschner, P. A., & Lai, K.-W. (2007). Online communities of practice in education. *Technology, Pedagogy and Education*, *16*(2), 127–131. https://doi.org/10.1080/14759390701406737

Kiyokawa, K., Billinghurst, M., Hayes, S., Gupta, A., Sannohe, Y., & Kato, H. (2002). Communication behaviors of co-located users in collaborative AR interfaces.

Proceedings. International Symposium on Mixed and Augmented Reality, 139–148. https://doi.org/10.1109/ISMAR.2002.1115083

Klein, J. H., & Connell, C. (2008). The identification and cultivation of appropriate communities of practice in higher education. In C. Kimble, P. Hildreth, & I. Bourdon (Eds.). Information Age Publishing.

Klein, J. H., Connell, N. A. D., & Meyer, E. (2005). Knowledge characteristics of communities of practice. *Knowledge Management Research & Practice*, *3*(2), 106–114. https://doi.org/10.1057/palgrave.kmrp.8500055

Klinker, G., Dutoit, A., Bauer, M., Bayer, J., Novak, V., & Matzke, D. (2002). Fata Morgana - a presentation system for product design. *Proceedings. International Symposium on Mixed and Augmented Reality*, 76–85. https://doi.org/10.1109/ISMAR.2002.1115076

Kloepper, L. (2023). Making group-work work: Navigating conflict and encouraging collaborative revision. *The Journal of the Acoustical Society of America*, *153*(3_supplement), A273–A273. https://doi.org/10.1121/10.0018821

Knowles, E. S. (1982). From Individuals to Group Members: A Dialectic for the Social Sciences. In W. Ickes & E. S. Knowles (Eds.), *Personality, Roles, and Social Behavior* (pp. 1–32). Springer. https://doi.org/10.1007/978-1-4613-9469-3_1

Kowalski, M., & Christensen, A. (2019). "No One Wants to be a Loser:" High School Students' Perceptions of Academic Competition. *Mid-Western Educational Researcher*, *31*(4). https://scholarworks.bgsu.edu/mwer/vol31/iss4/2

Kramer, R., & Tyler, T. (1996). *Trust in Organizations: Frontiers of Theory and Research*. SAGE Publications, Inc. https://doi.org/10.4135/9781452243610

Krathwohl, D. R. (2002). A Revision of Bloom's Taxonomy: An Overview. *Theory Into Practice*, *41*(4), 212–218. https://doi.org/10.1207/s15430421tip4104_2

Kropp, S., & Dodd, D. (2024). Accountability, Ownership, and Satisfaction: An Innovative Approach to Teamwork in Engineering Education. *2024 ASEE Annual Conference & Exposition Proceedings*, 46516. https://doi.org/10.18260/1-2--46516

Laal, M., & Ghodsi, S. M. (2012). Benefits of collaborative learning. *Procedia - Social and Behavioral Sciences*, *31*, 486–490. https://doi.org/10.1016/j.sbspro.2011.12.091

Lave, J., & Wenger, E. (1991). *Situated learning: Legitimate peripheral participation*. Cambridge University Press.

Lee, I.-J. (2021). Kinect-for-windows with augmented reality in an interactive roleplay system for children with an autism spectrum disorder [Publisher: Routledge _eprint: https://doi.org/10.1080/10494820.2019.1710851]. *Interactive Learning Environments*, *29*(4), 688–704. https://doi.org/10.1080/10494820.2019.1710851

Lehtinen, E., Hakkarainen, K., Lipponen, L., Rahikainen, M., & Muukkonen, H. (1999). Computer supported collaborative learning: A review. *The JHGI Giesbers reports on education*, *10*, 1999.

Lim, K. Y. T., & Lim, R. (2020). Semiotics, memory and augmented reality: History education with learner-generated augmentation. *British Journal of Educational Technology*, *51*(3), 673–691. https://doi.org/10.1111/bjet.12904

Lingard, R., & Barkataki, S. (2012). Teaching teamwork in engineering and computer science. https://doi.org/10.1109/FIE.2011.6143000

Liu, F., Xu, Y., Yang, T., Li, Z., Dong, Y., Chen, L., & Sun, X. (2022). The mediating roles of time management and learning strategic approach in the relationship between

smartphone addiction and academic procrastination. *Psychology Research and Behavior Management*, 2639–2648.
Llorens, A., Berbegal-Mirabent, J., & Llinàs-Audet, X. (2017). Aligning professional skills and active learning methods: An application for information and communications technology engineering. *European Journal of Engineering Education*, *42*(4), 382–395. https://doi.org/10.1080/03043797.2016.1189880
Lo, H.-C. (2009). Utilizing Computer-mediated Communication Tools for Problem-based Learning. *Educational Technology & Society*, *12*(1), 205–213.
Long, P. D., & Ehrmann, S. C. (2005). The future of the learning space: Breaking out of the box. *EDUCAUSE review*, *40*(4), 42–58.
Lu, J., & Churchill, D. (2013). Creating personal learning environments to enhance learning engagement. *2013 IEEE 63rd Annual Conference International Council for Education Media (ICEM)*, 1–8. https://doi.org/10.1109/CICEM.2013.6820194
Madden, L., Beyers, J. E., & Stanton, N. (2020). Future elementary teachers' perspectives on the importance of stem. In *Critical questions in stem education* (pp. 211–225). Springer.
Marasi, S. (2019). Team-building: Developing Teamwork Skills in College Students Using experiential Activities in a Classroom Setting. *Organization Management Journal*, *16*(4). https://scholarship.shu.edu/omj/vol16/iss4/9
Margaryan, A., Littlejohn, A., & Vojt, G. (2011). Are digital natives a myth or reality? University students' use of digital technologies. *Computers & Education*, *56*(2), 429–440. https://doi.org/10.1016/j.compedu.2010.09.004
Marques, B., Silva, S., Alves, J., Araújo, T., Dias, P., & Santos, B. S. (2022). A Conceptual Model and Taxonomy for Collaborative Augmented Reality. *IEEE Transactions on Visualization and Computer Graphics*, *28*(12), 5113–5133. https://doi.org/10.1109/TVCG.2021.3101545
McCale, C. (2008). It's Hard Work Learning Soft Skills: Can Client Based Projects Teach the Soft Skills Students Need and Employers Want? [ERIC Number: EJ1055586]. *Journal of Effective Teaching*, *8*(2), 50–60. Retrieved June 29, 2025, from https://eric.ed.gov/?id=EJ1055586
McDonald, J., & Cater-Steel, A. (Eds.). (2017). *Communities of Practice: Facilitating Social Learning in Higher Education*. Springer Singapore. https://doi.org/10.1007/978-981-10-2879-3
McInnerney, J. M., & Roberts, T. S. (2004). Online learning: Social interaction and the creation of a sense of community. *Journal of Educational Technology & Society*, *7*(3), 73–81.
Milgram, P., Takemura, H., Utsumi, A., & Kishino, F. (1995). Augmented reality: A class of displays on the reality-virtuality continuum. *Telemanipulator and Telepresence Technologies*, *2351*, 282–292. https://doi.org/10.1117/12.197321
Moro, C., Phelps, C., Redmond, P., & Stromberga, Z. (2021). HoloLens and mobile augmented reality in medical and health science education: A randomised controlled trial. *British Journal of Educational Technology*, *52*(2), 680–694. https://doi.org/10.1111/bjet.13049
Murphy, K. L., & Cifuentes, L. (2001). Using Web tools, collaborating, and learning online. *Distance Education*, *22*(2), 285–305. https://doi.org/10.1080/0158791010220207

Murray, M., & Perez, J. (2014). Unraveling the digital literacy paradox: How higher education fails at the fourth literacy. *Issues in Informing Science and Information Technology*, *11*, 85–100. https://digitalcommons.kennesaw.edu/facpubs/3438

Nadarajah, J. (2021). MEASURING THE GAP IN EMPLOYABILITY SKILLS AMONG MALAYSIAN GRADUATES. *International Journal of Modern Trends in Social Sciences*, *4*(15), 81–87. https://doi.org/10.35631/IJMTSS.415007

Newton, P. M., Da Silva, A., & Peters, L. G. (2020). A Pragmatic Master List of Action Verbs for Bloom's Taxonomy. *Frontiers in Education*, *5*, 107. https://doi.org/10.3389/feduc.2020.00107

Ng, P. M., Chan, J. K. Y., Wut, T. M., Lo, M. F., & Szeto, I. (2021). What makes better career opportunities for young graduates? Examining acquired employability skills in higher education institutions. *Education + Training*, *63*(6), 852–871. https://doi.org/10.1108/ET-08-2020-0231

Nissen, H. A., Evald, M. R., & Clarke, A. H. (2014). Knowledge sharing in heterogeneous teams through collaboration and cooperation: Exemplified through Public–Private-Innovation partnerships. *Industrial Marketing Management*, *43*(3), 473–482. https://doi.org/10.1016/j.indmarman.2013.12.015

Nørgård, R. T., & Hilli, C. (2022). Hyper-Hybrid Learning Spaces in Higher Education. In *Hybrid Learning Spaces* (pp. 25–41).

Nuere, S., Suz, A. A., & Álvarez, L. d. M. (2021). Reflections on the adaptation to the COVID-19 pandemic in higher education. In *Remote Learning in Times of Pandemic*. Routledge.

Odunaike, S., Olugbara, O., & Ojo, S. (2013). E-learning implementation critical success factors. *innovation*, *3*(4).

Özgen, D. S., Afacan, Y., & Sürer, E. (2021). Usability of virtual reality for basic design education: A comparative study with paper-based design. *International Journal of Technology and Design Education*, *31*(2), 357–377. https://doi.org/10.1007/s10798-019-09554-0

Paechter, M., & Maier, B. (2010). Online or face-to-face? Students' experiences and preferences in e-learning. *The Internet and Higher Education*, *13*(4), 292–297. https://doi.org/10.1016/j.iheduc.2010.09.004

Page, M. J., Moher, D., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., … McKenzie, J. E. (2021). PRISMA 2020 explanation and elaboration: Updated guidance and exemplars for reporting systematic reviews [Publisher: British Medical Journal Publishing Group Section: Research Methods & Reporting]. *BMJ*, *372*, n160. https://doi.org/10.1136/bmj.n160

Passey, D. (2019). Technology-enhanced learning: Rethinking the term, the concept and its theoretical background. *British Journal of Educational Technology*, *50*(3), 972–986. https://doi.org/10.1111/bjet.12783

Penn, M., & Ramnarain, U. (2023). A Systematic Review of Pedagogy Related to Mixed Reality in K-12 Education. *Mixed Reality for Education*, 85–108.

Peppler, K. A., & Solomou, M. (2011). Building creativity: Collaborative learning and creativity in social media environments. *On the horizon*, *19*(1), 13–23.

Petchamé, J., Iriondo, I., Villegas, E., Riu, D., & Fonseca, D. (2021). Comparing Face-to-Face, Emergency Remote Teaching and Smart Classroom: A Qualitative Exploratory Research Based on Students' Experience during the COVID-19 Pandemic. *Sustainability*, *13*(12), 6625. https://doi.org/10.3390/su13126625

Petrucco, C. (2021). Smartphones in and Outside Classroom. In *Teaching and Mobile Learning*. CRC Press.

Pooryousef, V., Cordeil, M., Besançon, L., Bassed, R., & Dwyer, T. (2024). Collaborative Forensic Autopsy Documentation and Supervised Report Generation Using a Hybrid Mixed-Reality Environment and Generative AI. *IEEE Transactions on Visualization and Computer Graphics*, *30*(11), 7452–7462. https://doi.org/10.1109/TVCG.2024.3456212

Popov, V., Brinkman, D., Biemans, H. J., Mulder, M., Kuznetsov, A., & Noroozi, O. (2012). Multicultural student group work in higher education. *International Journal of Intercultural Relations*, *36*(2), 302–317. https://doi.org/10.1016/j.ijintrel.2011.09.004

Radmehr, F., & Drake, M. (2017). Revised Bloom's taxonomy and integral calculus: Unpacking the knowledge dimension. *International Journal of Mathematical Education in Science and Technology*, *48*(8), 1206–1224. https://doi.org/10.1080/0020739X.2017.1321796

Radu, I., & Schneider, B. (2023). How Augmented Reality (AR) Can Help and Hinder Collaborative Learning: A Study of AR in Electromagnetism Education. *IEEE Transactions on Visualization and Computer Graphics*, *29*(9), 3734–3745. https://doi.org/10.1109/TVCG.2022.3169980

Regenbrecht, H., Wagner, M., & Baratoff, G. (2002). MagicMeeting: A Collaborative Tangible Augmented Reality System. *Virtual Reality*, *6*(3), 151–166. https://doi.org/10.1007/s100550200016

Reinius, H., Korhonen, T., & Hakkarainen, K. (2021). The design of learning spaces matters: Perceived impact of the deskless school on learning and teaching. *Learning Environments Research*, *24*(3), 339–354. https://doi.org/10.1007/s10984-020-09345-8

Rezaei, A. R. (2018). Effective Groupwork Strategies: Faculty and Students' Perspectives [ERIC Number: EJ1182589]. *Journal of Education and Learning*, *7*(5), 1–10. Retrieved February 9, 2025, from https://eric.ed.gov/?id=EJ1182589

Richards, T. (2023). Reimagining the classroom : Creating new learning spaces and connecting with the world /.

Riverin, S., & Stacey, E. (2008). Sustaining an Online Community of Practice: A Case Study. *International Journal of E-Learning & Distance Education / Revue internationale du e-learning et la formation à distance*, *22*(2).

Rivoltella, P. C. (2008). Introducing Multimedia in the Classroom. In *Empowerment Through Media Education: An Intercultural Dialogue* (pp. 201–210). The International Clearinghouse on Children, Youth; Media.

Rodden, T. (1991). A survey of CSCW systems. *Interacting with Computers*, *3*(3), 319–353. https://doi.org/10.1016/0953-5438(91)90020-3

Roschelle, J., & Teasley, S. D. (1995). The Construction of Shared Knowledge in Collaborative Problem Solving. In C. O'Malley (Ed.), *Computer Supported*

Collaborative Learning (pp. 69–97). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-85098-1_5

Saidin, N. F., Abd Halim, N., & Yahaya, N. (2015). A Review of Research on Augmented Reality in Education: Advantages and Applications. *International Education Studies*, *8*. https://doi.org/10.5539/ies.v8n13p1

Salmons, J. (2019). *Learning to collaborate, collaborating to learn: Engaging students in the classroom and online* (First edition). Stylus.

Sanabria, J. C., & Arámburo-Lizárraga, J. (2017). Enhancing 21st Century Skills with AR: Using the Gradual Immersion Method to develop Collaborative Creativity. *EURASIA Journal of Mathematics, Science and Technology Education*, *13*(2). https://doi.org/10.12973/eurasia.2017.00627a

Schaper, M.-M., Santos, M., Malinverni, L., Zerbini Berro, J., & Pares, N. (2018). Learning about the past through situatedness, embodied exploration and digital augmentation of cultural heritage sites. *International Journal of Human-Computer Studies*, *114*, 36–50. https://doi.org/10.1016/j.ijhcs.2018.01.003

Schiffeler, N., Stehling, V., Hees, F., & Isenhardt, I. (2019). Effects of Collaborative Augmented Reality on Communication and Interaction in Learning Contexts – Results of a Qualitative Pre-Study. *2019 ASEE Annual Conference & Exposition Proceedings*, 32694. https://doi.org/10.18260/1-2--32694

Schmalstieg, D. (2016). *Augmented reality : Principles and practice /*. Addison-Wesley,

Schmalstieg, D., Fuhrmann, A., Hesina, G., Szalavári, Z., Encarnação, L. M., Gervautz, M., & Purgathofer, W. (2002). The *Studierstube* Augmented Reality Project. *Presence: Teleoperators and Virtual Environments*, *11*(1), 33–54. https://doi.org/10.1162/105474602317343640

Schott, D., Kunz, M., Mandel, J., Schwenderling, L., Braun-Dullaeus, R., & Hansen, C. (2024). An AR-Based Multi-User Learning Environment for Anatomy Seminars. *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, 949–950. https://doi.org/10.1109/VRW62533.2024.00271

Selwyn, N. (2009). The digital native – myth and reality. *Aslib Proceedings*, *61*(4), 364–379. https://doi.org/10.1108/00012530910973776

Shank, P. (2005). The Value of Multimedia in Learning. *ADOBE DESIGN CENTER*, (501), 2–12.

Shen, Y., Ong, S., & Nee, A. (2010). Augmented reality for collaborative product design and development. *Design Studies*, *31*(2), 118–145. https://doi.org/10.1016/j.destud.2009.11.001

Shinya, O., Yusuke, I., & Hiroki, T. (2016). The Condition for Generous Trust (N. Georgantzis, Ed.). *PLOS ONE*, *11*(11), e0166437. https://doi.org/10.1371/journal.pone.0166437

Siddaway, A. P., Wood, A. M., & Hedges, L. V. (2019). How to Do a Systematic Review: A Best Practice Guide for Conducting and Reporting Narrative Reviews, Meta-Analyses, and Meta-Syntheses [_eprint: https://doi.org/10.1146/annurev-psych-010418-102803]. *Annual Review of Psychology*, *70*(1), 747–770. https://doi.org/10.1146/annurev-psych-010418-102803

Smaldone, F., Ippolito, A., Lagger, J., & Pellicano, M. (2022). Employability skills: Profiling data scientists in the digital labour market. *European Management Journal*, *40*(5), 671–684. https://doi.org/10.1016/j.emj.2022.05.005

Smith, K. A., Johnson, D. W., & Johnson, R. T. (1992). Cooperative Learning and Positive Change in Higher Education. In *Collaborative Learning: A Sourcebook for Higher Education* (pp. 46–56). National Center on Postsecondary Teaching, Learning; Assessment, University Park, PA.

Sohmen, V. (2015). Leadership and Teamwork: Two Sides of the Same Coin. https://doi.org/10.13140/RG.2.1.4241.7766

Stanny, C. (2016). Reevaluating Bloom's Taxonomy: What Measurable Verbs Can and Cannot Say about Student Learning. *Education Sciences*, *6*(4), 37. https://doi.org/10.3390/educsci6040037

Sutipitakwong, S., & Jamsri, P. (2020). Pros and Cons of Tangible and Digital Wireframes [ISSN: 2377-634X]. *2020 IEEE Frontiers in Education Conference (FIE)*, 1–5. https://doi.org/10.1109/FIE44824.2020.9274234

Swan, K. (2002). Building Learning Communities in Online Courses: The importance of interaction. *Education, Communication & Information*, *2*(1), 23–49. https://doi.org/10.1080/1463631022000005016

Tang, A., Owen, C., Biocca, F., & Mou, W. (2003). Comparative effectiveness of augmented reality in object assembly. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 73–80. https://doi.org/10.1145/642611.642626

Tauer, J. M., & Harackiewicz, J. M. (2004). The Effects of Cooperation and Competition on Intrinsic Motivation and Performance. *Journal of Personality and Social Psychology*, *86*(6), 849–861. https://doi.org/10.1037/0022-3514.86.6.849

Thai, N. T. T., De Wever, B., & Valcke, M. (2020). Face-to-face, blended, flipped, or online learning environment? Impact on learning performance and student cognitions. *Journal of Computer Assisted Learning*, *36*(3), 397–411. https://doi.org/10.1111/jcal.12423

Thomas, A. (2005). Children Online: Learning in a Virtual Community of Practice. *E-Learning and Digital Media*, *2*(1), 27–38. https://doi.org/10.2304/elea.2005.2.1.27

Tulgan, B. (2023). *Bridging the soft skills gap: How to teach the missing basics to the new hybrid workforce* (Second edition). Wiley.

Turnbull, M., Littlejohn, A., & Allan, M. (2010). Creativity and Collaborative Learning and Teaching Strategies in the Design Disciplines. *Industry and Higher Education*, *24*(2), 127–133. https://doi.org/10.5367/000000010791191029

UNESCO. (2014). Skills for holistic human development. Retrieved February 3, 2025, from https://unesdoc.unesco.org/ark:/48223/pf0000245064

Upadhyay, B., Brady, C., Chalil Madathil, K., Bertrand, J., McNeese, N. J., & Gramopadhye, A. (2024). Collaborative augmented reality in higher education: A systematic review of effectiveness, outcomes, and challenges. *Applied Ergonomics*, *121*, 104360. https://doi.org/10.1016/j.apergo.2024.104360

Van Der Vegt, G. S., De Jong, S. B., Bunderson, J. S., & Molleman, E. (2010). Power Asymmetry and Learning in Teams: The Moderating Role of Performance Feedback. *Organization Science*, *21*(2), 347–361. https://doi.org/10.1287/orsc.1090.0452

Van Wyk, J., & Haffejee, F. (2017). Benefits of Group Learning As a Collaborative Strategy in a Diverse Higher Education Context. *International Journal of Educational Sciences*, *18*(1-3), 158–163. https://doi.org/10.1080/09751122.2017.1305745

Vangen, S., & Huxham, C. (2012). The Tangled Web: Unraveling the Principle of Common Goals in Collaborations. *Journal of Public Administration Research and Theory*, *22*(4), 731–760. https://doi.org/10.1093/jopart/mur065

Vassigh, S., Davis, D., Behzadan, A. H., Mostafavi, A., Rashid, K., Alhaffar, H., Elias, A., & Gallardo, G. (2020). Teaching Building Sciences in Immersive Environments: A Prototype Design, Implementation, and Assessment. *International Journal of Construction Education and Research*, *16*(3), 180–196. https://doi.org/10.1080/15578771.2018.1525445

Wagner, S. M., Eggert, A., & Lindemann, E. (2010). Creating and appropriating value in collaborative relationships. *Journal of Business Research*, *63*(8), 840–848. https://doi.org/10.1016/j.jbusres.2010.01.004

Walas-Trębacz, J., Krzyżak, J., Herdan, A., Setyohadi, D. B., Jeyanathan, J. S., & Nair, A. (2025). Social interactions and attitudes toward remote learning vs perceptions of its effectiveness in the view of Education 4.0. *The TQM Journal*, *ahead-of-print*(ahead-of-print). https://doi.org/10.1108/TQM-09-2024-0335

Wang, Q. (2009). Design and evaluation of a collaborative learning environment. *Computers & Education*, *53*(4), 1138–1146. https://doi.org/10.1016/j.compedu.2009.05.023

Wang, X., & Dunston, P. S. (2011). Comparative Effectiveness of Mixed Reality-Based Virtual Environments in Collaborative Design. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, *41*(3), 284–296. https://doi.org/10.1109/TSMCC.2010.2093573

Warburton, S. (2009). Second Life in higher education: Assessing the potential for and the barriers to deploying virtual worlds in learning and teaching. *British Journal of Educational Technology*, *40*(3), 414–426. https://doi.org/10.1111/j.1467-8535.2009.00952.x

Watters, A. (2021). *Teaching machines*. The MIT Press.

Weiser, M. (1991). The Computer for the 21st Century. *Scientific American*, *265*(3), 94–104. https://doi.org/10.1038/scientificamerican0991-94

Wenger, E. (1998, July). *Communities of Practice: Learning, Meaning, and Identity* (1st ed.). Cambridge University Press. https://doi.org/10.1017/CBO9780511803932

Wenger, E., McDermott, R. A., & Snyder, W. (2002). *Cultivating communities of practice: A guide to managing knowledge*. Harvard Business School Press.

Widayati, C. C., Arijanto, A., Magita, M., & Septiana, D. (2022). The Effect of Emotional Intelligence, Teamwork, Organizational Culture and Empathy on Employee Performance: https://doi.org/10.2991/assehr.k.220407.120

Wisotzky, E. L., Rosenthal, J.-C., Eisert, P., Hilsmann, A., Schmid, F., Bauer, M., Schneider, A., & Uecker, F. C. (2019). Interactive and Multimodal-based Augmented Reality for Remote Assistance using a Digital Surgical Microscope [ISSN: 2642-5254]. *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 1477–1484. https://doi.org/10.1109/VR.2019.8797682

Xefteris, S., Palaigeorgiou, G., & Tsorbari, A. (2019). A Learning Environment for Geography and History Using Mixed Reality, Tangible Interfaces and Educational

Robotics. In M. E. Auer & T. Tsiatsos (Eds.), *The Challenges of the Digital Transformation in Education* (pp. 106–117, Vol. 917). Springer International Publishing. https://doi.org/10.1007/978-3-030-11935-5_11

Yasojima, E. K. K., Meiguins, B. S., & Meiguins, A. S. G. (2012). Collaborative Augmented Reality Application for Information Visualization Support [ISSN: 2375-0138]. *2012 16th International Conference on Information Visualisation*, 164–169. https://doi.org/10.1109/IV.2012.37

Yu, K., Roth, D., Strak, R., Pankratz, F., Reichling, J., Kraetsch, C., Weidert, S., Lazarovici, M., Navab, N., & Eck, U. (2023). Mixed Reality 3D Teleconsultation for Emergency Decompressive Craniotomy: An Evaluation with Medical Residents [ISSN: 2473-0726]. *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 662–671. https://doi.org/10.1109/ISMAR59233.2023.00081

Zand, D. E. (1972). Trust and Managerial Problem Solving. *Administrative Science Quarterly*, *17*(2), 229. https://doi.org/10.2307/2393957

Zandvakili, E., Washington, E., Gordon, E. W., Wells, C., & Mangaliso, M. (2019). Teaching Patterns of Critical Thinking: The 3CA Model—Concept Maps, Critical Thinking, Collaboration, and Assessment. *SAGE Open*, *9*(4), 215824401988514. https://doi.org/10.1177/2158244019885142

Zhang, H., Khaskheli, A., Raza, S. A., & Masood, A. (2024). Linkage between Students' Skills and Employability: Moderating Influence of University Reputation. *Corporate Reputation Review*, *27*(4), 229–248. https://doi.org/10.1057/s41299-023-00169-9

Zubiri-Esnaola, H., Vidu, A., Rios-Gonzalez, O., & Morla-Folch, T. (2020). Inclusivity, participation and collaboration: Learning in interactive groups. *Educational Research*, *62*(2), 162–180. https://doi.org/10.1080/00131881.2020.1755605